

New developments in sequential change point detection for time series and spatio-temporal analysis

by

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A PhD Dissertation

Submitted to the Faculty

of the

WORCESTER POLYTECHNIC INSTITUTE

In partial fulfillment of the requirements for the

Degree of Doctor of Philosophy

in

Statistics

May, 2023

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Abstract

Abrupt aberrations in stochastic systems often result from external factors of interest, such as changes in trading intensity patterns or outbreaks of infectious diseases. These factors can introduce abnormal observations into the corresponding data collection systems. However, the data being monitored typically involve multiple sources, high dimensionality, and convoluted mutual dependence. To promptly detect any change points within complex streaming data, my dissertation research focuses on developing efficient methods for sequential change point detection and multivariate time series modeling.

First, we focus on the study of online structural break detection in financial durations. We propose an ensemble non-parametric methodology that leverages asymptotic theories and re-sampling approaches for robust structural break detection, integrated with semi-parametric model inference techniques. By detecting changes in the pattern of financial durations, practitioners can take advantage of short-term profit opportunities through volatility-related option trading or adjust their position to mitigate the impact of sell-offs in the high-frequency financial market.

Second, we develop a Bayesian hierarchical framework with bivariate temporal effect and latent level-correlated effect for multivariate discrete-valued financial time series. Our framework enables the analysis of how count data relates to relevant covariates and provides forecasts for future individual count data. Additionally, it establishes a connection between time-varying observational correlation and latent correlations to more accurately quantify the association between transaction counts at various risk levels. The INLA implementation of this framework grants computational efficiency and flexibility for large-scale numerical studies.

Third, to address the complexity of the surveillance data, such as the spatio-temporal interdependence, we synthesize relevant techniques from the previous two research projects and propose an iterative sequential outbreak detection procedure for online spatio-temporal daily count data. Specifically, we develop a Bayesian online spatio-temporal outbreak detection with prior updating and p-value adaptation (BOSTON-PUPA) procedure. This iterative procedure involves the generalized Poisson distribution (GPD) model and supports synchronous surveillance over multiple locations with a controlled false detection rate as well as high sensitivity against outbreaks in a wide range of signal-to-noise ratios.

Our research tackles various sequential change point problems across different scenarios, providing efficient modeling for multivariate time series and corresponding sequential change point detection techniques for time-dependent and spatio-temporal data. These methodologies

have been successfully applied in real-world applications such as finance and public health, where they offer high-quality statistical inference in an online fashion and can be easily extended to other domains using a similar framework.

Acknowledgements

Having my five-year Ph.D. journey was both a challenging and rewarding experience. Seeking research-oriented solutions to advanced real-world problems required not only individual tenacity but also a significant amount of support and guidance from different great people along this road.

Firstly, I want to express my sincere gratitude towards my Ph.D. advisor, Professor Jian Zou. Not only does he share a wealth of research experience and communication skills with me professionally, but also demonstrates his exemplary work ethic and family commitment in a well coordinated manner. As his Ph.D. student, I was lucky to have his heartfelt cheers for my research as well as career achievements and I also felt blessed to receive his encouragement during my hardships. With his advice and feedback, the progress of my dissertation was made much smoother and more efficient. Professor Zou set an excellent example to me about being an outstanding statistician with a combined trait of decent statistical background and high interpersonal intelligence, which I will admire for my own career development in the future.

Secondly, I also want to thank the rest of my dissertation committee members, Professor Nalini Ravishanker, Professor Fangfang Wang, Professor Zheyang Wu and Professor Qingshuo Song for their generous inputs to my dissertation research. It was a very fulfilling experience for me to collaborate with and learn from Professor Ravishanker since her high-standard requirements for research deliverable and manuscript writing inspired me to take extra miles to justify and refine my work. Expertise of Professor Wang and Professor Wu was a key motivating factor for me to delve into advanced research areas such as multivariate time series analysis in spatial econometric research and correlated p-value combination methods in genetic study. With Professor Song's affluent financial modeling experience and unique insights into article organization, numerous edits of my dissertation were stemming from his constructive suggestions. Their significant impacts on my dissertation accomplishment are invaluable and are deeply appreciated.

I am also grateful for having Professor Balgobin Nandram as my independent study instructor for advanced Bayesian statistics. His dedication and mastery towards Bayesian statistics expedited my understanding of this field in a solid fashion. My special appreciation goes to Ziji Yu, Jianchang Lin from Takeda and Yaohua Zhang from Vertex as my external career influencers, since their contribution to solving real world problems as prestigious industrial biostatisticians broadened my

horizon over the real application of statistics. In addition, I want to send my thankfulness to other faculty members and staff who helped me along my journey, especially Rhonda Podell for her administrative support and Mike Malone for his technical support, when I was a Teaching Assistant. I also appreciated the quality time with my Ph.D. peers from various majors taking advanced courses, brainstorming project problems, discussing career plans, and having fun together.

At last, I want to give my genuine thanks to my Dad Tanggui Wang, my Mom Jijuan Hu, and my cousin Cheng Wang for their unconditional love and support, which is a crucial driving force for me to persevere in the face of various obstacles and barriers.

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1 Introduction

1.1 Sequential change point detection

In real life, abrupt behavioral changes in a stochastic system are usually related to some triggering events and people need to take prompt action at the occurrence of such events. Therefore, online change point detection in many fields is an important research topic. Nowadays, with the advancement of technology, it is common for streaming data to have high-dimensional structures with time-dependence and intercorrelation. To make timely and accurate inference on change point, analyses of such modern streaming data with increased complexity and sizes are in great demand of methodological innovation beyond traditional statistical techniques.

Early online change point detection method originated from Abraham Wald’s sequential design (Wald, 1947), which aimed for early termination of an unpromising experiment with fewer observations. Page (1954) made further improvement and proposed CUSUM-type method for quality control application in industry, with hypothesis testing on the CUSUM-type test statistic under null hypothesis of no change points. Such hypothesis testing procedures are usually considered distribution-free and no model fitting is needed (Miklós and Lajos, 1997; Bai, 1999; Banerjee and Urga, 2005; Döring, 2011; Robbins et al., 2016). Despite its easy implementation, such testing procedures are conducted under the assumption of independent samples. For time-dependent data, there is a plethora of literature providing solutions within a Gaussian stochastic linear system framework, such as a combination of Kalman filter and generalized likelihood ratio (GLR) (Lai and Shan, 1999). Under some parametric assumptions, detector statistic can take the form of additive changes based on likelihood or innovations. Furthermore, non-likelihood-based detector statistics are discussed and implemented but such studies considered asymptotic Gaussian property for statistical inference. In the situations of change point detection in modern online data of elevated complexity, the current trend of corresponding solutions lies in a linear system or parametric framework, incorporating Gaussian property for inference on the detector statistics.

1.2 Multivariate time series analysis

Time series analysis is a popular research topic in various fields such as finance, econometrics, climatology, epidemiology, clinical trials as well as interdisciplinary areas because the statistical

models under the independent assumptions usually lack sufficient accountability for the corresponding chronological pattern. Due to the escalated demand of studying multiple measurements as a whole, multivariate time series analysis becomes a natural tendency for correlation study for multiple time-dependent data and draw universal inference on the latent dynamics from a collection of associated time series. In this dissertation, we propose our methods to address the research problems arising in the financial time series and public health surveillance data.

With the advent of high-frequency trading, technological advancement allows people to record and store a larger amount of transaction data within a short time interval. Transaction arrival times as well as measurements of interest can be studied as time series. However, market signals are not salient among the high volume of raw data. For example, within a short period of time, numerous transactions on an asset can be made but the asset price fluctuations are negligible. To study the price change in a high-frequency setting, one can easily reach some biased conclusion from a statistical model fitted with excessive insignificant price changes. Another feature of high-frequency financial data is irregularity. Unlike conventional time series data with standard time index, the arrival times in high-frequency data can be randomly spaced. We need to take special care of data processing in high-frequency financial market to study its micro-structure. Additionally, assets prices usually are not only dependent on its own historical observations but also by some other factors such as the overall market performance or other affiliated assets. Therefore, multivariate time series analysis is an important tool to reveal the interrelations among multiple high-frequency financial time series and unveil both macro-structure of the market behavior via the associated individual assets and its micro-structure in granulated time period.

Public health surveillance analysis is a traditional research field, and its recent development brings up new statistical challenges. Syndromic surveillance is one of the useful tools in public health systems, which supports the monitoring of public health impacts and raise speedy alarms at their occurrences. Preliminary diagnosis from syndromic surveillance uncovers potential onsets of infectious outbreaks prior to a monitoring regime requiring laboratory confirmation (Hughes et al., 2020). A decent syndromic system incorporates covariate information, spatial and temporal dynamics of a disease, as well as integration of data from multiple sources. Therefore, the real-time monitoring of an infectious disease such as COVID-19 in a large geographical area requires unique accommodation for the spatio-temporal intricacy. Conventional methods either don't account for

the spatial and temporal dynamics such as CUSUM (Page, 1954) or lack measure of uncertainty for identified disease cluster (Kulldorff, 1997). Within a spatio-temporal framework, one needs to not only overcome more complex configuration of graph structure in the model, but also keep the implementation at a reasonable computational cost. Thanks to the advancement of technology, Bayesian inference gains its popularity in complex modeling problems. Bayesian technique grants the flexibility of incorporating historical information into the priors so that this approach can alleviate the computational burden for complex Bayesian model inference. With a few stationarity assumptions on the time series data, one is able to aim for historical information integration via prior distributions.

1.3 Fast Bayesian inference approximation

Bayesian statistical analysis regained its popularity with the advancements in Markov chain Monte Carlo (MCMC) sampling method during 1950s. Traditional Bayesian applications rely on the joint posterior distribution of the model parameters. One of common concerns for the MCMC method is about its computational efficiency. When the complexity of the model increases with the number of parameters, the joint posterior distribution in a high dimension parameter space raises the difficulty level for practitioners to draw samples from. In addition, posterior distribution without a closed form will also lead an intensive computation for the traditional Bayesian inference.

However, in many situations, individual posterior marginals are often sufficient for adequate statistical inference (Gómez-Rubio, 2020). Rue et al. (2009) proposed an approximate Bayesian inference for latent Gaussian models by Integrated Nested Laplace Approximation (INLA). Not only is the inference based on posterior marginals as good as the inference based on the joint posterior distribution, but also the computational cost can be enhanced to a large extent, especially for a sparse Gaussian Markov Random Field (GMRF) in a high dimensional parameter space. Since our Bayesian model frameworks involve GMRF, we will be able to implement customized GMRF random effects using INLA and enjoy the benefit of speedy computation.

1.4 Summary

For the two prevalent research directions, new challenges are burgeoning in their combinations, where practitioners need adequate modeling for the complicated real-time data to proactively make

decisions on the basis of the exhibited behavioral change therefrom. Innovative methodologies are expected to support elaborate statistical models and enjoy easy implementation with an agreeable running time.

To solve these problems in sequential change point detection as well as multivariate time series analysis in finance and public health, the dissertation is organized as follows. In Chapter 2, we deal with the online change point detection problem in high-frequency market and propose an ensemble framework for sequential analysis within univariate time series. In Chapter 3, we introduce a Bayesian framework to account for the association among multiple assets, as well as capture the behavior of the market from multivariate time series analysis. In Chapter 4, we conduct the sequential analysis of infectious outbreak via multivariate time series in the public health surveillance data. We conclude our work and discuss future work in Chapter 5.

2 Online structural break point detection

2.1 Background

Automated high-frequency trading deals with a large number of trading orders. For each asset, prices at these transactions form an irregularly spaced time series (Wang and Zou, 2014). Therefore, a good awareness of the volatility alternation of an asset is crucial to market participants, especially in a high-frequency setting. An important practical focus for investors will be monitoring the volatile financial market quickly and assessing accurately any behavioral change in an online fashion, so that reliable inference can be maintained with updated model parameters.

A financial duration is defined as the difference in arrival time between two consecutive events of interest. An event can be defined as, but is not limited to, a single transaction, a return beyond a certain percentage, or a price change exceeding a certain amount. The i -th duration can be observed as $x_i = t_i - t_{i-1}$, where t_i denotes the timing of the i -th event. The irregularity of arrival times of transactions can provide useful information for the market participants. For example, if transactions happen rapidly for an asset, then the durations between transactions will be in the form of numerous short time intervals, while infrequent transactions will provide a sequence of longer durations. Further, higher frequency of the transactions implies higher volatility in price fluctuations, Easley and O'hara (1992) discussed the link between the existence of information, the timing of trading and price movement. Thus the negative influence of long durations on returns and variances of asset price has predictive power for ultra-high-frequency volatility jointly with a GARCH model for the price movement (Engle and Russell, 1997; Engle, 2000). When a structural break is detected, practitioners can update their time series model and modify their trading algorithm with financial domain knowledge. Therefore, We address this problem of structural breaks in time series of durations and describe an online approach for detection but the embedding of detected breaks into a specific trading strategy is upon the practitioners.

2.2 Literature review

With the definition of inter-trade durations, the objective is to detect structural breaks in the duration time series from financial market, in a fast online fashion. In this subsection, we are going to introduce the literature on the approaches for duration time series modeling and various

contributions in change point detection topic, which lay the foundation of our method.

2.2.1 Log ACD model

Modeling inter-event duration is a popular direction for statistical application in high-frequency trading (Dutta et al., 2022). One of the prevalent frameworks to model time series of durations is via conditional duration models. Starting from the seminal work of (Engle and Russell, 1998) on auto-regressive conditional duration (ACD) models, there is a rich literature extension of ACD models from different perspectives. Linear ACD modeling is rather common but restrictive, and include the Exponential ACD (EACD), Weibull ACD (WACD), and Gamma ACD (GACD), etc. Non-linear ACD models include the Log ACD models (Bauwens and Giot, 2000), the Stochastic Conditional Duration, or SCD (p, q) models (Bauwens and Veredas, 2004) and Augmented ACD (AACD) models through Box-Cox transformation (Fernandes and Grammig, 2006). In order to address the long memory dependence exhibited in durations, Jasiak (1999) proposed a class of fractionally integrated ACD (FIACD) models, and Deo et al. (2010) introduced a parametric, latent-variable, long memory stochastic duration (LMSD) to handle the long memory feature of inter-trade durations better than ordinary ACD models. For extreme value modeling in durations, Zheng et al. (2016) incorporated Fréchet innovations in the usual ACD model, and demonstrated a better fit with Fréchet ACD model than the Weibull ACD model. Pacurar (2008) discusses detailed theoretical properties and applications of duration models. A plethora of literature has showed the versatility of duration models in the fields of Finance, Econometrics, etc. Our work is hence established under log ACD model framework for online structural break detection problems in the financial durations.

Useful approaches for parameter estimation for the ACD models are maximum likelihood estimation (MLE), generalized method of moments (GMM) estimation, quasi-maximum likelihood estimation (QMLE) and estimating function (EF) estimation (Engle and Russell, 1998; Grammig and Wellner, 2002; Allen et al., 2008; Liang et al., 2011). Thavaneswaran et al. (2015) proposed a combined martingale EF approach for recursive parameter estimation of generalized duration models. Zhang et al. (2019) proposed a penalized estimating function (PEF) for the aforementioned recursive parameter estimation for the log ACD models. To enhance the efficiency of parameter estimation, we will adopt a distribution-free approach using the penalized estimation function for

a robust outcome.

The key idea of ACD model can be described as following. Let $x_i = t_i - t_{i-1}$, where $i = 1, 2, \dots$ denote the financial duration time series, and let $\mathcal{F}_{i-1}^x$ denote the information set of past durations. The ACD model explains the time dependence between x_i and past observations through the conditional expectation of x_i on $\mathcal{F}_{i-1}^x$. Then the ACD(p, q) model is

$$x_i = \psi_i \varepsilon_i, \text{ where } \psi_i = \omega + \sum_{j=1}^p \alpha_j x_{i-j} + \sum_{j=1}^q \beta_j \psi_{i-j}. \quad (2.1)$$

In the Eq (2.1), the parameters are $\boldsymbol{\theta} = (\omega, \alpha_1, \dots, \alpha_p, \beta_1, \dots, \beta_q)$. The weak stationarity of this non-negative process is guaranteed by the conditions of $\omega > 0, \alpha_j \geq 0$ for $j = 1, \dots, p, \beta_j \geq 0$ for $j = 1, \dots, q$ and $\sum_{j=1}^p \alpha_j + \sum_{j=1}^q \beta_j < 1$. The innovations ε_i are assume to be independent and indentially distributed non-negative random variables with $E(\varepsilon) = 1$ and density $f_\varepsilon(\cdot)$ and independent of $\mathcal{F}_{i-1}^x$. Log ACD model takes a similar form except that $x_i = \exp(\psi_i) \varepsilon_i$, and its elaboration will be discussed in the unified E-PEF framework subsection.

2.2.2 Change point detection

There is a plethora of literature on change point detection, which can be classified into retrospective analysis and sequential (or online) analysis. As for the detection approach for the specific monitoring process, online change point detection fits the needs of high-frequency traders in a real-time setting because they need to make decisions promptly with streaming data. Some viewpoints from retrospective detection (with complete data) inspire us in our endeavor. The seminal work of change point detection originated in quality control (Page, 1954, 1955), with the main consideration as hypothesis testing on the CUSUM-type test statistic under the null hypothesis of no break points. Such hypothesis testing procedures are usually considered distribution-free and require no model fitting (Miklós and Lajos, 1997; Bai, 1999; Banerjee and Urga, 2005; Döring, 2011; Robbins et al., 2016). In financial applications, Zhang et al. (2001) proposed a threshold ACD model and improved several inadequacies of the original ACD models. Zhang et al. (2018) implemented a penalized estimating function (PEF) approach for recursive estimation of log ACD model parameters and applied a *FindPeaks* procedure to detect the structural breaks on the trajectory of the recursively estimated parameter values. However, this retrospective study requires that the complete

observations are available.

In the online change detection field, the Kalman filter and its extensions have gained its popularity for Gaussian linear dynamic systems. Besides innovation-based methodologies, generalized likelihood ratio (GLR) based testing is another direction to achieve online detection (Willsky and Jones, 1976; Lai and Shan, 1999). Chu et al. (1996) discussed two sequential tests for the online monitoring of economic behavior via linear regression models. The break detection was determined by the fluctuation of the online sequential parameter estimates. However, the aforementioned online detection methodologies handle the additive changes through either likelihood or innovations under a parametric modeling framework with specific distributional assumptions.

In addition, an asymptotic Gaussian non-likelihood statistic is developed for online detection in signal processing and adaptive control (Benveniste et al., 1987, 2012). Berkes et al. (2004) extended online sequential change point detection via a quasi-likelihood function based approach, and applied it to GARCH (p, q) models in a financial application. The test statistic for a structural break consists of a standardized partial sum of the quasi-likelihood score function; The critical value can be approximated via a Wiener process (Heyde, 1997). Hušková et al. (2007) investigated the limiting behavior of test statistics based on various functionals of the partial sums of weighted residuals. Aue et al. (2009) put forward a change point testing procedure for the volatility and cross-volatility of multivariate time series via a vectorized CUSUM-type statistic. Horváth and Rice (2014); Xie et al. (2021) give a broad review of change point detection methodologies across different areas and their extensions.

2.3 E-PEF detection

The literature mentioned before addressed the change point detection in sequential data, under a linear system or parametric framework, or through the asymptotic Gaussian property of the detector statistics. Therefore, we propose an innovative ensemble penalized estimating function (E-PEF) approach to solve the detection problem. Our contribution is a non-Gaussian detector statistic based on the penalized estimation function for the log ACD models when the asymptotic property is not well obtained from finite samples. This ensemble approach complements a non-Gaussian distribution with the empirical bootstrap distribution to control the false detection rate in the online setting, while still enjoying its computational speed. Based on our straightforward and

well-grounded detection mechanism, practitioners can effortlessly monitor online transaction-level financial data, and make strategic investment with sound statistical support.

2.3.1 Model Framework

For a duration time series $x_i, i = 1, 2, \dots$, we aim to detect the first break point (τ) such that $x_1, x_2, \dots, x_\tau$ are generated from one process and $x_{\tau+1}, x_{\tau+2}, \dots$, are generated from a different porcess. Consider the log ACD framework (Thavaneswaran et al., 2015), in a high-frequency setting, a penalized log ACD $(p, 0)$ model with large p generally provides better computational speed and precision gain of parameter estimation than a penalized log ACD (p, q) framework does. For simplicity, we take the log ACD $(p, 0)$ model as an illustration.

$$\begin{aligned} x_i &= \frac{\exp(\psi_i)}{\mu_\varepsilon} \varepsilon_i, i = 1, 2, \dots, \\ E(x_i | \mathcal{F}_{i-1}^x) &= \frac{\exp(\psi_i)}{\mu_\varepsilon}, \\ \psi_i &= \omega + \sum_{j=1}^p \alpha_j \log x_{i-j}, \end{aligned} \tag{2.2}$$

where ε_i is the non-negative innovation term with mean μ_ε , $\mathcal{F}_{i-1}^x$ denotes the information set associated with $\{x_1, x_2, \dots, x_{i-1}\}$, $\omega, \alpha_1, \dots, \alpha_p$ are the AR coefficients in the log ACD model. For weak stationarity of the log ACD process, we let $\sum_{j=1}^p \alpha_j < 1$. To complete the detection of any potential structural break point τ in the streaming time series of durations, there are three steps to take: parameter estimation, detector statistic monitoring, and hypothesis testing, as described below.

2.3.2 Parameter estimation

We apply the quasi-likelihood-type method (Berkas et al., 2004) to detect the structural break on the level of parameter change in the log ACD model. The structural break point detection is based upon the PEF (Zhang et al., 2019) for the parameters in the log ACD model. Let $\boldsymbol{\theta} = (\omega, \alpha_1, \alpha_2, \dots, \alpha_p)'$ be the parameters in the log ACD $(p, 0)$ model and p be the order of the log ACD model. p is chosen to be a large number (to capture potential AR structure for the underlying log ACD model as well as to approximate the MA structure of the log ACD model, if any). The total number of

parameters involved is $d = p + 1$.

For a comprehensive procedure of recursive martingale PEF estimation, we need to introduce a few required quantities involved in the PEF, with detailed definitions from Zhang et al. (2019). Suppose $x_i, i = 1, 2, \dots, n$ is a realization of a duration process with parameters in $\boldsymbol{\theta}$, define linear and quadratic martingale differences respectively as,

$$\begin{aligned} m_i(\boldsymbol{\theta}) &= x_i - \mu_i(\boldsymbol{\theta}), \\ Q_i(\boldsymbol{\theta}) &= m_i^2(\boldsymbol{\theta}) - \sigma_i^2(\boldsymbol{\theta}), \end{aligned} \tag{2.3}$$

where $\mu_i(\boldsymbol{\theta})$ and $\sigma_i^2(\boldsymbol{\theta})$ are conditional mean and variance of x_i correspondingly. Quadratic variations and covariation of m_i and Q_i are defined as,

$$\begin{aligned} \langle m \rangle_i &= E[m_i^2(\boldsymbol{\theta}) | \mathcal{F}_{i-1}^x] = \sigma_i^2(\boldsymbol{\theta}), \\ \langle Q \rangle_i &= E[m_i^4(\boldsymbol{\theta}) | \mathcal{F}_{i-1}^x] - (E[m_i^2(\boldsymbol{\theta}) | \mathcal{F}_{i-1}^x])^2 = \kappa_i(\boldsymbol{\theta}) - \sigma_i^4(\boldsymbol{\theta}), \\ \langle m, Q \rangle_i &= E[m_i^3(\boldsymbol{\theta}) | \mathcal{F}_{i-1}^x] = \gamma_i(\boldsymbol{\theta}), \end{aligned} \tag{2.4}$$

where $\kappa_i(\boldsymbol{\theta})$ and $\gamma_i(\boldsymbol{\theta})$ are the third and fourth central moments of x_i respectively. Next, Thavaneswaran et al. (2015) derived the optimal (Godambe-Durbin) combined martingale EF,

$$g_C^*(\boldsymbol{\theta}) = \sum_{i=1}^n g_i^*(\boldsymbol{\theta}) = \sum_{i=1}^n (a_{i-1}^* m_i(\boldsymbol{\theta}) + b_{i-1}^* Q_i(\boldsymbol{\theta})), \tag{2.5}$$

where

$$\begin{aligned} a_{i-1}^*(\boldsymbol{\theta}) &= \rho_i^2 \left(-\frac{\partial \mu_i(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \frac{1}{\langle m \rangle_i} - \left(E \left[\frac{\partial(m_i^2(\boldsymbol{\theta}) - \sigma_i^2(\boldsymbol{\theta}))}{\partial \boldsymbol{\theta}} \middle| \mathcal{F}_{i-1}^x \right] - \frac{\partial E[(m_i^2(\boldsymbol{\theta}) - \sigma_i^2(\boldsymbol{\theta})) | \mathcal{F}_{i-1}^x]}{\partial \boldsymbol{\theta}} \right) \eta_i \right), \\ b_{i-1}^*(\boldsymbol{\theta}) &= \rho_i^2 \left(\frac{\partial \mu_i(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} \eta_i + \left(E \left[\frac{\partial(m_i^2(\boldsymbol{\theta}) - \sigma_i^2(\boldsymbol{\theta}))}{\partial \boldsymbol{\theta}} \middle| \mathcal{F}_{i-1}^x \right] - \frac{\partial E[(m_i^2(\boldsymbol{\theta}) - \sigma_i^2(\boldsymbol{\theta})) | \mathcal{F}_{i-1}^x]}{\partial \boldsymbol{\theta}} \right) \frac{1}{\langle Q \rangle_i} \right), \end{aligned} \tag{2.6}$$

with

$$\rho_i^2 = \left(1 - \frac{\langle m, Q \rangle_i^2}{\langle m \rangle_i \langle Q \rangle_i} \right)^{-1}, \quad \eta_i = \frac{\langle m, Q \rangle_i}{\langle m \rangle_i \langle Q \rangle_i}. \tag{2.7}$$

Based on (2.5), the PEF is,

$$g_{C,\lambda}^*(\boldsymbol{\theta}) = g_C^*(\boldsymbol{\theta}) - np'_\lambda(\boldsymbol{\theta}), \quad (2.8)$$

with the first derivative of the SCAD penalty (Fan and Li, 2001) for $a > 2$ and penalty parameter λ ,

$$p'_\lambda(|\boldsymbol{\theta}|) = \lambda \{I(|\boldsymbol{\theta}| \leq \lambda) + \frac{(a\lambda - |\boldsymbol{\theta}|)_+}{(a-1)\lambda} I(|\boldsymbol{\theta}| > \lambda)\}. \quad (2.9)$$

The choice depends on the liquidity of a stock which is proportional to n , and details on how to select a and the tuning parameter λ are discussed in (Zhang et al., 2019). For (2.3) - (2.8), $\mu_i(\boldsymbol{\theta})$, $\sigma_i^2(\boldsymbol{\theta})$, $\gamma_i(\boldsymbol{\theta})$, $\kappa_i(\boldsymbol{\theta})$, $\langle m \rangle_i$, $\langle M \rangle_i$, $\langle m, M \rangle_i$, ρ_i^2 , η_i are scalars, while $\boldsymbol{\theta}$, $g_{C,\lambda}^*(\boldsymbol{\theta})$, $a_{i-1}^*(\boldsymbol{\theta})$, $b_{i-1}^*(\boldsymbol{\theta})$ are $d \times 1$ vectors. In the parameter estimation stage, streaming time series of durations recursively updates the estimation of $\boldsymbol{\theta}$.

2.3.3 Detector statistic

Consider the observed durations $x_1, x_2, \dots, x_{M_2}$ for model training, the estimation of $\boldsymbol{\theta}$ is recursively updated to $\hat{\boldsymbol{\theta}}_{M_2}$. As the new observations $x_{M_2+1}, x_{M_2+2}, \dots, x_{M_2+k}$ arrive, we will sequentially evaluate their behavioral pattern and assess the occurrence of any structural breaks through a detector statistic.

Similar to Berkes et al. (2004), our proposed detector statistic relies on the PEF. It is a d -dimensional column vector,

$$\mathbf{G}^{M_2}(k) = \left(G_1^{M_2}(k), G_2^{M_2}(k), \dots, G_d^{M_2}(k) \right)', \quad (2.10)$$

where

$$G_j^{M_2}(k) = \sum_{M_2 < i \leq M_2+k} \left(g_{i,\lambda}^*(\hat{\boldsymbol{\theta}}_{j,M_2}) \right) / \sqrt{\hat{\mathbf{D}}_{j,j}^{(M_1,M_2)}}, k \in \mathbb{N}^+, j = 1, 2, \dots, d, \quad (2.11)$$

with

$$\begin{aligned} \hat{\mathbf{D}}^{(M_1,M_2)} &= \frac{1}{M_2 - M_1} \sum_{M_1 < i \leq M_2} \left(g_{i,\lambda}^*(\hat{\boldsymbol{\theta}}_{M_2}) \right) \left(g_{i,\lambda}^*(\hat{\boldsymbol{\theta}}_{M_2})' \right), \\ g_{i,\lambda}^*(\hat{\boldsymbol{\theta}}_{M_2}) &= g_i^*(\hat{\boldsymbol{\theta}}_{M_2}) - p'_\lambda(|\hat{\boldsymbol{\theta}}_{M_2}|). \end{aligned} \quad (2.12)$$

From (2.10) to (2.12), the detector is in the form of a standardized PEF, with the scaling matrix being $\hat{\mathbf{D}}^{(M_1, M_2)}$. $\hat{\mathbf{D}}^{(M_1, M_2)}$ is regarded as the sample covariance matrix of $g_{i, \lambda}^*(\hat{\boldsymbol{\theta}}_{M_2})$. M_2 is the training sample size of the durations. M_1 ($M_1 < M_2$) serves as a length of burn-in period, which is part of the training period, because PEF approach takes a certain number of iterations to reach to a relative stable stage of $\boldsymbol{\theta}$ estimation. Truncating the burn-in period of the parameter estimation stabilizes the scaling matrix $\hat{\mathbf{D}}^{(M_1, M_2)}$. Consequently, after the training period M_2 , the detector statistic $\mathbf{G}^{M_2}(k)$ at the time $M_2 + k$ is accessible as a function value of estimated parameters from training period, $\hat{\boldsymbol{\theta}}_{M_2}$, new observed data $\{x_{M_2+1}, x_{M_2+2}, \dots, x_{M_2+k}\}$, and tuning parameter λ .

2.3.3.1 Spillover effect The detector statistic in (2.10) contains the partial derivatives of the martingale estimating function for each parameters. However, any parameter change doesn't coincide with just the change of an individual component of the detector statistic due to the inter-correlation among the components, known as the spillover effect.

Spillover effect theoretical justification: For the two major terms aside from the penalty expression $p'_\lambda(|\boldsymbol{\theta}|)$ in the penalized estimating function in (2.5), each element of the two vectors is involved with the terms $\langle m \rangle_i, \langle Q \rangle_i$ and η_i , and they can't be canceled in the computation. In the structure of ψ_i in (2.2), any parameter change will lead to a change on the value of ψ_i . In a log ACD $(p, 0)$ framework (2.2), the derivative of ψ_i with respect to $\boldsymbol{\theta} = (\omega, \alpha_1, \alpha_2, \dots, \alpha_p)^T$ becomes,

$$\frac{\partial \psi_i}{\partial \boldsymbol{\theta}} = (1, \log x_{i-1}, \log x_{i-2}, \dots, \log x_{i-p})'. \quad (2.13)$$

In the expressions of $a_{i-1}^*(\boldsymbol{\theta})$ and $b_{i-1}^*(\boldsymbol{\theta})$ in (2.6), the conditional mean of x_i is,

$$\mu_i(\boldsymbol{\theta}) = E[x_i | \mathcal{F}_{i-1}^x] = \exp\{\psi_i\} \cdot \frac{\mu_\epsilon}{\mu_\epsilon}, \quad (2.14)$$

and the derivative of $\mu_i(\boldsymbol{\theta})$ with respect to $\boldsymbol{\theta} = (\omega, \alpha_1, \alpha_2, \dots, \alpha_p)^T$ can be derived using the chain rule,

$$\frac{\partial \mu_i(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \frac{\partial \mu_i(\boldsymbol{\theta})}{\partial \psi_i} \cdot \frac{\partial \psi_i}{\partial \boldsymbol{\theta}} = \exp\{\psi_i\} (1, \log x_{i-1}, \log x_{i-2}, \dots, \log x_{i-p})'. \quad (2.15)$$

For the quadratic variation and covariation, they can also be expressed in terms of ψ_i

$$\begin{aligned}\langle m \rangle_i &= \sigma_i^2(\boldsymbol{\theta}) = \exp\{2\psi_i\} \frac{\sigma_\epsilon^2}{\mu_\epsilon^2}, \\ \langle Q \rangle_i &= \kappa_i(\boldsymbol{\theta}) - \sigma_i^4(\boldsymbol{\theta}) = \exp\{4\psi_i\} \frac{\kappa_\epsilon}{\mu_\epsilon^4}, \\ \langle m, Q \rangle_i &= \gamma_i(\boldsymbol{\theta}) = \exp\{3\psi_i\} \frac{\gamma_\epsilon}{\mu_\epsilon^3},\end{aligned}\tag{2.16}$$

where $\mu_\epsilon, \sigma_\epsilon^2, \gamma_\epsilon, \kappa_\epsilon$ are the mean, variance, third and fourth moments of the error ϵ_i in (2.2), respectively.

Inserting the quantities into (2.6), we have,

$$\begin{aligned}a_{i-1}^*(\boldsymbol{\theta}) &= \left(1 - \frac{\gamma_\epsilon^2}{\sigma_\epsilon^2 \cdot \kappa_\epsilon}\right)^{-1} \left[-\frac{\partial \psi_i}{\partial \boldsymbol{\theta}} \cdot \frac{1}{\exp\{\psi_i\} \cdot \frac{\sigma_\epsilon^2}{\mu_\epsilon^2}} + 2 \exp\{2\psi_i\} \frac{\sigma_\epsilon^2}{\mu_\epsilon^2} \cdot \frac{\partial \psi_i}{\partial \boldsymbol{\theta}} \cdot \frac{\gamma_\epsilon \cdot \mu_\epsilon^2}{\exp\{3\psi_i\} \cdot \sigma_\epsilon^2 \cdot \kappa_\epsilon} \right] \\ &= \left(1 - \frac{\gamma_\epsilon^2}{\sigma_\epsilon^2 \cdot \kappa_\epsilon}\right)^{-1} \cdot \frac{\partial \psi_i}{\partial \boldsymbol{\theta}} \left[-\frac{1}{\exp\{\psi_i\} \cdot \frac{\sigma_\epsilon^2}{\mu_\epsilon^2}} + 2 \frac{\gamma_\epsilon}{\exp\{\psi_i\} \cdot \kappa_\epsilon} \right], \\ b_{i-1}^*(\boldsymbol{\theta}) &= \left(1 - \frac{\gamma_\epsilon^2}{\sigma_\epsilon^2 \cdot \kappa_\epsilon}\right)^{-1} \left[\frac{\partial \psi_i}{\partial \boldsymbol{\theta}} \frac{\gamma_\epsilon \cdot \mu_\epsilon^2}{\exp\{3\psi_i\} \cdot \sigma_\epsilon^2 \cdot \kappa_\epsilon} - 2 \exp\{2\psi_i\} \frac{\sigma_\epsilon^2}{\mu_\epsilon^2} \cdot \frac{\partial \psi_i}{\partial \boldsymbol{\theta}} \cdot \frac{1}{\exp\{4\psi_i\} \cdot \frac{\kappa_\epsilon}{\mu_\epsilon^4}} \right] \\ &= \left(1 - \frac{\gamma_\epsilon^2}{\sigma_\epsilon^2 \cdot \kappa_\epsilon}\right)^{-1} \cdot \frac{\partial \psi_i}{\partial \boldsymbol{\theta}} \left[\frac{\gamma_\epsilon \cdot \mu_\epsilon^2}{\exp\{3\psi_i\} \cdot \sigma_\epsilon^2 \cdot \kappa_\epsilon} - \frac{2\sigma_\epsilon^2}{\exp\{2\psi_i\} \cdot \frac{\kappa_\epsilon}{\mu_\epsilon^2}} \right].\end{aligned}\tag{2.17}$$

Therefore, $g_{i,\lambda}^*(\boldsymbol{\theta})$ can be found as

$$\begin{aligned}g_{i,\lambda}^*(\boldsymbol{\theta}) &= \left(1 - \frac{\gamma_\epsilon^2}{\sigma_\epsilon^2 \cdot \kappa_\epsilon}\right)^{-1} \cdot \frac{\partial \psi_i}{\partial \boldsymbol{\theta}} \left[-\frac{1}{\exp\{\psi_i\} \cdot \frac{\sigma_\epsilon^2}{\mu_\epsilon^2}} + 2 \frac{\gamma_\epsilon}{\exp\{\psi_i\} \cdot \kappa_\epsilon} \right] \cdot (x_i - \exp\{\psi_i\}) + \\ &\quad \left(1 - \frac{\gamma_\epsilon^2}{\sigma_\epsilon^2 \cdot \kappa_\epsilon}\right)^{-1} \cdot \frac{\partial \psi_i}{\partial \boldsymbol{\theta}} \left[\frac{\gamma_\epsilon \cdot \mu_\epsilon^2}{\exp\{3\psi_i\} \cdot \sigma_\epsilon^2 \cdot \kappa_\epsilon} - \frac{2\sigma_\epsilon^2}{\exp\{2\psi_i\} \cdot \frac{\kappa_\epsilon}{\mu_\epsilon^2}} \right] \cdot \left((x_i - \exp\{\psi_i\})^2 - \exp\{2\psi_i\} \cdot \frac{\sigma_\epsilon^2}{\mu_\epsilon^2} \right) \\ &\quad - p'_\lambda(|\boldsymbol{\theta}|),\end{aligned}\tag{2.18}$$

where we note that the existence of ψ_i will have an impact on every element of the penalized estimating function. This helps to justify the spillover effect. As such, our online detection algo-

rithm will mainly focus on the detection of the existence of parameter change instead of specifying any particular varying parameters.

2.3.4 Hypothesis testing

In the hypothesis test of the detector statistic, we need to set up thresholds to decide if the break point has occurred. Consequently, some alarm will be raised when $|G_j^{M_2}(k)| > T_\alpha$ where T_α is a threshold based on the significance level α .

Wiener process threshold: According to (Heyde, 1997), the estimating functions preserve asymptotic normality property and $(M_2 - M_1)^{-1/2} G_j^{M_2}(k)$ can be approximated by $\left((1+k)W_j\left(\frac{k}{1+k}\right)\right)$, $k \in [0, \infty)$ under the null hypothesis, where $W_j(\cdot)$, $j = 1, 2, \dots, d$ are independent standard Wiener process. The threshold for $|G_j^{M_2}(k)|$ can be chosen as $T_{\alpha, \mathcal{W}}(k) = (M_2 - M_1)^{\frac{1}{2}} (1 + k / (M_2 - M_1)) g(k, M_2)$ and (Berkes et al., 2004) proposed that a constant boundary function $g(k, M_2) = c$ is sufficient. As an asymptotic result, Wiener process threshold has a loss of accuracy if the sample size is limited. To accommodate scenarios with small sample sizes, we will consider the empirical distribution of the detector statistic under the null hypothesis.

Bootstrap threshold: Note that $\mathbf{G}^{M_2}(k)$ is a standardized partial sum of $g_{i,\lambda}^*(\hat{\boldsymbol{\theta}}_{M_2})$. When $\hat{\boldsymbol{\theta}}_{M_2}$ is calculated from the training period, $g_{i,\lambda}^*(\hat{\boldsymbol{\theta}}_{M_2})$ becomes a function of only x_i , with λ being given. Therefore, $\mathbf{G}^{M_2}(k)|\hat{\boldsymbol{\theta}}_{M_2}$ can be regarded as a function $f(x_{M_2+1}, x_{M_2+2}, \dots, x_{M_2+k}|\hat{\boldsymbol{\theta}}_{M_2}, \lambda)$. Under the null hypothesis, we will implement a bootstrap method to sample from the stationary time series in the training period, $\{x_1, x_2, x_3, \dots, x_{M_2}\}$. Inserting the bootstrap sampled time series into the $\mathbf{G}$ function provides the empirical distribution of the quasi-score function $\mathbf{G}^{M_2}(k)$, and its empirical quantiles become accessible as bootstrap thresholds. There is an abundance of literature of the bootstrap method on time series analysis (Bühlmann, 2002; Politis, 2003). We will apply the block bootstrap resampling approach with a random block length (Politis and Romano, 1994) to obtain resampled time series for the monitoring period from the training period and summarize a distribution for $\mathbf{G}^{M_2}(k)|\hat{\boldsymbol{\theta}}_{M_2}, \lambda$. The procedure is illustrated as follows: 1) Based on the training period $\{x_1, x_2, \dots, x_{M_2}\}$, we implement the block bootstrap sampling method to collect the time series samples $x_{(M_2+k)b^*}^*$, $k = 1, 2, \dots, b^* = 1, 2, \dots, n_b$, of sample size, n_b , for the monitoring period as shown in Table 1. 2) Through insertion of the time series samples, we obtain the samples of $G_j^{M_2}(k)$, $j = 1, 2, \dots, d$ in the monitoring period, i.e., $\{G_{j,b^*}^{M_2}(k)\}$, $j = 1, 2, \dots, d$, $b^* = 1, 2, \dots, n_b$.

t \ B	1	2	3	...	n_b
$M_2 + 1$	$x_{(M_2+1)1}^*$	$x_{(M_2+1)2}^*$	$x_{(M_2+1)3}^*$	$\cdots$	$x_{(M_2+1)n_b}^*$
$M_2 + 2$	$x_{(M_2+2)1}^*$	$x_{(M_2+2)2}^*$	$x_{(M_2+2)3}^*$	$\cdots$	$x_{(M_2+2)n_b}^*$
$M_2 + 3$	$x_{(M_2+3)1}^*$	$x_{(M_2+3)2}^*$	$x_{(M_2+3)3}^*$	$\cdots$	$x_{(M_2+3)n_b}^*$
$M_2 + 4$	$x_{(M_2+4)1}^*$	$x_{(M_2+4)2}^*$	$x_{(M_2+4)3}^*$	$\cdots$	$x_{(M_2+4)n_b}^*$
$M_2 + 5$	$x_{(M_2+5)1}^*$	$x_{(M_2+5)2}^*$	$x_{(M_2+5)3}^*$	$\cdots$	$x_{(M_2+5)n_b}^*$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$M_2 + k$	$x_{(M_2+k)1}^*$	$x_{(M_2+k)2}^*$	$x_{(M_2+k)3}^*$	$\cdots$	$x_{(M_2+k)n_b}^*$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Table 1: Sampled time series from block bootstrap

See Table 2. 3) The bootstrap threshold, $T_{\alpha,j,B}(k)$ at time $t = M_2 + k$, can be evaluated as the empirical α -quantile of $G_j^{M_2}(k), j = 1, 2, \dots, d$.

k \ j	1	2	3	...	d
1	$\{G_{1,b^*}^{M_2}(1)\}_{b^*=1}^{b^*=n_b}$	$\{G_{2,b^*}^{M_2}(1)\}_{b^*=1}^{b^*=n_b}$	$\{G_{3,b^*}^{M_2}(1)\}_{b^*=1}^{b^*=n_b}$	$\cdots$	$\{G_{d,b^*}^{M_2}(1)\}_{b^*=1}^{b^*=n_b}$
2	$\{G_{1,b^*}^{M_2}(2)\}_{b^*=1}^{b^*=n_b}$	$\{G_{2,b^*}^{M_2}(2)\}_{b^*=1}^{b^*=n_b}$	$\{G_{3,b^*}^{M_2}(2)\}_{b^*=1}^{b^*=n_b}$	$\cdots$	$\{G_{d,b^*}^{M_2}(2)\}_{b^*=1}^{b^*=n_b}$
3	$\{G_{1,b^*}^{M_2}(3)\}_{b^*=1}^{b^*=n_b}$	$\{G_{2,b^*}^{M_2}(3)\}_{b^*=1}^{b^*=n_b}$	$\{G_{3,b^*}^{M_2}(3)\}_{b^*=1}^{b^*=n_b}$	$\cdots$	$\{G_{d,b^*}^{M_2}(3)\}_{b^*=1}^{b^*=n_b}$
4	$\{G_{1,b^*}^{M_2}(4)\}_{b^*=1}^{b^*=n_b}$	$\{G_{2,b^*}^{M_2}(4)\}_{b^*=1}^{b^*=n_b}$	$\{G_{3,b^*}^{M_2}(4)\}_{b^*=1}^{b^*=n_b}$	$\cdots$	$\{G_{d,b^*}^{M_2}(4)\}_{b^*=1}^{b^*=n_b}$
5	$\{G_{1,b^*}^{M_2}(5)\}_{b^*=1}^{b^*=n_b}$	$\{G_{2,b^*}^{M_2}(5)\}_{b^*=1}^{b^*=n_b}$	$\{G_{3,b^*}^{M_2}(5)\}_{b^*=1}^{b^*=n_b}$	$\cdots$	$\{G_{d,b^*}^{M_2}(5)\}_{b^*=1}^{b^*=n_b}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
k	$\{G_{1,b^*}^{M_2}(k)\}_{b^*=1}^{b^*=n_b}$	$\{G_{2,b^*}^{M_2}(k)\}_{b^*=1}^{b^*=n_b}$	$\{G_{3,b^*}^{M_2}(k)\}_{b^*=1}^{b^*=n_b}$	$\cdots$	$\{G_{d,b^*}^{M_2}(k)\}_{b^*=1}^{b^*=n_b}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Table 2: Detector statistics computed from resampled time series from block bootstrap

Mahalanobis distance threshold: In the computation of the bootstrap threshold procedure above, there is one concern when we collect the bootstrap samples for the j -th parameter, $\{G_{j,b^*}^{M_2}(k)\}_{b^*=1}^{b^*=n_b}, j = 1, 2, \dots, d$, at time point $M_2 + k$. Due to the correlation among the vector $\mathbf{G}^{M_2}(k)$ known as the spillover effect, the computation of bootstrap threshold for individual $G_j^{M_2}(k)$ doesn't take this correlation structure into account, and results in a conservative threshold from individual $\{G_{j,b^*}^{M_2}(k)\}$ independently. To remedy the lack of association among the individual bootstrap thresholds, we will use a Mahalanobis distance based threshold to incorporate the spillover effect within the detector statistics $\mathbf{G}^{M_2}(k)$. We regard the bootstrap samples of the vectors $\{(G_{1,b^*}^{M_2}(k), G_{2,b^*}^{M_2}(k), \dots, G_{d,b^*}^{M_2}(k))'\}_{b^*=1}^{b^*=n_b}$ as the empirical multivariate distribution of the

detector statistic $\mathbf{G}^{M_2}(k)$ at monitoring timestamp $M_2 + k$ and compute the Mahalanobis distance (Aggarwal, 2017). The Mahalanobis distance threshold can be found in the following steps,

Step 1 At time stamp $t = M_2 + k$, collect bootstrap samples of the vector of detector statistics

$$\{\mathbf{G}_{b^*}^{M_2}(k)\}_{b^*=1}^{b^*=n_b} = \{(G_{1,b^*}^{M_2}(k), G_{2,b^*}^{M_2}(k), \dots, G_{d,b^*}^{M_2}(k))^T\}_{b^*=1}^{b^*=n_b}.$$

Step 2 Compute the sample mean $\boldsymbol{\mu}_G$ and the sample covariance matrix $\mathbf{S}_G$ of $\{\mathbf{G}_{b^*}^{M_2}(k)\}_{b^*=1}^{b^*=n_b}$, and use them to approximate population mean and covariance matrix of the empirical multivariate distribution for $\mathbf{G}^{M_2}(k)$.

Step 3 Compute the squared Mahalanobis distances $d_{\mathcal{M},b^*}^2(k) = (\mathbf{G}_{b^*}^{M_2}(k) - \boldsymbol{\mu}_G)^T \mathbf{S}_G^{-1} (\mathbf{G}_{b^*}^{M_2}(k) - \boldsymbol{\mu}_G)$, where $\mathbf{G}_{b^*}^{M_2}(k)$ is the vector of detector statistics on the b^* -th bootstrap sample time series.

Step 4 Compute the desired quantile, e.g, the 95% quantile, from $\{d_{\mathcal{M},b^*}^2(k)\}_{b^*=1}^{b^*=n_b}$ to compare with the observed Mahalanobis distance $d_{\mathcal{M}}^2(k) = (\mathbf{G}_{obs}^{M_2}(k) - \boldsymbol{\mu}_G)^T \mathbf{S}_G^{-1} (\mathbf{G}_{obs}^{M_2}(k) - \boldsymbol{\mu}_G)$ where $\mathbf{G}_{obs}^{M_2}(k)$ is the observed detector statistic at $M_2 + k$ computed from the real-time data stream in the monitoring period.

Remark: We opt for the empirical threshold of the Mahalanobis distance instead of using a Chi-square distribution quantile as an approximation, because the normality assumption for the multivariate bootstrap samples is not always satisfied in real applications. See Figure 1 and 2 from a single realization example of some simulated time series for illustration. We note that not all the marginal bivariate distributions have the characteristics of bivariate Gaussian. We can also see that when the monitoring period (starting at $M_2 + 1 = 2501$) is before the true break point $\tau = 3500$, the observed detector statistics $G_j^{M_2}(k)$ fall within the clusters of bootstrap samples of $G_{j,b^*}^{M_2}(k)$. However, when the monitoring process is past the true break point $\tau = 3500$, some of the observed detector statistics $G_j^{M_2}(k)$ will deviate from the clusters of bootstrap samples of $G_{j,b^*}^{M_2}(k)$. The Mahalanobis distance takes the observed vector detector statistic as an entirety to detect the outlier at a multi-dimensional level.

2.3.5 Ensemble detection

2.3.5.1 Motivation of ensemble detection We've listed three types of thresholds for hypothesis testing and each type of threshold has its own pros and cons.

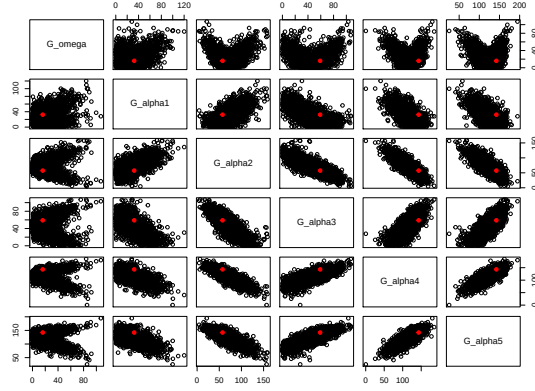


Figure 1: Scatter plot of detector statistics at time stamp $M_2 + k = 3000$ when $\tau = 3500$. The red dots are observed detector statistic and black dots are bootstrap samples of the detector statistics

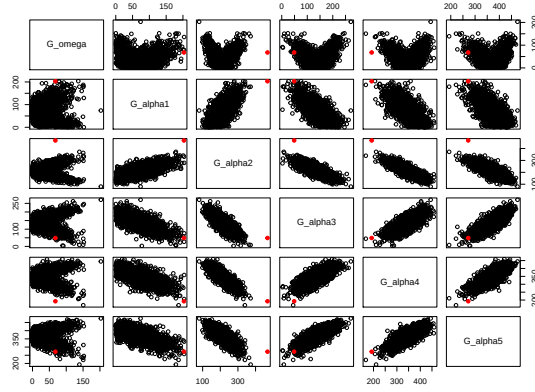


Figure 2: Scatter plot of detector statistics at time stamp $M_2 + k = 4000$ when $\tau = 3500$. The red dots are observed detector statistic and black dots are bootstrap samples of the detector statistics

1. Wiener process thresholds are convenient for quick computation and have a good control of the type I error at the onset of the monitoring period but it can be quickly exceeded by the detector statistics in the absence of a structural break.
2. Bootstrap thresholds are conservative and provide delayed detection of structural break due to the lack of the consideration for the correlation among the individual detector statistics $G_j^{M_2}(k)$.
3. Mahalanobis distance threshold is able to account for the correlation among the individual detector statistics. However, the distribution of Mahalanobis distance has a centroid and spread approximated by bootstrap sample mean vector and covariance matrix. Using thresh-

olds biased by sample outliers will impact the accuracy of break point detection performance.

To improve the overall performance of detection, we propose an ensemble detection scheme to combine the three aforementioned thresholds in a unified framework. By consolidating all the advantages of the three methods, we aim to maintain a high power of detection with a well-controlled type I error. Prior to a detailed description of our method, we would like to clarify the objectives of handling type I error and delay of detection in online financial duration time series.

Type I error: In the context of break point detection, the type I error is false detection before the true break point, which means the break point is incorrectly declared while the structure of the time series still remains identical to the training period. The type I error can be related to several cases: 1) The detection scheme is too sensitive to noise in the monitoring period. 2) The stationarity of the time series before the true break point doesn't hold. 3) The monitoring period is too long before the true break point occurs, so the accumulative noise in the long monitoring period eventually drives the detector statistic to pass the threshold. Our ensemble scheme can make adaptations to the extreme fluctuation in the first case scenario whereas the other two cases raise concerns for general online methods. For this research, our ensemble scheme will provide a viable solution to address the type I error issue.

Delay: The most ideal performance of any detection methods will declare a change point as soon as the structure of the time series changes. We can achieve quick and accurate detection when there is a dramatic change or jump at the break point. In general, for a mixture of signal and noise in the time series, its signal-to-noise ratio determines the speed of a break point detection. The bootstrap threshold is based upon the resampled time series from the training period, so data from a different generating process will eventually trigger the bootstrap threshold in the monitoring period. Due to the irregularity of the duration time series in our study, the delay between detection and actual change points differs from real calendar time difference. For example, if the true break is $\tau = 3500$ and the detected break point is $\hat{\tau} = 3700$, the time index lag is 200 but the calculation of real delay is $\mathcal{D}_{\text{delay}} = \sum_{i=\tau}^{\hat{\tau}} x_i = x_{3501} + x_{3502} + \cdots + x_{3700}$, which is data dependent. See Figure 3. In real applications, when the market is active and trading intensity is high, the online financial durations are observed in small values with high frequency, if the average durations between events are roughly 0.1 seconds, the delay of 200 durations will result in a 20-second delay in the real

calendar time. Similarly, the trading intensity shifts from high to low, the delay in calendar time can be longer but this type of transition of the financial durations are of less interest to the decision-makers and delay can be more tolerable.

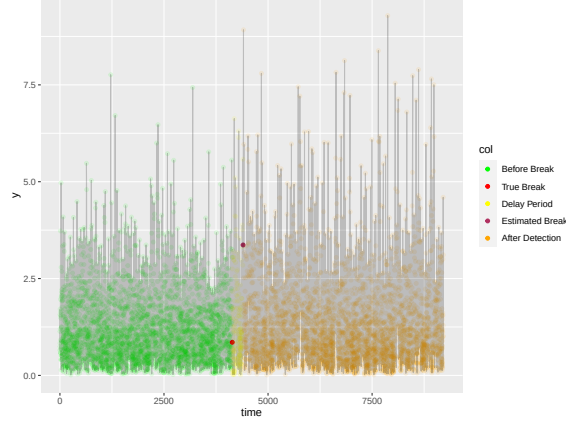


Figure 3: The trace plot of duration of duration in the calendar time. The x-axis is calendar time (in seconds). The y-axis denotes the value of duration. The lag between time indices of true break and detected break is 200 and the delay time in the calendar time is 261.1051 s.

2.3.5.2 Ensemble detection scheme This detection scheme is described as a pseudo algorithm below. The break point alarms are recorded in a sequence of $\delta(k), k = 1, 2, \dots$

Algorithm E-PEF detection

Input: $\{x_k\}_{k=M_2+1}^{\infty}$

Output: $\{\delta(k)\}_{k=M_2+1}^{\infty}$

Compute observed $\mathbf{G}_{obs}^{M_2}(k)$, Mahalanobis distance $d_{\mathcal{M}}^2(k)$. Find out the corresponding thresholds, $d_{1-\alpha_{\mathcal{M}}}^2(k)$, $\{T_{1-\alpha_{\mathcal{W}}, \mathcal{W}}(k)\}$ and $\{T_{\alpha_{\mathcal{B}}/2, j, \mathcal{B}}(k), T_{1-\alpha_{\mathcal{B}}/2, j, \mathcal{B}}(k)\}, j = 1, 2, \dots, d$, through the bootstrap samples respectively.

if $d_{\mathcal{M}}^2(k) > d_{1-\alpha_{\mathcal{M}}}^2(k)$ and $\exists j \in \{1, \dots, d\}, s.t., I(|G_j^{M_2}(k)| > T_{1-\alpha_{\mathcal{W}}, \mathcal{W}}(k)) = 1$ and $I(T_{\alpha_{\mathcal{B}}/2, j, \mathcal{B}}(k) < G_j^{M_2}(k) < T_{1-\alpha_{\mathcal{B}}/2, j, \mathcal{B}}(k)) = 0$ **then**

$\delta(k) = 1, k \leftarrow k+1$

else

$\delta(k) = 0, k \leftarrow k+1$

end if

Remarks:

1. As mentioned before, the reason of using bootstrap threshold of Mahalanobis distance is that the asymptotic normality of the $\mathbf{G}^{M_2}(k)$ may not be a valid assumption, although each individual $G_j^{M_2}(k)$ preserves a roughly symmetric and bell-shaped distribution. The critical

value of a Chi-squared distribution will give a lower threshold comparing with a heavier-tailed distribution.

2. Structural break at $M_2 + k$ will be declared only when both the observed Mahalanobis distance $d_{\mathcal{M}}^2(k)$ exceeds the Mahalanobis distance threshold $d_{1-\alpha_{\mathcal{M}}}^2$, and at least one of the individual $G_j^{M_2}(k)$ exceeds its Wiener process and bootstrap thresholds.
3. The algorithm contains multiple hypothesis testings, we take the Bonferroni's method to control the overall false discovery rate.

To have a detailed illustration of the type I error rate from the aforementioned algorithm, we denote the following events at monitoring time $t = M_2 + k$:

- $A(k) : d_{\mathcal{M}}^2(k) > d_{1-\alpha_{\mathcal{M}}}^2(k)$
- $B_j(k) : I(|G_j^{M_2}(k)| > T_{1-\alpha_{\mathcal{W}}, \mathcal{W}}(k)) \cdot \left(1 - I(T_{\alpha_{\mathcal{B}}/2, j, \mathcal{B}}(k) < G_j^{M_2}(k) < T_{1-\alpha_{\mathcal{B}}/2, j, \mathcal{B}}(k))\right) = 1$ ¹

We use $\delta(k)$ to denote the break point detection indicator at $t = M_2 + k$. Then the theoretical type I error under the null hypothesis of no break point can be derived and controlled as:

$$\begin{aligned}
Pr(\delta(k) = 1 | H_0) &= Pr \left\{ A(k) \cap \left\{ \bigcup_{j=1}^d B_j(k) \right\} \middle| H_0 \right\} < P(A(k) | H_0) + P\left(\bigcup_{j=1}^d B_j(k) | H_0\right) \\
&\leq \alpha_{\mathcal{M}} + \sum_{j=1}^d P(B_j(k) | H_0) \\
&\leq \alpha_{\mathcal{M}} + \sum_{j=1}^d \left(P(\{|G_j^{M_2}(k)| > T_{1-\alpha_{\mathcal{W}}, \mathcal{W}}(k)\} \cap \{T_{\alpha_{\mathcal{B}}/2, j, \mathcal{B}}(k) < G_j^{M_2}(k) < T_{\alpha_{\mathcal{B}}/2, j, \mathcal{B}}(k)\}^c) \right) \\
&\leq \alpha_{\mathcal{M}} + \sum_{j=1}^d (\alpha_{\mathcal{W}} + \alpha_{\mathcal{B}}) = \alpha_{\mathcal{M}} + d \cdot \alpha_{\mathcal{W}} + d \cdot \alpha_{\mathcal{B}} \tag{2.19}
\end{aligned}$$

From E.q (2.19), we can control the type I error by choosing the proper $\alpha_{\mathcal{M}}$, $\alpha_{\mathcal{W}}$, and $\alpha_{\mathcal{B}}$. In the simulation study, we set $\alpha_{\mathcal{M}} = d \cdot \alpha_{\mathcal{W}} = d \cdot \alpha_{\mathcal{B}} = \alpha_{overall}/3$ to control the type I error under $\alpha_{overall} = 0.05$. Given the motivation and theoretical support, the performance of our E-PEF method will be evaluated in the next subsections.

¹ $I(\cdot)$ is the indicator of an event.

2.4 Numerical study

In this section, the performance of E-PEF method is evaluated through extensive simulation study. Some empirical guidelines can be provided for practical implementation. The real application of E-PEF method is demonstrated through monitoring inter-trade duration time series from WRDS Trade and Quote data.

2.4.1 Simulation

For simulation studies, we will demonstrate E-PEF method through three different aspects. The first aspect addresses the common limitation of online detection studies, i.e., the false detection rate control can only be controlled within a finite monitoring window. We also show this feature of our method. Second aspect deals with the operating characteristics in terms of detection probabilities for different structural breaks. The detection probabilities is regarded as false detection (type I error) rate before the true break, and detection power after the true break. The last aspect is associated with real application. We will show the robustness of E-PEF method when the data is non-stationary to address the non-stationarity concern for the real data.

The simulations are based upon the following model:

$$x_i = \exp(\psi_i^{(j)}) \frac{\epsilon_i^{(j)}}{\mu_\epsilon^{(j)}}, \quad (2.20)$$

where

$$\psi_i^{(j)} = \omega^{(j)} + \sum_{k=1}^{p_j} \alpha_k^{(j)} \log x_{i-k} + \sum_{k=1}^{q_j} \beta_k^{(j)} \psi_{i-k}^{(j)}, j = 1, 2, \quad (2.21)$$

$j = 1$ is the model before the true break point τ and $j = 2$ is the model after the true break point.

To illustrate the performance of our algorithm, we apply our method according to different types of the model alteration. Below are the results for the ensemble detection outcomes for the four scenarios of 500 simulations with different true break points ($\tau = 2700, 3000, 3500, 4000$). The tuning parameter λ is determined by grid search. Among a list of λ candidates from 0.1 to 100, we select the model with the smallest Mean Absolute Deviation measure, i.e, $MAD = \frac{1}{M_2} \sum_{i=1}^{M_2} |x_i - \hat{x}_i|$. Then we conduct the E-PEF method upon the model of our choice.

Scenario 1:

$$\psi_i = \begin{cases} 0.2 + 0.1 \log x_{i-1} + 0.2 \log x_{i-2}, & 1 < i \leq \tau \\ 0.2 + 0.1 \log x_{i-1} + 0.5 \log x_{i-2}, & \tau + 1 \leq i \leq 7500 \end{cases} \quad (2.22)$$

$\epsilon \sim \text{Weibull}(.6, .7)$ for both segments.

Scenario 2:

$$\psi_i = \begin{cases} 0.2 + 0.1 \log x_{i-1}, & 1 < i \leq \tau \\ 0.2 + 0.4 \log x_{i-1}, & \tau + 1 \leq i \leq 7500 \end{cases} \quad (2.23)$$

$\epsilon \sim \text{Gamma}(.5, .5)$ for both segments.

Scenario 3:

$$\psi_i = \begin{cases} 0.2 + 0.1 \log x_{i-1} + 0.2 \psi_{i-1}, & 1 < i \leq \tau \\ 0.2 + 0.1 \log x_{i-1} + 0.5 \log x_{i-2}, & \tau + 1 \leq i \leq 7500 \end{cases} \quad (2.24)$$

$\epsilon \sim \text{Gamma}(.5, .5)$ for both segments.

Scenario 4:

$$\psi_i = \begin{cases} 0.6 + 0.1 \log x_{i-1} + 0.2 \log x_{i-2}, & 1 < i \leq \tau \\ 0.2 + 0.1 \log x_{i-1} + 0.2 \log x_{i-2}, & \tau + 1 \leq i \leq 7500 \end{cases} \quad (2.25)$$

$\epsilon \sim \text{Weibull}(.8, .9)$ for both segments.

For Scenarios 1 and 2, transitions before and after the break point are only based upon one of the AR coefficients within the log ACD framework of the same order. The different distributions (Weibull and Gamma distributions) assigned on the innovations are used to justify the robustness of our algorithm. Scenario 3 shows that when the model framework changes from an ARMA model to an AR model, our method is also able to detect the change point in a timely manner. We use Scenario 4 to mimic some of real data scenarios where the durations change from a long and infrequent pattern to a short and volatile state, which is similar to the market with increasing

trading intensity. We select a higher intercept before the structural break point and a lower one afterwards.

2.4.1.1 Monitoring horizon We first investigate the performance of E-PEF detection under the circumstance of no break for different scenarios. We also implement our method using different sizes of training data. Monitoring horizon (Berkes et al., 2004) stands for the maximum length of the monitoring window if there is no structural break in the data generating process, i.e., the detection probability is below the desired false detection (type I error) rate under the H_0 of no break point in the data. Therefore, as long as there is no structural break detected within the monitoring horizon, practitioner can use the ongoing time series model to make statistical inference, with the confidence that the parameters of the current model remain stable. However, extended monitoring period beyond the monitoring horizon will inflate the false detection rate because cumulative noise from the prolonged monitoring distorts the structural break detection. The monitoring horizon is summarized in Table 3.

There are three main takeaways in the table, 1) the false detection rate can only be controlled under a desired level within a monitoring window of finite length. Longer monitoring period leads to a greater inflation of false detection rate, though the false detection rate increases slowly as the monitoring window length increases. 2) a longer training period tends to prolong the monitoring horizon in general, except for some tiny fluctuation in the detection probability at different k , which is caused by the randomness for the bootstrap sampling. 3) As an empirical guidance for choosing a practical length of monitoring period, a monitoring window of length 250-500 is preferred for a desired false detection rate control when the training data is small ($M_2 < 2500$), while practitioners can expect a longer monitoring horizon of about 750-1000 if $M_2 \geq 2500$. Further trade-offs between false detection rate and power will also influence the monitoring window, according to specific demands from the practitioners.

In addition, it can be observed in Table 3 that ARMA framework (Scenario 3) as a data generating process is more robust against false detection, so it has a longer monitoring horizon. That means approximation using a high-order AR model works better when the online data are generated under an ARMA framework than under a low-order AR framework.

Scenarios	M_2	($k =$) 250	500	750	1000	1250	1500	1750	2000	2250	2500
1	2000	0.004	0.018	0.03	0.024	0.03	0.034	0.054	0.06	0.072	0.078
	2500	0.012	0.008	0.014	0.022	0.026	0.032	0.04	0.044	0.058	0.062
	3000	0.01	0.016	0.016	0.014	0.016	0.03	0.026	0.026	0.034	0.042
	3500	0.006	0.014	0.018	0.024	0.02	0.026	0.026	0.03	0.046	0.052
2	2000	0.004	0.014	0.018	0.034	0.04	0.046	0.072	0.082	0.088	0.102
	2500	0.006	0.016	0.022	0.026	0.034	0.036	0.042	0.044	0.06	0.08
	3000	0	0.022	0.022	0.026	0.03	0.032	0.044	0.046	0.058	0.07
	3500	0.016	0.01	0.012	0.012	0.014	0.02	0.026	0.038	0.054	0.048
3	2000	0.006	0.02	0.028	0.036	0.04	0.056	0.086	0.092	0.102	0.116
	2500	0.012	0.016	0.018	0.028	0.038	0.046	0.052	0.052	0.058	0.064
	3000	0.006	0.008	0.024	0.018	0.028	0.034	0.038	0.04	0.044	0.054
	3500	0.016	0.014	0.022	0.022	0.026	0.026	0.036	0.032	0.038	0.038
4	2000	0.028	0.036	0.034	0.048	0.064	0.066	0.076	0.074	0.086	0.098
	2500	0.012	0.02	0.038	0.026	0.03	0.038	0.046	0.062	0.064	0.084
	3000	0.01	0.016	0.018	0.022	0.03	0.024	0.036	0.044	0.054	0.046
	3500	0.01	0.014	0.016	0.012	0.02	0.024	0.032	0.044	0.056	0.068

Table 3: Detection probabilities in the monitoring horizon study. M_2 is the length of training period. k is the monitoring time point in the monitoring period after the training. The false detection rate control $\alpha = 0.05$.

2.4.1.2 Detection probability and Delay According to the result from monitoring horizon summary, we choose the training period to be $M_2 = 2500$, with the burn-in period $M_1 = 1500$ for the detection simulation study. The performance of E-PEF is presented through two measures, 1) detection probabilities at different monitoring times. The false detection and power refer to the detection probabilities before and after the true break points correspondingly. 2) Summary statistics of average delay. Average delay reflects the sensitivity of the detection algorithm after the occurrence of a structural break. These results are exhibited in Table 4 and Table 5.

From Table 4, E-PEF controls the false detection rate under the nominal level, $\alpha = 0.05$ and increases the detection power rapidly after the true structural breaks occur in the monitoring procedure. The E-PEF method has an overall satisfactory performance of detection, and a significant detection power can be expected when there is a structural break involved with framework change, e.g., from ARMA to AR in scenario 3. Detection power spikes up to 1 in a short period.

From Table 5, E-PEF method has different sensitivity levels for the different kinds of structural breaks. For Scenario 1, 2, and 4, parameter changes within the same AR framework are detected slower than the framework changes as in Scenario 3. In scenario 4, belated true break associated

Scenarios	τ	$(M_2 + k =)$	2750	3000	3250	3500	3750	4000	4250	4500	4750	5000
1	2700	0.02	0.636	0.962	0.99	1	1	1	1	1	1	1
	3000	0.012	0.014	0.324	0.854	0.982	1	1	1	1	1	1
	3500	0.01	0.012	0.012	0.018	0.228	0.724	0.932	0.982	0.998	1	1
	4000	0.012	0.008	0.006	0.026	0.02	0.03	0.2	0.59	0.876	0.968	1
2	2700	0.02	0.548	0.924	0.98	0.998	1	1	1	1	1	1
	3000	0.006	0.016	0.192	0.724	0.922	0.978	1	0.998	1	1	1
	3500	0.006	0.018	0.02	0.028	0.134	0.492	0.812	0.936	0.986	0.994	1
	4000	0.004	0.018	0.024	0.036	0.036	0.046	0.132	0.368	0.698	0.87	1
3	2700	0.06	0.996	1	1	1	1	1	1	1	1	1
	3000	0.01	0.012	0.882	1	1	1	1	1	1	1	1
	3500	0.01	0.014	0.02	0.028	0.636	0.998	1	1	1	1	1
	4000	0.01	0.012	0.02	0.024	0.044	0.042	0.486	0.952	0.998	1	1
4	2700	0.008	0.442	0.918	0.996	1	1	1	1	1	1	1
	3000	0.012	0.026	0.184	0.686	0.946	0.998	1	1	1	1	1
	3500	0.014	0.022	0.034	0.026	0.152	0.538	0.844	0.962	0.994	1	1
	4000	0.014	0.02	0.028	0.028	0.032	0.038	0.13	0.442	0.746	0.924	1

Table 4: Detection probabilities at different monitoring time under different scenarios. τ is the true break point. k is the monitoring time point in the monitoring period.

with a prolonged monitoring period, tends to give more false detection. For the application of E-PEF, the delay in calendar time is data-dependent, so its conversion from the delay in the time index doesn't demonstrate the timeliness of the E-PEF detection without the summation of durations between true break and the detected break. On the other hand, detection of minor parameter changes within the same framework takes more streaming data for sufficient evidence but the consequence of drawing inference from a slightly biased model is not too grave if practitioners are just using the same model framework with subtle differences between the parameters. Since our method has a high sensitivity to the model framework change, timely parameters update can be made at the occurrence of such significant structural breaks.

2.4.1.3 Robust performance for non-stationary scenarios Since E-PEF method involves stationary log ACD models, it is difficult for data in real application to abide by the stationarity assumption. Therefore, we will demonstrate the robust performance of E-PEF method for non-stationary data. Although non-stationary data can come in various forms, local non-stationarity can be still described by a linear trend or quadratic trend. For a small size of monitoring horizon in

Scenarios	τ	(Delay) Mean	SD	Min	Q1	Q2	Q3	Max
1	2700	264.61	126.30	97	173.5	240	328.25	825
	3000	326.45	159.08	109	211.5	301.5	412.75	943
	3500	391.14	200.63	71	251.75	372	501.25	1257
	4000	445.36	245.48	65	269.75	424	585.25	1368
2	2700	294.25	147.70	106	185.75	266.5	372.25	1167
	3000	387.09	184.97	135	251	359	485.75	1191
	3500	511.92	261.74	136	325.75	480.5	656	1780
	4000	570.34	312.38	82	332	545.5	749	1685
3	2700	116.46	49.02	48	82	109	143	325
	3000	162.77	72.36	60	109	153.5	206.25	463
	3500	214.87	96.01	63	140.5	202	283.25	548
	4000	249.72	118.72	37	167	247.5	327	651
4	2700	329.49	141.60	131	221	309	422	900
	3000	411.03	182.23	130	278.5	390	531.25	1032
	3500	490.85	242.63	117	311.25	459	643.25	1344
	4000	561.95	287.58	142	341.75	519.5	743	1596

Table 5: Summary statistics of average delay under different scenarios for different break points. τ is the true break point. False detection rate control is $\alpha = 0.05$. The length of training period is $M_2 = 2500$.

practice, it will sufficient to display the robust performance of E-PEF against non-stationary data with a linear trend or quadratic trend.

The linear time trend and quadratic time trend are introduced in both segments before and after the structural breaks as follows,

Scenario 1:

$$\psi_i = \begin{cases} 0.2 + 0.05 \log(i) + 0.1 \log x_{i-1} + 0.2 \log x_{i-2}, & 1 < i \leq \tau \\ 0.2 + 0.05 \log(i) + 0.1 \log x_{i-1} + 0.5 \log x_{i-2}, & \tau + 1 \leq i \leq 7500 \end{cases} \quad (2.26)$$

$\epsilon \sim \text{Weibull}(.6, .7)$ for both segments.

Scenario 2:

$$\psi_i = \begin{cases} 0.2 + 0.1 \log(i) + 0.1 \log x_{i-1}, & 1 < i \leq \tau \\ 0.2 + 0.1 \log(i) + 0.4 \log x_{i-1}, & \tau + 1 \leq i \leq 7500 \end{cases} \quad (2.27)$$

$\epsilon \sim \text{Gamma}(.5, .5)$ for both segments.

Scenario 3:

$$\psi_i = \begin{cases} 0.2 + 0.05 \log(i) - 0.01 \log^2(i) + 0.1 \log x_{i-1} + 0.2 \psi_{i-1}, & 1 < i \leq \tau \\ 0.2 + 0.05 \log(i) - 0.01 \log^2(i) + 0.1 \log x_{i-1} + 0.5 \log x_{i-2}, & \tau + 1 \leq i \leq 7500 \end{cases} \quad (2.28)$$

$\epsilon \sim \text{Gamma}(.5, .5)$ for both segments.

Scenario 4:

$$\psi_i = \begin{cases} 0.6 + 0.1 \log(i) - 0.01 \log^2(i) + 0.1 \log x_{i-1} + 0.2 \log x_{i-2}, & 1 < i \leq \tau \\ 0.2 + 0.1 \log(i) - 0.01 \log^2(i) + 0.1 \log x_{i-1} + 0.2 \log x_{i-2}, & \tau + 1 \leq i \leq 7500 \end{cases} \quad (2.29)$$

$\epsilon \sim \text{Weibull}(.8, .9)$ for both segments.

For the non-stationary settings of 2.26,2.27,2.28,2.29, we consider assigning small coefficients for the linear trend and the quadratic trend for two reasons, 1) large coefficient for a time trend can quickly shrink the durations to zero or cause the overflow of the durations. 2) Small coefficients for the trend terms can generate similar ranges of the durations, which are consistent with the ones for real data.

Since our method performs universally well in all four scenarios with or without stationarity assumption, we demonstrate the model results in Figure 4 (Scenario 1) for brevity. All the monitoring processes start at $M_2 = 2500$ and the structural break detection probabilities can be accessed at different phases in the monitoring period. For the scenarios with different structural breaks, our algorithm shows a good control of type I error under 0.05 before the true structural breaks occur. Meanwhile, the detection probability increases quickly after the occurrences of the true structural

breaks.

For illustration, the intermediate steps are shown in the Figure 5 by a single time series realization example from Scenario 1. (a) shows the observations x_i with true break point at 3500 (at the red vertical dashed line). (b) and (c) depict the trajectory of the observed detector statistic $\mathbf{G}^{M_2}(k)$ and the Mahalanobis distance. The green shaded area in (b) is the combined confidence interval based on the theoretical Wiener process threshold and the empirical bootstrap sample threshold. The red horizontal line in (c) is the empirical bootstrap sample threshold for the Mahalanobis distance. (d) gives the binary results of break detection at each time point in the monitoring process from $i = 2500$. From (d), the detected break point is around 3700, which is close to the true break point $\tau = 3500$.

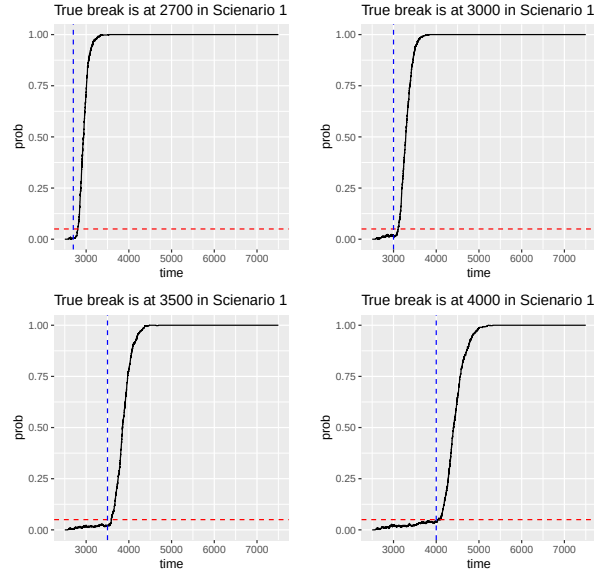


Figure 4: Break detection outcome from 500 simulations of in a monitoring period of length 5000. X-axis stands for the monitoring period starting at $M_2 = 2500$. Y-axis stands for the empirical detection probability at a specific time point. The blue dashed lines stand for the true break point $\tau = 2700, 3000, 3500$, and 4000 . The red dashed lines stand for the significance level $\alpha_{overall} = 0.05$.

For practical implementation, these results provide useful guidance for the decision-makers to adjust their tolerance of the type I error and the preference of the sensitivity of the break detection. These results also show that in Scenarios 1 and 2 where there is only one parameter change within the same log ACD model framework, our algorithm is able to detect the structural break rapidly and accurately. In Scenario 3 where there is a change on the type of the log ACD model, the detection probability increases more quickly to 100% after the structural break occurs. Finally, since the

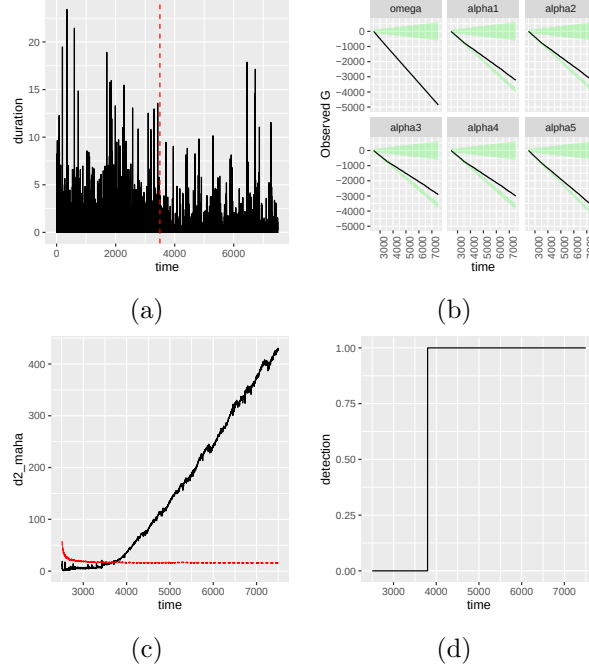


Figure 5: (a) A single realization of Scenario 1 with true break at $\tau = 3500$; (b) The trace plot for the individual $G_j^{M2}(k)$ with the confidence intervals based on the Wiener process and bootstrap thresholds in green bands; (c) The trace plot for the Mahalanobis distance $d_M^2(k)$, with bootstrap threshold $d_{1-\alpha_M}^2(k)$ (red horizontal line); (d) The trace plot for the binary indicator of break detection $\delta(k)$

detector statistic is based upon the quasi-score change instead of prediction on the observational level, for Scenario 4 we are able to implement the structural break detection if there is an increasing trading intensity between two phases of financial duration time series. Additional simulation results for Scenarios 2, 3, and 4 are provided in the supplemental material.

2.4.2 Real application

In addition to the extensive simulation study to illustrate our method in different scenarios, we also implement it for some real financial duration data. We are able to detect the structural breaks and associate them with some financial information that may trigger these structural breaks. Nonetheless, real data applications of our method need attentions to different aspects of implementation. To get a meaningful monitoring process and identify outbreaks of interest, practitioners need to consider the general features of the financial durations and idiosyncratic characteristics of different assets.

In existing retrospective studies of structural break detection, the raw time series data can be adjusted universally to filter out some common structures and enhance the model's adequacy. For

example, Zhang et al. (2018) performed structural break detection in a retrospective fashion, and the complete intra-day financial durations were adjusted through a linear regression to accommodate the diurnal effect, which is displayed as higher trading intensity in the opening and closing period of the stock market. However, since our ensemble method processes real-time streaming data in practice, we cannot replicate the same retrospective adjustment for online break detection. If the aim is to detect some non-diurnal-effect-related structural breaks, we suggest discarding the first and last 30-minute time windows and applying the algorithm to diminish the diurnal effect.

Due to the flexible definition of a financial event, users can feel free to choose the price or return changes as the target events to collect the online duration time series based on individual risk tolerance and trading preferences. Besides, we also recommend market participants customize their monitoring process according to the characteristics of individual assets and set proper tuning parameters for the monitoring such as the length of the training period.

As illustrated in the examples in real applications, we apply our break detection method to four specific stocks: IBM, BAC, MMM, and GE, and explore the dynamic structures underlying the financial durations. We choose the price changes between two successive transactions exceeding a pre-specified amount, δ , as the financial events. The financial durations are computed from the WRDS Trade and Quote data in June 2013 (Zou et al., 2015). Following a common approach used by financial professionals, we used the data from the previous month (May 2013) to calculate the average turnover ratio as average daily volume divided by total shares outstanding. Therefore, $\delta = 0.00377, 0.004, 0.00376, 0.00408$ for GE, BAC, IBM, and MMM respectively.

When it comes to the tuning parameters, a meaningful monitoring process needs to accommodate the liquidity of the assets. GE and BAC are liquid stocks with high trading volumes (100,000 to 200,000 transactions per day on average) and a relatively low price spread of about \$ 0.5 per day on average. Meanwhile, IBM is also a liquid stock with a medium number of average transactions per day (around 20,000) with an absolute price spread between \$3 and \$4. MMM has variable low numbers of transactions (between 4000 and 15,000 per day) and absolute price spread of about \$1.5. In practice, for the liquid assets, we choose a longer training period for better model adequacy. But for the less liquid assets such as MMM, the number of transactions can be very small. Therefore, we suggest choosing a shorter training period and λ candidates of large values in the penalty term $p'_\lambda(|\boldsymbol{\theta}|)$ of the penalized estimating function for grid search, in order to enhance model stability in

the recursive estimation procedure during the training period.

To reflect the market behaviour and trading intensity through the pattern change of financial durations, we apply the ensemble detection method to a log ACD (20, 0) as the training model for the break detection. All transactions are selected from 10:00 a.m. and 3:30 p.m.. We are able to detect some structural breaks, and benchmark with published financial news for these companies on the same day, or during off hours before. With the detected breaks shown in Figure 6, We list the break detection results as well as their related financial news for stocks IBM, BAC, MMM, and GE respectively.

On June 4th, the structural break of IBM was detected at around 1:01 p.m.. A piece of financial news about IBM was released on June 4th at 9:11 a.m.. From The New York Times, “*I.B.M. announced on Tuesday that it had agreed to buy SoftLayer Technologies, a cloud computing company, in an effort to strengthen I.B.M.’s position in the fast-growing market for computing sold to businesses as a service delivered over the Internet.*”.

On June 5th, the structural break of BAC was detected at around 10:33 a.m.. We have also found a piece of financial news about Bank of America on June 4th at 9:47 p.m. was as follows. From Reuters, “*NEW YORK, June 4 (Reuters) - American International Group Inc argued on Tuesday that a proposed \$8.5 billion settlement between Bank of America Corp and investors in Countrywide Financial Corp mortgage-backed securities was not big enough.*”.

On June 10th, the structural break of MMM was detected at around 13:10 p.m.. The CEO of MMM reclaimed the company was research-driven and invested a large amount of money in its research development on June 10th at 6:53 a.m. From CNBC, “*In a lab in St. Paul, Minnesota, engineers test solar panels and connectors. A half a mile away, a technician smashes the windows of a car coated with 3M protective film, watching to see if the glass shatters or holds together. Elsewhere, there are scientists working on more aerodynamic products for airplanes, software systems to run a municipality’s department of motor vehicles, and light-sensor technology to create new crowns for teeth in under 2 hours.*”.

On June 24th, the structural break of GE was detected at around 12:54 p.m., A recent event that happened to GE incorporation was on June 22th (Saturday). From military aerospace, “*TransDigm Group Inc. (NYSE:TDG) has entered into a definitive agreement to acquire the assets of GE Aviation’s Electromechanical Actuation Division for approximately \$150 million in cash.*”.

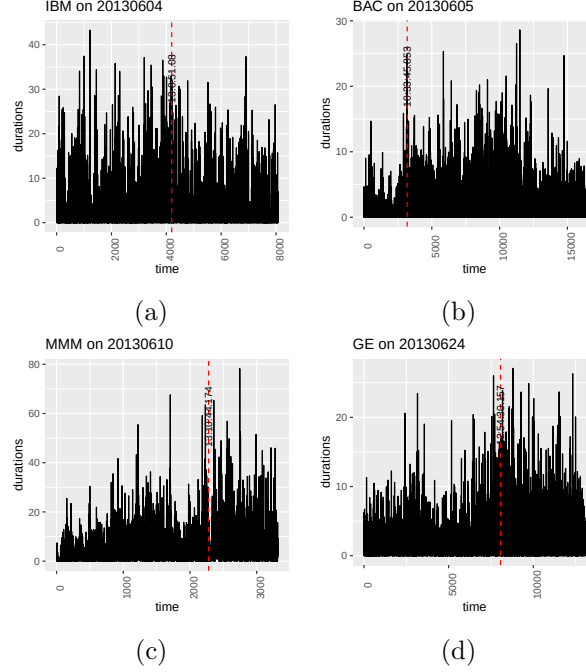


Figure 6: Structural break points detected (red vertical dash lines) in the duration time series of IBM, BAC, MMM and GE respectively.

From real applications, our E-PEF method is able to detect break points in duration time series and offers an adaptive framework incorporating different asset features. With detected change points from online data, the practitioners can act proactively and seek for more evidence from different sources of information such as market news to evaluate the asset or market behaviors. With the combination of statistically detected break points and real world information, E-PEF method can reassure practitioners to use adequate models for the online data and make subsequent transactional decisions.

2.5 Summary

In a high-frequency trading setting, the financial duration models can give a reasonable interpretation on the volatile behavior of the market in terms of trading intensity. Therefore, accurate awareness of the structural breaks will provide helpful insights for the market participants to update the statistical model as soon as a structural break is detected. In this article, we propose an innovative E-PEF method in the online detection strategy for the log ACD model, which is a prevalent tool for modeling financial durations. In the parameter estimation stage for the training period, we take advantage of the PEF recursive estimation approach for the log ACD model regard-

less of the distribution of the innovations. In the online monitoring stage, we propose an ensemble algorithm combining three different types of thresholds for the quasi-score functions to facilitate structural break detection in the financial duration time series. We illustrate the performance of our ensemble approach by extensive simulations under different scenarios. We implement our method in real data applications to monitor the duration time series and detect some structural breaks with various sources of financial information for a better interpretation of the market behavior.

Our contribution lies in the innovative ensemble online detection framework for financial duration time series. This framework enjoys a distribution-free statistical inference on the log ACD model with penalized estimating functions. The block bootstrap method provides the empirical distribution of PEF from the observed training time series without additional parametric assumptions. Resampled data further lead to the incorporation of two new thresholds, the bootstrap threshold and the Mahalanobis distance threshold, into the monitoring procedure. The detection rule is formulated in an ensemble fashion integrating all three types of thresholds. The combined advantages of both asymptotic and empirical properties of the online detector test statistic facilitate the reduction of the false discovery rate. The E-PEF approach shows that the type I error can be well controlled under a specific level while preserving a high power to detect any true breakpoints. The merit of our method focuses on timely structural break detection under parameter changes in an online fashion.

3 Multivariate latent level correlation model (LCM) for financial high frequency count time series

3.1 Background

Volatility analysis helps investors navigate their asset positions to optimize their gains in the financial market. As high-frequency, intra-day asset pricing data become increasingly available, exploration and explanation for the micro-structure of the financial market behavior forms an important and popular research direction. The volatility study of the general financial data involved with irregularity and intricacy already requires non-trivial methods for delicate statistical inference. The advancement of high frequency trading escalates the challenge with increased data size as well as dimension. Prompt inference on high frequency financial data is in great demand for the market participants. In a high frequency trading setting, the raw transactions are made within milliseconds, and the corresponding price changes occur with various magnitudes of fluctuation. Therefore, log returns for the asset price can be calculated based on these price changes. In addition to other different research-related definitions, our study use the count of the transactions with extreme log return to describe the market volatility, and the extremeness of a log return is determined by practitioners according to their interest. Within a given time interval, an asset with higher volatility tend to have a larger count, which motivates one to implement discrete time series model to account for the underlying market volatility.

Specifically, we convert transaction-by-transaction log returns to a multivariate count time series by using a threshold $\tau > 0$. During a given period of time, the transactions with the absolute log return exceeding τ are categorized as high risk level, while the other are categorized as low risk level. Therefore, for a specific asset, two types of count will be observed and studied simultaneously. Under general circumstance of high frequency trading trading, the price changes don't have significant fluctuation thus the transactions of low risk level are expected to have a large number of count, and such count has its own time dependent property. The count for transactions of high risk level provide more insights on the volatility but the count may not be as abundant as the former, due to the asset idiosyncrasy or the choice of τ . With the access to a plenitude of low-risk-level transaction data, practitioners could have more options to adjust and refine their

investment strategy if they have a sound comprehension of the co-movement between these two types of counts.

3.2 Literature review

3.2.1 Discrete time series modeling

Count time series arise naturally in numerous applied scientific fields such as finance, epidemiology, agriculture, etc., and the analysis of such non-Gaussian data has been an active research area for a long time. There is a rich literature on univariate modeling of count time series, both in the frequentist and Bayesian frameworks. The frequentist approaches include discrete auto-regressive moving average (DARMA) models (Jacobs and Lewis, 1983), integer-valued auto-regressive (INAR) models (Al-Osh and Alzaid, 1987), and integer-valued GARCH (INGARCH) models (Ferland et al., 2006). West et al. (1985) and Gamerman et al. (2013) have discussed Bayesian dynamic generalized linear models (DGLM). Financial applications of univariate count time series models include Heinen (2003) who implemented the double exponential family GLM of Efron (1986) by introducing the auto-regressive conditional Poisson model. They applied the model for volatility modeling of price change count time series of IBM returns, exploiting the ability of the model to explain the autocorrelation and overdispersion in the data. Liesenfeld et al. (2006) used an integer count hurdle model with a negative binomial sampling distribution for stock transaction price changes. However, it is now widely accepted that financial data are time-dependent across assets and markets. Although the aforementioned methodologies deal with univariate time series, their perspectives of approaching the univariate time series modeling still facilitate the extension of multivariate time series framework.

For analyzing multivariate count time series, we need models that not only explain the temporal correlation within each series, while accommodating possible overdispersion, but also explain cross-sectional correlation between the components. Pedeli and Karlis (2013) extended the INAR(1) framework to the bivariate case. Karlis and Meligkotsidou (2005) discussed a multivariate Poisson (MVP) distribution with a two-way covariance structure, while Karlis and Meligkotsidou (2007) described a finite mixture of multivariate Poisson distributions which allows for both positive and negative covariances between components. Ravishanker et al. (2014) used MVP sampling distribu-

tions and Gibbs sampling for hierarchical dynamic modeling for multivariate time series of counts of gastropod abundance. Jung et al. (2011) proposed a parameter-driven approach for multivariate count financial time series through a dynamic factor model incorporating both common and idiosyncratic factors in the conditional Poisson mean. Quoreshi (2017) fit a bivariate integer-valued fractionally integrated moving average model to time series of counts of transactions in equally spaced time intervals, and managed to accommodate the long-memory pattern of the integer-valued time series. Aktekin et al. (2018) combined dynamic temporal parameters and individual static parameters into a product as the Poisson means in the multivariate Poisson-scaled beta model, and used forward filtering backward sampling (FFBS) and particle learning algorithms setting in the dynamic online Bayesian updating. See Soyer and Zhang (2021) for an excellent review of recent advances in multivariate count time series modeling. Despite the existent methodologies for multivariate count time series, the concern about computational feasibility arises when it comes to the high-dimension time series modeling, whether or not in frequentist or Bayesian approach. For multivariate INAR or MVP, the computation for the probability mass function of a multivariate discrete random variable requires a large number of summations from inverse mapping. As a consequence, Expectation-Maximization (EM) algorithm for maximum likelihood estimation will be slowed down significantly due to the increase of dimension even though the utilisation of recursive relationships to calculate the conditional expectation can alleviate some of the computational burden. For Bayesian approach, exact inference for joint posterior by MCMC sampling is also of great difficulty because it is not uncommon for the posteriors to be intractable. Hence, the computation of high dimensional integrals suffers from the posteriors that are not in a closed form.

A few alternative ideas have been proposed to carry out fast Bayesian modeling. West (2020) proposed a decouple/recouple idea in DGLM for computational efficiency and model adequacy for node-node dependency based upon the complete sets of posterior samples of the dynamic coefficients. Lavine et al. (2020) introduced copula modeling to multi-step, multi-scale, and dynamic latent factors, using variational Bayes' (VB) optimization (Blei et al., 2017). Being appealing to provide comparable accuracy to Gibbs sampling at greater speed, VB approach often requires a large amount of work to derive the set of equations for iterative parameter update. Therefore, An alternative to VB for fast, approximate Bayesian modeling can be considered for statistical inference, which is the integrated nested Laplace approximation, see Rue et al. (2009) or Rue et al.

(2017).

3.2.2 Integrated Nested Laplace Approximation (INLA)

A wide variety of models can be fit by INLA (Gómez-Rubio, 2020). In general, for a vector of n observations $\mathbf{y} = (y_1, \dots, y_n)$, mean μ_i of observations y_i can be related to the linear predictor η_i through different link functions,.

$$\eta_i = \alpha + \sum_{j=1}^{n_\beta} \beta_j z_{ji} + \sum_{k=1}^{n_f} f^{(k)}(u_{ki}) + \varepsilon_i; i = 1, \dots, n, \quad (3.1)$$

where α is the intercept, $\beta_j, j = 1, \dots, n_\beta$, are coefficients of the covariates $\{\mathbf{z}_j\}_{j=1}^{n_\beta}$, functions $f^{(k)}(u)$ define n_f random effects on the covariates $\{\mathbf{u}_k\}_{k=1}^{n_f}$. ε_i is the error term, and it can be missing depending on the likelihood. The vector of latent effect $\mathbf{x}$ is defined as,

$$\mathbf{x} = (\eta_1, \dots, \eta_n, \alpha, \beta_1, \dots). \quad (3.2)$$

The distributions of $\mathbf{y}$ are dependent on the latent effect $\mathbf{x}$ and some hyperparameters $\boldsymbol{\theta}_1$, and the precision matrix for the Gaussian Markov Random Field (GMRF) of $\mathbf{x}$ is determined by some other hyperparameters $\boldsymbol{\theta}_2$. With the assumptions that $\mathbf{x}$ has a sparse structure of GMRF (Rue and Held, 2005) as well as that y_i 's are independent to each other given $\mathbf{x}$, INLA will take advantage of the sparse structure and conditional independence properties of GMRF to enhance computational efficiency. Let $\boldsymbol{\theta} = (\boldsymbol{\theta}_1, \boldsymbol{\theta}_2)$. Instead of obtaining the inference from the joint posterior distribution of $(\mathbf{x}, \boldsymbol{\theta})$, INLA focuses on the marginal inference on the latent effects and hyperparameters. Starting with the form of joint posterior density of $\mathbf{x}$ and $\boldsymbol{\theta}$,

$$\begin{aligned} \pi(\mathbf{x}, \boldsymbol{\theta} | \mathbf{y}) &\propto \pi(\boldsymbol{\theta}) |\mathbf{Q}(\boldsymbol{\theta})|^{1/2} \exp \left\{ -\frac{1}{2} \mathbf{x}^T \mathbf{Q}(\boldsymbol{\theta}) \mathbf{x} \right\} \prod_{i \in \mathcal{I}} \pi(y_i | x_i, \boldsymbol{\theta}) = \\ &\pi(\boldsymbol{\theta}) |\mathbf{Q}(\boldsymbol{\theta})|^{1/2} \exp \left\{ -\frac{1}{2} \mathbf{x}^T \mathbf{Q}(\boldsymbol{\theta}) \mathbf{x} + \sum_{i \in \mathcal{I}} \log(\pi(y_i | x_i, \boldsymbol{\theta})) \right\}. \end{aligned} \quad (3.3)$$

In order to get the posterior marginals, the approximation of the joint posterior of $\boldsymbol{\theta}$, $\tilde{\pi}(\boldsymbol{\theta} | \mathbf{y})$, is proposed as,

$$\tilde{\pi}(\boldsymbol{\theta} | \mathbf{y}) \propto \frac{\pi(\mathbf{x}, \boldsymbol{\theta}, \mathbf{y})}{\tilde{\pi}_G(\mathbf{x} | \boldsymbol{\theta}, \mathbf{y})} \Big|_{\mathbf{x}=\mathbf{x}^*(\boldsymbol{\theta})}, \quad (3.4)$$

where $\tilde{\pi}_G(\mathbf{x}|\boldsymbol{\theta}, \mathbf{y})$ can be a Gaussian approximation or Laplace approximation at the model of the full conditiona, $\mathbf{x}^*(\boldsymbol{\theta})$. Hence, the marginals for the individual latent effect x_l and hyperparameter θ_k become available,

$$\begin{aligned}\pi(x_l|\mathbf{y}) &= \int \pi(x_l|\boldsymbol{\theta}, \mathbf{y}) \tilde{\pi}(\boldsymbol{\theta}|\mathbf{y}) d\boldsymbol{\theta}, \\ \pi(\theta_k|\mathbf{y}) &= \int \tilde{\pi}(\boldsymbol{\theta}|\mathbf{y}) d\boldsymbol{\theta}_{-k}.\end{aligned}\tag{3.5}$$

INLA has been widely used in several applications of time series. Ruiz-Cárdenas et al. (2012) discussed a variety of state-space dynamic models, including count time series modeling. Schrödle and Held (2011) described disease count data modeling through incidence rates, while Salmon et al. (2015) discussed a Bayesian approach for detecting outbreak in an infectious disease surveillance system. Sadykova et al. (2017) used zero-inflated and hurdle Poisson spatio-temporal models for predator-prey and competitor species habitat. Serhiyenko et al. (2018) implemented a latent level correlation model (LCM) using INLA for dynamic modeling of multivariate counts in a marketing application involving monthly prescriptions written by physicians for a pharmaceutical company's drugs. Riebler and Held (2017) used the INLA approach for age-period-cohort analysis. Raman et al. (2020) explored different univariate static and dynamic models using INLA for evaluating promotions of marketing schemes.

3.3 BVAR(1)-LCM model

In this chapter, we propose a fast and accurate Bayesian framework for correlated bi-variate count time series with latent level correlation (BVAR(1)-LCM) using the INLA approach. The aim is to describe the association between the two count time series across multiple assets and exhibit some model parameters of interest to better explain the microstructure of the financial market. Despite the elevated model complexity, our method enjoys computational speediness due to the sparsity of Gaussian Markov Random Field (GMRF) among the model parameters. Fast Bayesian inference approximation can be obtained with good quality, and practitioners can study the dynamics within and between the count time series in a higher dimension efficiently.

3.3.1 Model framework

Let $\{Y_{j,st}\}$ denote the count for j -th type of transaction for s -th asset in the t -th time interval, $j = 1, 2, \dots, J$, $s = 1, 2, \dots, S$ and $t = 1, 2, \dots, T$. Since we study the association between transaction count with a low-risk level and the one with a high-risk level, $J = 2$. S is the total number of assets involved in the framework and T is the total number of time intervals. To account for the temporal and cross-sectional association among the assets, we implement a Bayesian Poisson Lognormal framework conditioned on various sources of fixed or random effects:

$$\begin{aligned}
Y_{j,st} | \lambda_{j,st} &\overset{ind}{\sim} Pois(\lambda_{j,st}), \\
\eta_{j,st} = \log \lambda_{j,st} &= \mathbf{Z}_j \boldsymbol{\beta}_j + \gamma_{j,t} + \alpha_{j,st}, \\
\gamma_{j,t} &= \phi_j \gamma_{j,t-1} + \omega_{j,t}, \\
\boldsymbol{\omega}_t &= \begin{pmatrix} \omega_{1,t} \\ \omega_{2,t} \end{pmatrix} \sim N(\mathbf{0}, \boldsymbol{\Sigma}_\omega), \\
\boldsymbol{\Sigma}_\omega &= \begin{pmatrix} \sigma_{\omega_1}^2 & \rho_\omega \sigma_{\omega_1} \sigma_{\omega_2} \\ \rho_\omega \sigma_{\omega_1} \sigma_{\omega_2} & \sigma_{\omega_2}^2 \end{pmatrix}, \\
\boldsymbol{\alpha}_{st} &= \begin{pmatrix} \alpha_{1,st} \\ \alpha_{2,st} \end{pmatrix} \sim N(\mathbf{0}, \boldsymbol{\Sigma}_\alpha), \\
\boldsymbol{\Sigma}_\alpha &= \begin{pmatrix} \sigma_{\alpha_1}^2 & \rho_\alpha \sigma_{\alpha_1} \sigma_{\alpha_2} \\ \rho_\alpha \sigma_{\alpha_1} \sigma_{\alpha_2} & \sigma_{\alpha_2}^2 \end{pmatrix}.
\end{aligned} \tag{3.6}$$

In (3.6), $\mathbf{Z}_j$ and $\boldsymbol{\beta}_j$ are respectively covariates (intercept included) and coefficients. $\gamma_{j,t}$ is the temporal random effect with an AR process for j -th type of count. Since they are two types of count involved, j takes values of either 1 or 2. The innovation terms $\omega_{j,t}$ for the two AR processes have the correlation coefficient, ρ_ω . $\sigma_{\omega_1}^2$ and $\sigma_{\omega_2}^2$ are correspondingly the variances for the AR innovations. $\boldsymbol{\alpha}_{st}$ is the level correlated random effect, $\boldsymbol{\Sigma}_\alpha$ is its variance-covariance matrix.

The benefit of Poisson-Lognormal distribution has been discussion in Aitchison and Ho (1989). It not only retains the interpretability of the parameters in the model but also it well addresses the overdispersion issue encountered by a standard Poisson distribution. Considering a simple

univariate Poisson-Lognormal random variable:

$$\begin{aligned} Y|\lambda &\sim \text{Pois}(\lambda), \\ \log \lambda|\mu, \sigma^2 &\sim N(\mu, \sigma^2). \end{aligned} \tag{3.7}$$

Using the laws of total expectation and variance we can easily derive the conditional expectation $E(Y|\mu, \sigma^2)$ and the conditional variance $\text{Var}(Y|\mu, \sigma^2)$.

For $E(Y|\mu, \sigma^2)$, we have,

$$\begin{aligned} E(Y|\mu, \sigma^2) &= E_{\lambda|\mu, \sigma^2} (E(Y|\lambda, \mu, \sigma^2)) \\ &= E_{\lambda|\mu, \sigma^2} (E(Y|\lambda)) = E_{\lambda|\mu, \sigma^2} (\lambda) \\ &= E_{\lambda|\mu, \sigma^2} (e^{\log \lambda}) = e^{\mu + \sigma^2/2}. \end{aligned} \tag{3.8}$$

For $\text{Var}(Y|\mu, \sigma^2)$, we have,

$$\begin{aligned} \text{Var}(Y|\mu, \sigma^2) &= E_{\lambda|\mu, \sigma^2} (\text{Var}(Y|\lambda, \mu, \sigma^2)) + \text{Var}_{\lambda|\mu, \sigma^2} (E(Y|\lambda, \mu, \sigma^2)) \\ &= E_{\lambda|\mu, \sigma^2} (\text{Var}(Y|\lambda)) + \text{Var}_{\lambda|\mu, \sigma^2} (E(Y|\lambda)) \\ &= E_{\lambda|\mu, \sigma^2} (\lambda) + \text{Var}_{\lambda|\mu, \sigma^2} (\lambda) \\ &= E_{\lambda|\mu, \sigma^2} (e^{\log \lambda}) + \text{Var}_{\lambda|\mu, \sigma^2} (e^{\log \lambda}) \\ &= e^{\mu + \sigma^2/2} + (e^{2\mu + 2\sigma^2} - e^{2\mu + \sigma^2}) \\ &= E(Y|\mu, \sigma^2) + e^{2\mu} e^{\sigma^2} (e^{\sigma^2} - 1). \end{aligned} \tag{3.9}$$

From Equations 3.8 and 3.9, σ^2 can well account for the overdispersion or underdispersion of distribution through the sign of $e^{\sigma^2} - 1$.

In the d-dimensional multivariate setting for Poisson-Lognormal distribution denoted by $P\Lambda^d(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, with $\boldsymbol{\mu} = (\mu_1, \dots, \mu_d)'$ and $\boldsymbol{\Sigma} = (\sigma_{ii})_{d \times d}$ (Aitchison and Ho, 1989; Serhiyenko et al., 2018), another advantage of this distribution is that it can recover both positive and negative correlation between two count variables, which presents the bottleneck to a bivariate Poisson distribution. For a multivariate Poisson-Lognormal variable $\mathbf{Y} = (y_1, \dots, y_d)'$, its probability density function could be expressed through the integration of the latent variable $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_d)'$,

$$p(\mathbf{Y}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \int_{\mathcal{R}_+^d} \prod_{i=1}^d f(y_i|\lambda_i) \cdot \mathcal{N}^d(\boldsymbol{\lambda}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) d\boldsymbol{\lambda}. \quad (3.10)$$

The expectation, variance and correlation between each pair of the count variable are found as:

$$\begin{aligned} E(Y_i|\boldsymbol{\mu}, \boldsymbol{\Sigma}) &= \exp(\mu_i + \sigma_{ii}/2) = m_i, \\ Var(Y_i|\boldsymbol{\mu}, \boldsymbol{\Sigma}) &= m_i + m_i^2(\exp(\sigma_{ii}) - 1), \\ Corr(Y_i, Y_j|\boldsymbol{\mu}, \boldsymbol{\Sigma}) &= \frac{\exp(\sigma_{ij}) - 1}{\left[(\exp(\sigma_{ii}) - 1 + m_i^{-1})(\exp(\sigma_{jj}) - 1 + m_j^{-1})\right]^{1/2}}. \end{aligned} \quad (3.11)$$

For the count data, there is a lower bound zero thus leading to the difficulty of recovering a strong negative correlation between two count variables, i.e, when one approaches infinity, the other one will approach zero instead of exploding into negative infinity. However, Poisson-Lognormal distribution still exhibits a wide range of correlation coverage and works impressively well when there exists a positive correlation between two count variables.

Incorporating the latent level correlation structure, represented by $\boldsymbol{\Sigma}$, is inspired by the fact that in real life many other important covariates are not collected or even not observable but have a significant impact on the count variables. Therefore, neglecting the potential correlation structure can gravitate to a biased inference and prediction. With the interrelated multivariate Poisson-Lognormal model for each specific count variable, we can explain the inherent pattern across the different count data. In the noisy high-frequency financial data, the incorporation of latent level correlation helps reduce biased inference.

The applications of multivariate Poisson-Lognormal distribution have been demonstrated in transportation crash counts using MCMC approach for inference (Park and Lord, 2007; Ma et al., 2008) . However, their approach only handles the crash counts of different severity levels but with no temporal random effect. In other words, the size of crash count data is much smaller than the high-frequency count data due to the absence of time index. In volatility analysis, temporal dependence is commonly considered and so does our method. As a consequence, high-frequency count data with a larger size pose an even bigger computational challenge for MCMC approach especially when more parameters and latent effects are considered in the model. We will next

address this problem with Integrated Nested Laplace Approximation (INLA) approach.

3.3.2 INLA implementation

As mentioned in the previous work, despite the benefits of Poisson-Lognormal models, the complexity of such models influences their computational cost, especially when a larger number of parameters need to be incorporated. MCMC method can become very computationally expensive as it computes the joint posterior distribution of model parameters in a high dimension space. However, INLA approach instead targets the individual posterior marginals, which usually are sufficient for statistical inference on the model parameters and latent effect, and there is no need to handle complicated joint posterior distributions. Fortunately, the parameters in the Poisson-Lognormal framework can be regarded as a Gaussian Markov Random Field (GMRF), i.e, a finite-dimensional random vector following a multivariate Gaussian distribution (Rue and Held, 2005). With the conditional independence assumptions, INLA takes advantage of the sparsity of the precision matrix for the GMRF and use efficient Bayesian approximation inference for the parameters. As an illustration, we describe the detailed INLA implementation proposed by Rue et al. (2009) for the (3.6).

Let the vector of latent effect be $\mathbf{x} = (\boldsymbol{\beta}, \gamma_{1,1}, \gamma_{1,2}, \dots, \gamma_{1,T}, \gamma_{2,1}, \gamma_{2,2}, \dots, \gamma_{2,T}, \boldsymbol{\alpha}_{1,1}, \dots, \boldsymbol{\alpha}_{S,T})$ and the vector of hyperparameters be $\boldsymbol{\theta} = (\phi_1, \phi_2, \boldsymbol{\Sigma}_{\boldsymbol{\omega}}, \boldsymbol{\Sigma}_{\boldsymbol{\alpha}})$. Within framework, this latent structure is a GMRF of zero mean and precision matrix determined by $\boldsymbol{\theta}$. Based upon the joint probability density,

$$\begin{aligned} \pi(\mathbf{x}, \mathbf{Y}, \boldsymbol{\theta}) &\propto \pi(\boldsymbol{\theta}) \pi(\mathbf{x}|\boldsymbol{\theta}) \prod_{i \in \mathcal{I}} \pi(Y_i | x_i, \boldsymbol{\theta}) \\ &\propto \pi(\boldsymbol{\theta}) \pi(\boldsymbol{\beta}) \prod_{t=1}^T \left(f(\boldsymbol{\gamma}_t | \boldsymbol{\gamma}_{t-1}, \phi_{\omega_1}, \phi_{\omega_2}, \boldsymbol{\Sigma}_{\boldsymbol{\omega}}) \prod_{s=1}^S \mathcal{N}(\boldsymbol{\alpha}_{st}; \mathbf{0}, \boldsymbol{\Sigma}) \prod_{j=1}^2 \frac{\lambda_{j,st}^{Y_{j,st}} e^{-\lambda_{j,st}}}{Y_{j,st}!} \right), \end{aligned} \quad (3.12)$$

where $\pi(\boldsymbol{\theta})$ is the prior density for the hyperparameters $\boldsymbol{\theta}$, we use the default priors, Gaussian prior for the internal transformed hyperparameter, $\log\left(\frac{1+\phi_j}{1-\phi_j}\right)$, $j = 1, 2$. and Wishart priors for the precision matrices, $\boldsymbol{\Sigma}_{\boldsymbol{\omega}}^{-1}, \boldsymbol{\Sigma}_{\boldsymbol{\alpha}}^{-1}$. $\pi(\boldsymbol{\beta})$ is a Gaussian prior for the intercepts and coefficients $\boldsymbol{\beta}$, $\prod_{t=1}^T f(\boldsymbol{\gamma}_t | \boldsymbol{\gamma}_{t-1}, \phi_1, \phi_2, \boldsymbol{\Sigma}_{\boldsymbol{\omega}})$ is the Gaussian likelihood for the correlated temporal random effects $\gamma_j, j = 1, 2$, and $\mathcal{N}(\cdot)$ is the bivariate Gaussian density.

The posterior marginal density for the hyperparameters $\boldsymbol{\theta}$ can be obtained by the Laplace approximation,

$$\tilde{\pi}(\boldsymbol{\theta}|\mathbf{Y}) \propto \frac{\pi(\mathbf{x}, \mathbf{Y}, \boldsymbol{\theta})}{\tilde{\pi}_G(\mathbf{x}|\boldsymbol{\theta}, \mathbf{Y})} \Big|_{\mathbf{x}=\mathbf{x}^*(\boldsymbol{\theta})}, \quad (3.13)$$

where $\tilde{\pi}_G(\mathbf{x}|\boldsymbol{\theta}, \mathbf{Y})$ is the Laplace approximation of $\pi(\mathbf{x}|\boldsymbol{\theta}, \mathbf{Y})$, $\mathbf{x}^*(\boldsymbol{\theta})$ is the mode of the conditional posterior density $\pi(\mathbf{x}|\boldsymbol{\theta}, \mathbf{Y})$.

Next, using the approximation of $\pi(\boldsymbol{\theta}|\mathbf{Y})$, the discrete integration can provide us with the approximation of the posterior marginal densities of interest $\pi(x_i|\mathbf{Y})$,

$$\tilde{\pi}(x_i|\mathbf{Y}) = \sum_k \tilde{\pi}(x_i|\boldsymbol{\theta}_k, \mathbf{Y}) \tilde{\pi}(\boldsymbol{\theta}_k|\mathbf{Y}) \Delta_k, \quad (3.14)$$

Where $\tilde{\pi}(x_i|\boldsymbol{\theta}_k, \mathbf{Y})$ is the marginal Gaussian density for the approximation of $\pi(x_i|\boldsymbol{\theta}_k, \mathbf{Y})$ derived from $\tilde{\pi}_G(\mathbf{x}|\boldsymbol{\theta}, \mathbf{Y})$, Δ_k is the area weights in the discrete integration.

Since correlated bivariate AR process is not available in the current INLA package, we need to derive the precision matrix from $\prod_{t=1}^T f(\boldsymbol{\gamma}_t|\boldsymbol{\gamma}_{t-1}, \phi_1, \phi_2, \boldsymbol{\Sigma}_\omega)$ is the Gaussian likelihood for the correlated temporal random effects $\gamma_j, j = 1, 2$, and define this specific latent effect using `inla.rgeneric()` function in INLA. For the bivariate AR(1) process according to (3.6),

$$\begin{cases} \gamma_{1,t} = \phi_1 \gamma_{1,t-1} + \omega_{1,t} \\ \gamma_{2,t} = \phi_2 \gamma_{2,t-1} + \omega_{2,t} \end{cases}, -1 < \phi_1, \phi_2 < 1, \boldsymbol{\omega}_t = \begin{pmatrix} \omega_{1,t} \\ \omega_{2,t} \end{pmatrix} \sim N(\mathbf{0}, \boldsymbol{\Sigma}_\omega), \quad (3.15)$$

we have the analytical form of the Gaussian likelihood as,

$$\begin{aligned} \prod_{t=1}^T f(\boldsymbol{\gamma}_t|\boldsymbol{\gamma}_{t-1}, \phi_1, \phi_2, \boldsymbol{\Sigma}_\omega) &= f(\boldsymbol{\gamma}_1|\phi_1, \phi_2, \boldsymbol{\Sigma}_\omega) \prod_{t=2}^T f(\boldsymbol{\gamma}_t|\boldsymbol{\gamma}_{t-1}, \phi_1, \phi_2, \boldsymbol{\Sigma}_\omega) \\ &= \frac{\sqrt{(1-\phi_1^2)(1-\phi_2^2)}}{(2\pi\sqrt{\sigma_{\omega_1}^2\sigma_{\omega_2}^2(1-\rho_\omega^2)})^T} \exp \left\{ -\frac{1}{2(1-\rho_\omega^2)} \left(\frac{(1-\phi_1^2)\gamma_{1,1}^2}{\sigma_{\omega_1}^2} + \frac{(1-\phi_2^2)\gamma_{2,1}^2}{\sigma_{\omega_2}^2} - \frac{2\rho_\omega\gamma_{1,1}\gamma_{2,1}\sqrt{1-\phi_1^2}\sqrt{1-\phi_2^2}}{\sigma_{\omega_1}\sigma_{\omega_2}} \right) \right\} \\ &\cdot \prod_{t=2}^T \exp \left\{ -\frac{1}{2(1-\rho_\omega^2)} \left(\frac{(\gamma_{1,t} - \phi_1\gamma_{1,t-1})^2}{\sigma_{\omega_1}^2} + \frac{(\gamma_{2,t} - \phi_2\gamma_{2,t-1})^2}{\sigma_{\omega_2}^2} - \frac{2\rho_\omega(\gamma_{1,t} - \phi_1\gamma_{1,t-1})(\gamma_{2,t} - \phi_2\gamma_{2,t-1})}{\sigma_{\omega_1}\sigma_{\omega_2}} \right) \right\}, \end{aligned} \quad (3.16)$$

(3.16) can be treated as the Gaussian likelihood for the random vector,

$$\mathbf{V} = \begin{pmatrix} \sqrt{1 - \phi_1^2} \frac{\gamma_{1,1}}{\sigma_{\omega_1}} \\ \sqrt{1 - \phi_2^2} \frac{\gamma_{2,1}}{\sigma_{\omega_2}} \\ \frac{\gamma_{1,2}}{\sigma_{\omega_1}} - \phi_1 \frac{\gamma_{1,1}}{\sigma_{\omega_1}} \\ \frac{\gamma_{2,2}}{\sigma_{\omega_2}} - \phi_2 \frac{\gamma_{2,1}}{\sigma_{\omega_2}} \\ \vdots \\ \frac{\gamma_{1,T}}{\sigma_{\omega_1}} - \phi_1 \frac{\gamma_{1,T-1}}{\sigma_{\omega_1}} \\ \frac{\gamma_{2,T}}{\sigma_{\omega_2}} - \phi_2 \frac{\gamma_{2,T-1}}{\sigma_{\omega_2}} \end{pmatrix}, \quad (3.17)$$

with its precision matrix,

$$\mathbf{\Lambda} = \frac{1}{1 - \rho_\omega^2} \begin{pmatrix} 1 & -\rho_\omega & 0 & 0 & 0 & 0 \\ -\rho_\omega & 1 & 0 & 0 & \dots & 0 & 0 \\ 0 & 0 & 1 & -\rho_\omega & \dots & 0 & 0 \\ 0 & 0 & -\rho_\omega & 1 & & 0 & 0 \\ & \vdots & \vdots & & \ddots & \vdots & \\ 0 & 0 & 0 & 0 & \dots & 1 & -\rho_\omega \\ 0 & 0 & 0 & 0 & & -\rho_\omega & 1 \end{pmatrix}_{2T \times 2T} \quad (3.18)$$

To implement latent effect $\mathbf{V}$ with INLA, we also need to find the linear transformation between $\mathbf{V}$ and its internal representation $\mathbf{U}$,

$$\mathbf{V} = \mathbf{A}\mathbf{U}, \quad (3.19)$$

where

$$\mathbf{U} = \begin{pmatrix} \gamma_{1,1} \\ \gamma_{1,2} \\ \vdots \\ \gamma_{1,T} \\ \gamma_{2,1} \\ \gamma_{2,2} \\ \vdots \\ \gamma_{2,T} \end{pmatrix}, \quad (3.20)$$

$$\mathbf{A} = \begin{pmatrix} \frac{\sqrt{1-\phi_1^2}}{\sigma_{\omega_1}} & 0 & \dots & 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 & \frac{\sqrt{1-\phi_2^2}}{\sigma_{\omega_2}} & 0 & \dots & 0 \\ -\frac{\phi_1}{\sigma_{\omega_1}} & \frac{1}{\sigma_{\omega_1}} & \dots & 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 & -\frac{\phi_1}{\sigma_{\omega_2}} & \frac{1}{\sigma_{\omega_2}} & \dots & 0 \\ & & \vdots & & & & \vdots & \\ 0 & \dots & -\frac{\phi_1}{\sigma_{\omega_1}} & \frac{1}{\sigma_{\omega_1}} & 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 & 0 & \dots & -\frac{\phi_1}{\sigma_{\omega_2}} & \frac{1}{\sigma_{\omega_2}} \end{pmatrix}_{2T \times 2T} \quad (3.21)$$

Using sparse matrix arithmetic in R, we can easily compute the covariance matrix of $\mathbf{U}$ through matrix multiplication,

$$Cov(\mathbf{U}) = \mathbf{A}^{-1}Cov(\mathbf{V})(\mathbf{A}^{-1})^T,$$

where $Cov(\mathbf{V}) = \mathbf{\Lambda}^{-1}$. Then the complete implementation of bivariate AR(1) latent effect will be done through the *rgeneric()* function in the R package, INLA. The choice of the prior distribution can be customized via an appropriate reparameterization approach with its corresponding Jacobian matrix. Bivariate random walk of order 1 can be done similarly except for the distributional assumption on the initial $\boldsymbol{\gamma}_1$. The Gaussian likelihood part of RW(1) is exactly the same as the AR(1)'s, with $\phi_1 = \phi_2 = 1$. Implementation of multivariate AR temporal effect with higher orders in INLA is also feasible, as long as the precision matrix of the latent effect can be analytically specified, meanwhile the dimension of its precision matrix will increase quickly as a product of the latent vector dimension and the length of the time series, with an expected higher computational cost. However, that is beyond the scope of this paper. In the next section, the performance of

parameter recovery, in-sample prediction, and computational cost are demonstrated through an extensive simulation study before one can apply this method practically.

3.4 Numerical study

3.4.1 Simulation study: INLA v.s STAN

Bayesian hierarchical models focus on the inference of sampled parameters from their joint posterior distribution. However, computational efficiency can be of great concern if accurate statistical inference is needed shortly in real applications. In this section, we conduct a comparison simulation study through a simplified framework from (3.6), where no additional covariates are involved in the link function to the conditional Poisson means. The comparison aims at showing the competitive performance of INLA regarding parameter recovery and in-sample prediction at a much lower computational cost versus the traditional MCMC method. In addition, the results also will encourage the practitioners to implement customized latent effects via *rgeneric()* function with a correctly specified precision matrix for accurate parameter inference. We have 200 simulations for each combination of different numbers of assets ($S = 10, 15, 20$) and time intervals ($T = 50, 100, 150$) in three different scenarios. The parameter setup is summarized in Table 6.

Scenario	ϕ_1	ϕ_2	$\sigma_{\omega_1}^{-2}$	$\sigma_{\omega_2}^{-2}$	ρ_{ω}	$\sigma_{\alpha_1}^{-2}$	$\sigma_{\alpha_2}^{-2}$	ρ_{α}
1	0.5	0.8	6	7	0.6	10	10	0.6
2	0.5	0.8	6	7	0.6	10	10	0.2
3	0.5	0.8	6	7	0.6	10	10	-0.6

Table 6: Parameters for different Scenarios

Since the latent level correlation is of more interest to the practitioners for real application, we explore the model inference performance by using different values of $\rho_{\alpha} = 0.6, 0.2$, and -0.6 , which displays the different strengths and directions of the correlation between the level-correlated effects. As for the correlated temporal effect, we set a moderate correlation $\rho_{\omega} = 0.6$ to describe the overall market behavior. Practitioners can always use one identical or two different uncorrelated temporal effects to account for the extremely strong or weak temporal correlations via the default setting in INLA, but the BVAR(1) framework provides more flexibility to implement a wider range of latent temporal correlations. The AR coefficients are set to be $\phi_1 = .5$ and $\phi_2 = .8$ to indicate

a decent temporal dependence for each of the temporal effects. The precision parameters for the latent effects are set to be relatively large so that their variances are all controlled under 1 to avoid unreasonably extreme values generated in the simulation study. The conventional Bayesian inference is implemented in STAN because of its high-quality chain provided by NUTS Hamiltonian Monte Carlo (HMC) sampler, compared with JAGS or NIMBLE (Beraha et al., 2021), whose default samplers are the Gibbs samplers and Metropolis-Hastings sampler. We set the posterior sample size to be 1000 after 1000 burn-ins for the traditional MCMC method. The choice of the priors for the parameters between INLA and STAN are identical,

$$\begin{aligned}\log\left(\frac{1+\phi_i}{1-\phi_i}\right) &\sim N(0, 2), i = 1, 2, \\ \Sigma_{\omega} &\sim W_2(4, \mathbf{I}_{2 \times 2}), \\ \Sigma_{\alpha} &\sim W_2(4, \mathbf{I}_{2 \times 2}),\end{aligned}\tag{3.22}$$

where $\mathbf{I}_{2 \times 2}$ is an identity matrix. Between our framework and the traditional MCMC method, we will evaluate their performances including parameter recovery rate, Mean Square Error (MSE), Mean Absolute Error (MAE), Weighted Mean Absolute Percentage Error (WMAPE), and computational cost. The parameter recovery rate is defined as the frequency of the true parameter being captured in the default 95% credible interval out of these 200 simulations. MSE, MAE, and WMAPE measure the average in-sample deviation between true count $Y_{j,st}$ with predicted count $\hat{Y}_{j,st}$. MSE, MAE, and WMAPE are calculated as,

$$\begin{aligned}\text{MSE} &= \frac{1}{n} \sum_{i=1}^n (y_{\text{pred},i} - y_i)^2, \\ \text{MAE} &= \frac{1}{n} \sum_{i=1}^n |y_{\text{pred},i} - y_i|, \\ \text{WMAPE} &= \frac{\sum_{i=1}^n |y_{\text{pred},i} - y_i|}{\sum_{i=1}^n |y_i|}.\end{aligned}\tag{3.23}$$

The parameter recovery performances among different scenarios are summarized in Table 7 and 8. The parameter recovery rate reflects the credibility of the proposed framework in terms of the statistical inference on the parameters of interest. Regarding the in-sample prediction accuracy and computational cost, the comparison results are in Table 9.

Regarding the parameter recovery performance, our framework shows a competitive performance. First of all, the latent correlation recovery rate is comparable to the STAN's performance. For the temporal correlation, both of these two methods have a recovery rate as high as over 90% for most scenarios. For the latent level correlation, the parameter recovery is better (around 95%) when the true value is near 0 ($\rho_{\alpha} = 0.2$) but is less satisfactory when ρ_{α} is close to 1 (75% to 90% in Scenario 1 and Scenario 3). Since the temporal effects have slightly larger variances ($\sigma_{\alpha_1}^{-2} = 6, \sigma_{\alpha_2}^{-2} = 7$) than the level-correlated effects do ($\sigma_{\omega_1}^{-2} = 10, \sigma_{\omega_2}^{-2} = 10$) in the simulation setup, a larger sample size, i.e., a larger value of T will enhance the recovery of level correlation. Second, the recovery rates for the precision parameters for both latent effects are satisfactory regardless of ρ_{α} . The recovery rates of the AR coefficients are also okay with the INLA but we need to point out that the recovery of ϕ_{ω_2} is not as ideal as the MCMC approach when the latent level correlation ρ_{α} is negative. However, in real applications, a positive correlation between different types of counts is more common as both transactions of different risk levels are usually driven by the trading intensity of the corresponding asset in a positive manner. This scenario is just for the illustration of an overall satisfactory parameter recovery performance of the INLA implementation of the BVAR(1)-LCM framework. In addition, the temporal effects are explained as the overall market behavior and the direct interpretation of AR coefficients in the latent state is of less interest to the practitioners, as long as the AR coefficients have reasonable estimates and the overall model adequacy and prediction are satisfactory. Despite a few exceptions, the INLA method, as an approximate Bayesian inference tool, has a competitive parameter recovery performance in general compared with the full Bayesian inference approach based on the joint posterior distribution.

We've also considered the in-sample performance and computational time between the INLA method and the STAN method. In Table 9, There is no significant difference between the in-sample deviations from the true count data under various measurements. With similar in-sample prediction performance, the INLA method has superior computational efficiency over the STAN method. The current MCMC sampling size is 2000 including the 1000 burn-in samples. The MCMC sampling procedure can still take more than the ten-fold computational time consumed by the INLA method. As the dimension of financial data increases, higher computational costs can be expected by the STAN method. When dealing the financial data containing $S = 20$ assets with each count time series with length $T = 150$, the STAN method takes on average over 2000 seconds to make inferences

based on 1000 posterior samples. As for the irregularity of the computational time of the INLA method, the main reason could be caused by the randomness in the simulation study. In the internal computing procedure, when the generated data deviates too much from the prior distribution, the default initial value will impact the convergence speed of the model, and sometimes a problematic initial value can definitely cause a re-initialization of the computing procedure thus leading to a longer computational time. Even though the computational time is not proportionally related to the data dimension for the INLA method, its computational cost reduction is already significant enough that the STAN method is no match in this perspective.

In this simulation study, we compared the INLA method with the traditional MCMC method for model implementation in three aspects: parameter recovery rate, in-sample prediction accuracy, and computational efficiency. We have shown the INLA method provides satisfactory performance in parameter recovery except for a few cases, with similar in-sample prediction accuracy, and much lower computational cost. In real applications in the HFT market, practitioners need to model financial data on a large scale and make reasonable decisions promptly, so the INLA method can provide a better trade-off between computational efficiency and statistical inference accuracy. As for the evaluation of out-of-sample prediction, we will conduct the study on real data across different models for real applications. We will adopt the INLA method for model implementation due to the size of the real financial data to save computational time.

3.4.2 Real application

Before we illustrate how to apply our model to the high-frequency financial market, we will first give a brief background introduction to the financial market including how high-frequency trading (HFT) firms make profits and some empirical patterns of financial data. Then, we are going to apply our model framework to investigate the market dynamics of different stock sectors. Meanwhile, we will also compare additional existing model candidates for model selection. Next, we will account for the interrelation among the multiple stocks within the same sector, via correlated temporal effect and level-correlated effect, including exogenous covariates of interest. Finally, we will provide market behavior interpretations using our model and some perspectives on what benefits our model can bring to practitioners.

Scenario	S	T	$\sigma_{\omega_1}^{-2}$		$\sigma_{\omega_2}^{-2}$		ρ_{ω}		ϕ_{ω_1}		ϕ_{ω_2}	
			INLA	STAN	INLA	STAN	INLA	STAN	INLA	STAN	INLA	STAN
$1(\rho_{\alpha} = .6)$	10	50	0.78	0.93	0.8	0.93	0.92	0.91	0.82	0.94	0.88	0.98
		100	0.89	0.96	0.92	0.95	0.94	0.92	0.94	0.94	0.86	0.94
		150	0.89	0.96	0.89	0.92	0.98	0.95	0.94	0.96	0.9	0.96
	15	50	0.85	0.95	0.76	0.93	0.94	0.93	0.86	0.96	0.78	0.95
		100	0.89	0.94	0.87	0.94	0.95	0.93	0.94	0.96	0.89	0.94
		150	0.91	0.96	0.9	0.94	0.95	0.94	0.92	0.95	0.92	0.95
	20	50	0.8	0.94	0.82	0.92	0.94	0.92	0.84	0.95	0.82	0.98
		100	0.87	0.94	0.89	0.94	0.95	0.92	0.96	0.98	0.89	0.95
		150	0.92	0.96	0.9	0.92	0.96	0.94	0.92	0.93	0.89	0.96
$2(\rho_{\alpha} = .2)$	10	50	0.84	0.96	0.83	0.96	0.9	0.86	0.83	0.95	0.84	0.92
		100	0.9	0.92	0.86	0.92	0.96	0.94	0.94	0.94	0.84	0.94
		150	0.94	0.96	0.95	0.96	0.92	0.92	0.94	0.96	0.88	0.95
	15	50	0.86	0.96	0.77	0.94	0.92	0.9	0.8	0.94	0.82	0.96
		100	0.86	0.95	0.88	0.94	0.94	0.94	0.94	0.98	0.86	0.96
		150	0.91	0.94	0.89	0.92	0.91	0.92	0.94	0.94	0.86	0.94
	20	50	0.81	0.96	0.86	0.97	0.94	0.9	0.88	0.94	0.84	0.94
		100	0.9	0.96	0.9	0.92	0.96	0.93	0.92	0.96	0.84	0.96
		150	0.9	0.95	0.9	0.93	0.94	0.92	0.92	0.94	0.77	0.92
$3(\rho_{\alpha} = -.6)$	10	50	0.83	0.97	0.84	0.94	0.85	0.86	0.85	0.97	0.9	0.97
		100	0.88	0.94	0.88	0.9	0.91	0.88	0.94	0.95	0.88	0.95
		150	0.92	0.94	0.9	0.92	0.88	0.85	0.94	0.94	0.86	0.96
	15	50	0.83	0.95	0.84	0.92	0.88	0.84	0.88	0.93	0.85	0.95
		100	0.9	0.94	0.92	0.94	0.95	0.93	0.92	0.96	0.82	0.94
		150	0.94	0.96	0.86	0.92	0.92	0.91	0.9	0.94	0.78	0.94
	20	50	0.81	0.94	0.79	0.94	0.94	0.92	0.84	0.98	0.76	0.96
		100	0.88	0.94	0.88	0.97	0.92	0.9	0.92	0.97	0.69	0.94
		150	0.9	0.96	0.88	0.93	0.92	0.9	0.94	0.95	0.62	0.92

Table 7: Parameter recovery rate comparison between INLA and STAN for correlated temporal effects ω_1 and ω_2

Scenario	S	T	$\sigma_{\alpha_1}^{-2}$		$\sigma_{\alpha_2}^{-2}$		ρ_{α}	
			INLA	STAN	INLA	STAN	INLA	STAN
$1(\rho_{\alpha} = .6)$	10	50	0.9	0.9	0.96	0.94	0.74	0.72
		100	0.93	0.92	0.96	0.94	0.84	0.82
		150	0.92	0.91	0.96	0.96	0.84	0.84
	15	50	0.92	0.92	0.92	0.9	0.83	0.8
		100	0.95	0.92	0.92	0.9	0.84	0.84
		150	0.96	0.94	0.97	0.96	0.87	0.84
	20	50	0.92	0.93	0.94	0.9	0.81	0.8
		100	0.94	0.93	0.94	0.93	0.87	0.89
		150	0.92	0.92	0.95	0.95	0.88	0.86
$2(\rho_{\alpha} = .2)$	10	50	0.96	0.95	0.98	0.96	0.95	0.94
		100	0.95	0.94	0.96	0.94	0.93	0.94
		150	0.96	0.96	0.96	0.94	0.98	0.96
	15	50	0.95	0.97	0.94	0.96	0.92	0.95
		100	0.97	0.95	0.94	0.92	0.95	0.96
		150	0.94	0.96	0.96	0.96	0.94	0.95
	20	50	0.96	0.94	0.94	0.96	0.93	0.93
		100	0.96	0.96	0.92	0.95	0.97	0.96
		150	0.94	0.96	0.94	0.94	0.96	0.95
$3(\rho_{\alpha} = -.6)$	10	50	0.95	0.92	0.93	0.92	0.66	0.61
		100	0.94	0.94	0.96	0.96	0.79	0.75
		150	0.94	0.92	0.94	0.94	0.84	0.83
	15	50	0.92	0.92	0.91	0.9	0.78	0.74
		100	0.95	0.96	0.93	0.92	0.82	0.8
		150	0.94	0.94	0.93	0.91	0.9	0.84
	20	50	0.92	0.92	0.94	0.94	0.81	0.8
		100	0.93	0.92	0.96	0.96	0.86	0.83
		150	0.96	0.94	0.96	0.96	0.9	0.86

Table 8: Parameter recovery rate comparison between INLA and STAN for level-correlated effects α_1 and α_2

Scenario	S	T	MAE		MSE		WMAPE		CPU.used	
			INLA	STAN	INLA	STAN	INLA	STAN	INLA	STAN
$1(\rho_{\alpha} = .6)$	10	50	1.524	1.533	3.691	3.772	0.168	0.169	5.849	146.802
		100	1.573	1.576	3.947	4.005	0.173	0.175	19.458	402.746
		150	1.595	1.593	4.026	4.084	0.176	0.176	9.016	603.799
	15	50	1.558	1.569	3.876	3.979	0.176	0.178	6.742	249.312
		100	1.613	1.609	4.126	4.177	0.179	0.178	6.323	444.649
		150	1.63	1.626	4.217	4.268	0.18	0.18	11.162	1147.675
	20	50	1.594	1.599	4.062	4.121	0.177	0.177	16.811	358.517
		100	1.633	1.632	4.238	4.308	0.181	0.181	7.366	676.496
		150	1.646	1.642	4.316	4.364	0.182	0.182	13.754	1868.829
$2(\rho_{\alpha} = .2)$	10	50	0.98	0.985	1.444	1.504	0.094	0.094	11.92	232.927
		100	0.982	0.985	1.446	1.501	0.095	0.095	27.974	474.528
		150	1.008	0.989	1.508	1.511	0.095	0.093	21.403	1407.225
	15	50	0.984	0.994	1.457	1.551	0.094	0.096	9.118	260.872
		100	1.009	0.991	1.512	1.519	0.097	0.095	9.571	488.026
		150	1.01	0.992	1.515	1.521	0.096	0.095	16.254	1147.024
	20	50	0.988	1.014	1.466	1.631	0.095	0.097	28.657	507.624
		100	1.013	1.002	1.525	1.568	0.097	0.096	14.959	1093.478
		150	1.011	0.993	1.518	1.523	0.097	0.095	21.812	2124.168
$3(\rho_{\alpha} = -.6)$	10	50	1.476	1.483	3.586	3.651	0.124	0.125	6.962	319.741
		100	1.537	1.544	3.927	3.997	0.127	0.127	31.864	586.252
		150	1.585	1.583	4.156	4.213	0.124	0.124	10.757	1454.869
	15	50	1.55	1.651	3.977	19.226	0.126	0.133	13.785	1134.083
		100	1.601	1.603	4.257	4.345	0.127	0.127	9.615	1662.712
		150	1.625	1.624	4.399	4.457	0.129	0.129	11.927	2428.919
	20	50	1.572	1.577	4.119	4.173	0.132	0.132	26.201	1399.757
		100	1.639	1.639	4.49	4.554	0.129	0.129	15.304	2331.487
		150	1.66	1.659	4.616	4.673	0.128	0.128	14.68	4687.846

Table 9: In-sample prediction and computational time comparison between INLA and STAN

3.4.2.1 High-frequency trading (HFT) background According to Carrion (2013) and Dutta et al. (2022), one source of profit-making in HFT is related to intraday market time skills, buying when prices are temporarily low and selling when prices are temporarily high. This is not solely driven by very short-term signals or trading at fleeting prices, but by the existence of economically significant predictability in intraday prices. HFTs execute their trades at better prices than non-HFTs, have lower processing costs, and have some ability to avoid adverse selection costs on larger trades when supplying liquidity. To cover these fixed processing costs, the HFT firms earn very small profits per trade but place large volumes of orders, making profits based upon small amounts of predictive power on large trading volumes.

From the empirical data analysis of the raw transaction-level data, the price fluctuation regarding log price return within short time intervals only has a small range, and there are numerous transactions made with zero returns. Therefore, the practitioners in the market need to acknowledge that making a profit through a significantly high price return in HFT is not supported by the intrinsic market dynamics, and they can strategically place orders according to the numbers of transactions with zero return and non-zero return in a short term. For example, if the number of non-zero returns is large, it implies the asset price is volatile and short-term profit can be expected by supplying liquidity according to the price movement prediction with other models. Otherwise, buy-in opportunities can be considered at the occurrence of an abundance of zero returns since the price is temporarily stable.

In addition, HFT improves liquidity (Hendershott and Riordan, 2013), reduces volatility (Hasbrouck and Saar, 2013), and yields a different market microstructure than the conventional financial market without HFT (Ammar et al., 2020). There has been some research work conducted to account for the association between volatility and liquidity in the conventional financial market (Deuskar, 2006; Bedowska-Sójka and Kliber, 2019). Although there is only a general confirmation that liquidity and volatility are strongly associated, we can as well explore their relationship in the HFT setting with count data. Since there are various types of measurements to quantify liquidity and volatility, it is explicable that their relationship can take many forms. O’hara (1998) points out that liquidity is generally defined as the ability to trade large volumes quickly at low cost, and usually such trading doesn’t cause a drastic impact on price change. We can use the total number of transactions within a fixed short time interval as a surrogate measurement of volatility for count

data because the trading volumes among these short intervals are generally large and don't have a significant difference, the total trading volume can be roughly regarded as a multiple of the number of transactions. From this perspective, our method can account for this short-term association between liquidity and volatility in a time-dependent manner.

3.4.2.2 High-frequency count data description The transaction-level data for our analysis are retrieved from the Trade and Quote (TAQ) database from Wharton Research Data Services (WRDS). The TAQ database contains intraday transactions (trades and quotes) data for all securities listed on the New York Stock Exchange (NYSE) and American Stock Exchange (AMEX), as well as Nasdaq National Market System (NMS) and Small-Cap issues. Among all the HFT data, we focus only on the trading data from three GICS sectors in January 2013, Healthcare, Energy, and Industrials.

We first pre-process the raw HFT data. Let $t = 1, \dots, T$ denote the order of the fixed time intervals (e.g., 2-minute intervals in our study) within a trading day, let $i_t = 1, \dots, N_t$ be the order of the transactions within the time interval t due to the idiosyncrasy of high-frequency financial data and the fact that the number of transactions in a given interval is random. Let T_{i_t} , P_{i_t} , and R_{i_t} be the corresponding transaction time, price, and log returns at the i_t th transaction, with

$$R_{i_t} = \frac{P_{i_t} - P_{(i-1)_t}}{P_{(i-1)_t}}.$$

We construct two new variables, $Y_{H,t}$ and $Y_{L,t}$, i.e., transaction count of high-risk level and count of low-risk level respectively. For a user-defined threshold $\tau > 0$, let the incidence and count of the log return being greater than τ in the time interval t be

$$B_{i_t} = \begin{cases} 1 & \text{if } |R_{i_t}| > \tau, \\ 0 & \text{if } |R_{i_t}| \leq \tau, \end{cases} \quad \text{then} \quad (3.24)$$

$$\begin{cases} Y_{H,t} &= \sum_{i_t=1}^{N_t} B_{i_t} \\ Y_{L,t} &= N_t - Y_{H,t} \end{cases} \quad (3.25)$$

Choice of τ : Due to limited literature resources on HFT data, the threshold selection for

defining high-risk level counts is data-driven or of practitioners' interest. The larger the threshold τ , the sparser the count of high-risk level will be, and dealing with inflated zeros will require specific caution. In the following simple example of the stock ABT on 01/02/2013, in Table 10, $\tau = 0.0005$, or five basis points. The definition of extremeness can be determined by practitioners, for example, Brogaard et al. (2018) defined the extreme price movement in HFT using 99.9% quantile of the return distribution. Baron et al. (2019) selected 0.1% quantile of the distribution of response time (time-stamp difference) to define Decision-Latency to capture the fastest reaction time. Due to limited literature on the distribution of high-frequency price log returns, we are going to select $\tau = 0$ and categorize the transactions with non-zero log returns into the high-risk group and the ones with zero log returns into the low-risk group. Such a choice of $\tau = 0$ has a natural interpretation of the price volatility and can prevent imbalanced count data with excessive zero counts on high-risk levels if the threshold is not selected appropriately. Since the number of transactions with zero log returns takes a significant proportion under the high-frequency circumstance, the real applications in this paper will be based on $\tau = 0$ for demonstration. However, customization of τ for different assets is feasible as long as extra financial domain knowledge and the data empirical characteristics justify the threshold specification.

i	Time (T_i)	Price (P_i)	Log returns (R_i)
1	9:30:00.531	32.3300	NaN
2	9:30:03.167	32.3000	-0.000929
3	9:30:08.307	32.3500	0.000463
4	9:30:08.311	32.3250	0.000308
5	9:30:08.313	32.3525	-0.000001
...	...	...	...
652	9:31:58.736	32.2300	0.000310
653	9:31:58.886	32.2300	0.000000
654	9:31:59.287	32.2300	0.000000
655	9:31:59.611	32.2200	-0.000310
656	9:31:59.909	32.2300	0.000310

Table 10: An example of raw data for the stock ABT

Let $\mathbf{Y}_{st} = (Y_{H,st}, Y_{L,st})'$ be a 2-dimensional vector of count responses of high-risk transactions and low-risk ones for the s^{th} stock at equally spaced times t , for $s = 1, \dots, S$ and $t = 1, \dots, T$ ($T = 195$). The data includes $D = 21$ trading days in January 2013. For each trading day, there are 195 time intervals (2 minutes). Take the stock ABT on 01/02/2013 in the time window from

Stock	Count	Type	Time	Duration mean (in seconds)	logsize mean
ABT	413	High risk	9:30	0.283	5.241
ABT	306	High risk	9:32	0.392	5.168
ABT	269	High risk	9:34	0.446	5.194
ABT	244	Low risk	9:30	0.452	5.196
ABT	218	Low risk	9:32	0.553	5.189
ABT	234	Low risk	9:34	0.513	5.061

Table 11: Data structure for INLA modeling. count data for stock ABT between 9:30 and 9:34 a.m. on 01/02/2013.

9:30 to 9:34 for example as in Table 11:

In Table 11, the last two columns are the two covariates of our interest. Duration mean is the average duration between two consecutive transactions of the same type. The i -th financial duration, x_i , within an interval is defined as,

$$x_i = t_i - t_{i-1},$$

where t_i is the time point when the i -th financial event occurs. In our study, financial events are defined as transactions of different risk levels. Within a time interval, a large collection of successive short durations usually reflect a high trading intensity in the market thus leading to a large count of transactions. It would be useful to investigate and quantify the association between count and average duration. In addition, a common delineation for the intra-day periodic trading patterns is the phenomenon of high trading intensity (shorter durations) during the opening and closing periods of a trading day and relatively lower trading activity around noon. In Figure 7, the diurnal effect can be observed as a U-shape pattern in the intra-day count data and an upside-down U-shape pattern in the average duration data. Therefore, including average duration as a covariate in our model facilitates the accommodation for diurnal effect explanation. Another covariate of our interest is related to trade size. Due to the wide range of trade sizes for the transactions, we choose a logarithmic transformation on the trade size and calculate the average log trade size for the time intervals. Since price changes fluctuate significantly when a large number of shares are traded, it is also of our interest to account for this relationship between count and log trade size.

Similar stocks being in each sector, the co-movement of these time series is related to market

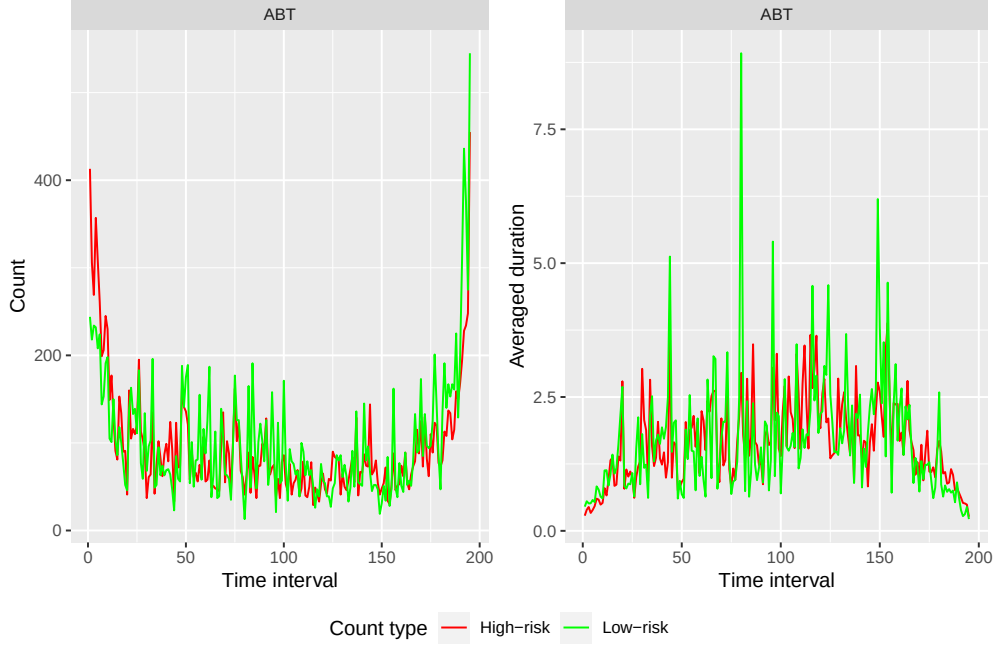


Figure 7: Diurnal pattern exhibited in observed counts and averaged duration in 2-min intervals for stock ABT on 01/02/2023.

behavior evaluation for the corresponding industry, the multi-asset pattern is shown in Figure 8. Before further investigation of the market micro-structure, we will implement different models on the real data, and evaluate model adequacy and out-of-sample prediction in the next step.

3.4.2.3 Model adequacy and prediction accuracy comparisons In addition to our proposed framework, we include a list of other model candidates to be considered. To begin with, the correlated bi-variate AR process with level correlation model (BVAR(1)-LCM) from (3.6) is one of the candidates. However, due to the non-stationary feature of count data, it is not uncommon to use a random walk (RW) process instead of an AR process to model the long memory temporal dependence. Therefore, the correlated bi-variate RW process with a level correlation model (BVRW(1)-LCM) with fewer parameters becomes another one of the model candidates. We also consider AR and RW processes with uncorrelated latent temporal effect with $\rho_{\omega} = 0$ as a special case. In addition, we include the last candidate, proposed by Ma et al. (2008), which can be regarded as an existing method using only the level correlation model (LCM) using the MCMC method, as a static LCM model without temporal latent effect $\gamma_{j,t}$ in (3.6). As for the fixed effects, we incorporate different fixed effects for the conditional mean of the count at different risk

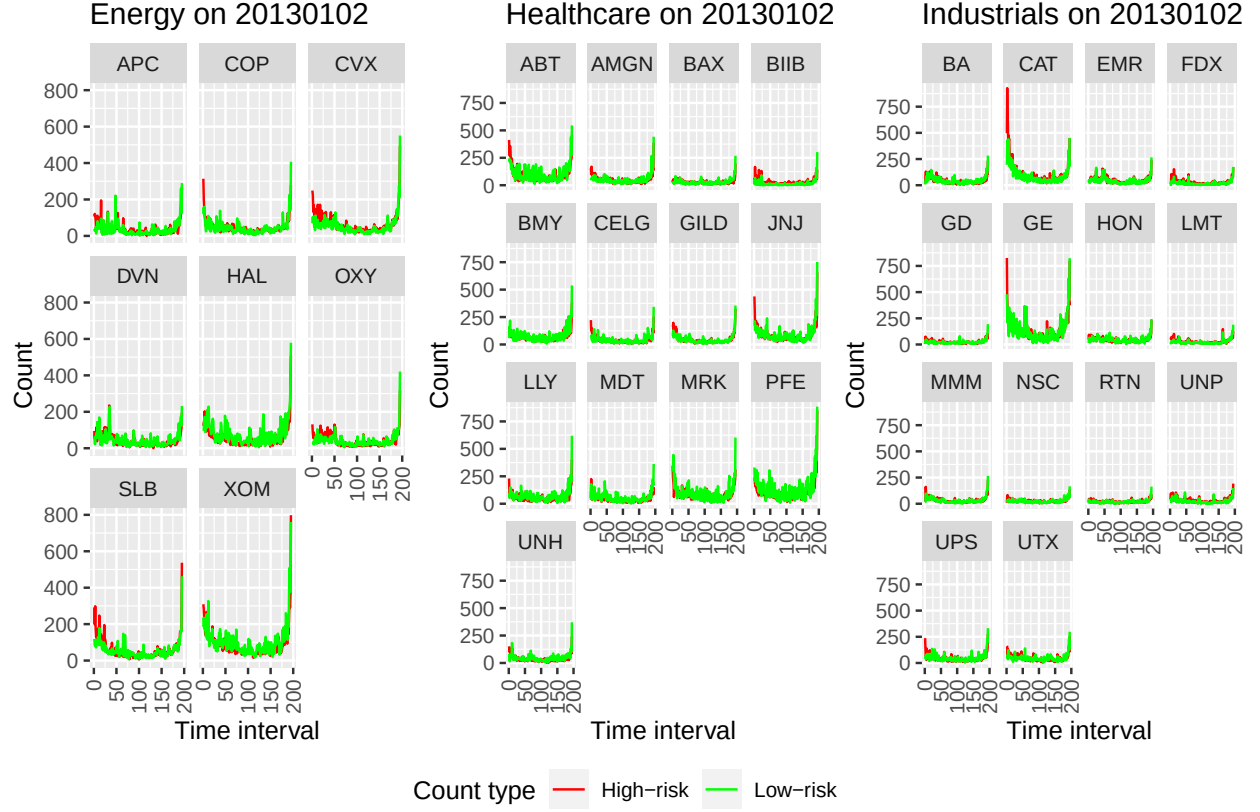


Figure 8: Count data for three GICS sectors: Energy, Health care, Industrials

levels. These model candidates will be compared from two aspects: in-sample model adequacy and out-of-sample prediction accuracy. To compare the in-sample model adequacy, we use the Watanabe-Akaike Information Criterion (WAIC) (Watanabe, 2010; Gelman et al., 2014) and Deviance Information Criterion (DIC). WAIC and DIC are predictive information criteria for Bayesian models, smaller values indicate a better model. WAIC (Watanabe-Akaike information criterion) is computed as,

$$\text{WAIC} = -2 \log p_{\text{post}}(y) + 2p_{\text{WAIC}}, \quad (3.26)$$

where p_{WAIC} is the correction for the effective number of parameters to adjust for overfitting

and there are two approaches available for the correction.

$$\begin{aligned} p_{\text{WAIC1}} &= 2 \sum_{i=1}^n (\log(E_{\text{post}} p(y_i|\theta)) - E_{\text{post}}(\log p(y_i|\theta))), \\ p_{\text{WAIC2}} &= \sum_{i=1}^n \text{var}_{\text{post}}(\log p(y_i|\theta)). \end{aligned} \quad (3.27)$$

DIC is computed as,

$$\text{DIC} = -2 \log p(y|\hat{\theta}_{\text{Bayes}}) + 2p_{\text{DIC}}, \quad (3.28)$$

where $\hat{\theta}_{\text{Bayes}}$ is the posterior mean of the parameter and p_{DIC} is the correction for the effective number of parameters,

$$p_{\text{DIC}} = 2(\log p(y|\hat{\theta}_{\text{Bayes}}) - E_{\text{post}}(\log p(y|\theta))). \quad (3.29)$$

For out-of-sample prediction, we use the first 180 observations as a training set to predict the last 15 observations, i.e., the count data in the last 30 minutes of a trading day. A smaller value of MAE, MSE, and WMAPE regarding the prediction and actual count indicates a better out-of-sample prediction performance.

There are three sectors of count data in our analysis, and each sector contains 21 days trading days. The total number of data sets will be $3 \times 21 = 63$. Applying the described criteria above, we select the best model under each measurement. Table 12 shows the model comparison results. The percentage in the table is the proportion out of the 63 data sets that favor the specific measurement regarding the model adequacy or out-of-sample performance accuracy. Around 80% of the data sets favor the BVAR(1)-LCM or BVRW(1)-LCM model regarding model adequacy and more than 90% regarding the out-of-sample prediction performance. Even though the other candidates perform better either in terms of model adequacy or prediction accuracy, our proposed model, especially the mode with correlated bivariate RW process with level-correlated (BVRW(1)-LCM) is favored by the majority of the data sets in both aspects. Next, we are going to use the BVRW(1)-LCM model to study the microstructure of the count data with long memory dependence.

Metric	BVAR(1)-LCM	BVRW(1)-LCM	AR(1)-LCM	RW(1)-LCM	Static LCM
WAIC	9.5%	69.8%	1.6%	19.1%	0.0%
DIC	19.0%	63.5%	1.6%	15.9%	0.0%
MAE	4.8%	88.8%	0.0%	4.8%	1.6%
MSE	4.8%	90.4%	0.0%	3.2%	1.6%
WMAPE	4.8%	88.8%	0.0%	4.8%	1.6%

Table 12: Percentage of the 63 data sets favoring each model regarding in-sample model adequacy and out-of-sample prediction accuracy

3.4.2.4 An illustration of BVRW(1)-LCM framework From the previous result of the model comparison, we will analyze the count data using the following model,

$$\begin{aligned}
Y_{j,st} | \lambda_{j,st} &\overset{ind}{\sim} Pois(\lambda_{j,st}), \\
\eta_{j,st} = \log \lambda_{j,st} &= \beta_{0,j} + \beta_{dur,j} dur_{j,st} + \beta_{size,j} size_{j,st} + \gamma_{j,t} + \alpha_{j,st}, \\
\gamma_{j,t} &= \phi_j \gamma_{j,t-1} + \omega_{j,t}, \\
\omega_t &= \begin{pmatrix} \omega_{1,t} \\ \omega_{2,t} \end{pmatrix} \sim N(\mathbf{0}, \Sigma_\omega), \\
\Sigma_\omega &= \begin{pmatrix} \sigma_{\omega_1}^2 & \sigma_{\omega_1} \sigma_{\omega_2} \\ \sigma_{\omega_1} \sigma_{\omega_2} & \sigma_{\omega_2}^2 \end{pmatrix}, \\
\alpha_{st} &= \begin{pmatrix} \alpha_{1,st} \\ \alpha_{2,st} \end{pmatrix} \sim N(\mathbf{0}, \Sigma_\alpha),
\end{aligned} \tag{3.30}$$

where $dur_{j,st}$ is the averaged durations between the transactions associated with risk levels of the j -th type for the s -th asset in the t -th time interval, and $size_{j,st}$ is the averaged logarithmic trading size (number of traded shares) in transactions associated with risk levels of the j -th type for the s -th asset in the t -th time interval. We apply the priors in (3.22) for the hyperparameters and weakly informative normal priors for the fixed effects. The statistical inference is summarized in terms of fixed effects and hyperparameters. We've also investigated the conditional posterior correlation between the count data by integrating the inference on the fixed effects and random effects. Finally, we provide some remarks for our model application.

Fixed effects of averaged durations and logarithmic trading size The fixed effects from

the averaged durations can provide some insights for the practitioners from four aspects in Figure 9. Firstly, We've found that averaged durations and different count data are negatively correlated because their coefficients are estimated to be below zero. The overall negative association reflects how the trading intensity influences the number of transactions. When the averaged durations are observed to be short, it means transactions are made frequently in the market, causing larger counts of transactions at different risk levels. Secondly, the averaged durations have a stronger impact on the high-risk counts across all sectors except for the Energy sector on Monday because the magnitude of the estimated coefficients for the high-risk counts is larger than the one for the low-risk counts. Since the low-risk counts tend to be driven by the liquidity feature of the corresponding asset, it is less impacted by the averaged durations. Therefore, the increased trading intensity will provide more high-risk counts than low-risk counts, and practitioners can be aware of the short-term proportion of transactions at both levels and evaluate the volatility of the market. Thirdly, the Energy and Healthcare sectors with a larger magnitude of the fixed effect are more sensitive to trading intensity than the Industrials sector with respect to the counts. The Healthcare sector has more consistent sensitivity to the averaged durations because the estimated fixed effect has a narrower range, while the counts from the other two sectors can react to the averaged durations to different degrees throughout the month. Such differences in price fluctuation related to trading intensity help practitioners design their option trading strategies for different sectors because option pricing depends heavily on how drastically the asset price fluctuates. Fourthly, in terms of the Day of Week effect, the counts tend to be impacted more from Tuesday to Thursday with a larger magnitude of the fixed effect from averaged duration, than Monday and Friday.

In addition to averaged durations, we will also describe and explain the impact of the averaged log trading size on the counts with several aspects in Figure 10. To start with, a higher averaged trading size is associated with a higher number of counts, since the fixed effects are significantly larger than zero. The average trading size is the average number of traded shares per transaction which is positively associated with the trading volume, which is the product of the number of transactions and their corresponding trading size. Hence, such a relationship infers that the asset with a higher trading volume tends to have a larger count of transactions. Secondly, we've also noticed that the averaged trading size within a time interval has a greater impact on the low-risk counts, which means orders put to trade more shares don't necessarily correlate with the high-risk

counts as strongly as with the low-risk counts. It can be related to the behavior of the market makers, quoting both a buy and sell price of the asset in the inventory, with the purpose of making a profit on a tiny margin of bid-ask spread via a high trading volume. Nevertheless, the positive impact of trading size on the high-risk counts still aligns with the common sense that the price change tends to be affected to a certain degree by the trading volume. Thirdly, among the three sectors, the counts in the Industrials sector are more sensitive to the trading volume and such sensitivity is more consistent over the month than the other two sectors. Practitioners can expect more volatile behavior in the Industrial sector than in the other two sectors. Lastly, there is no clear pattern regarding the Day of Week effect but we've seen that the high-risk counts on Fridays are impacted more by the trading volume on Fridays, especially in the Healthcare sector and the Industrials sector.

Since the fixed effects are directly related to the conditional mean of count data, such influences on the count data can be quantified by its coefficients. Holding other effects unchanged in the t -th time interval for the s -th asset, the difference of the $\beta_{dur,1}$ and $\beta_{dur,2}$ provides the estimation of ratios of conditional means of different types of count as $\exp(\beta_{dur,1}dur_{1,st} - \beta_{dur,2}dur_{2,st})$. The fixed effect of the averaged log trading size can be applied in this manner as well. With a better understanding of the ratio between different counts within a given interval, practitioners may evaluate their probability of making a profit regarding the dynamic volatility and customize their trading strategy in different time intervals. As for the prediction on the count data with averaged durations and log trading sizes, one can either use lagged observations in the past with the stationarity assumption or use non-negative time series models, such as log ACD models, to make predictions on future observations in the upcoming time intervals.

Hyperparameters: latent correlation and variance Since the hyperparameters describe the latent patterns of the random effect, they don't contribute to the direct estimation of the conditional mean of the count but they explain the uncertainty and association of the conditional mean functions. The hyperparameters we are going to study are the correlation between the latent temporal effects, ρ_{ω} , and the correlation between the level random effects, ρ_{α} , as well as the precision parameters for the random effects.

According to our model results, the latent temporal effects are highly correlated in this month. In Figure 11, the correlation coefficients present a strong relationship ($\hat{\rho}_{\omega} > .75$) between the latent

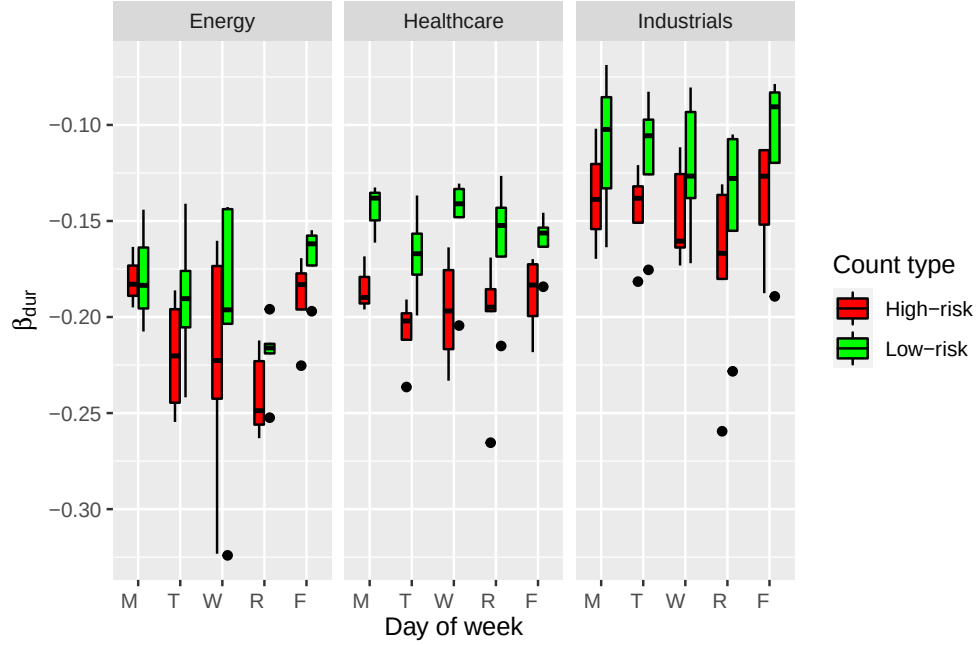


Figure 9: The association between averaged durations and counts

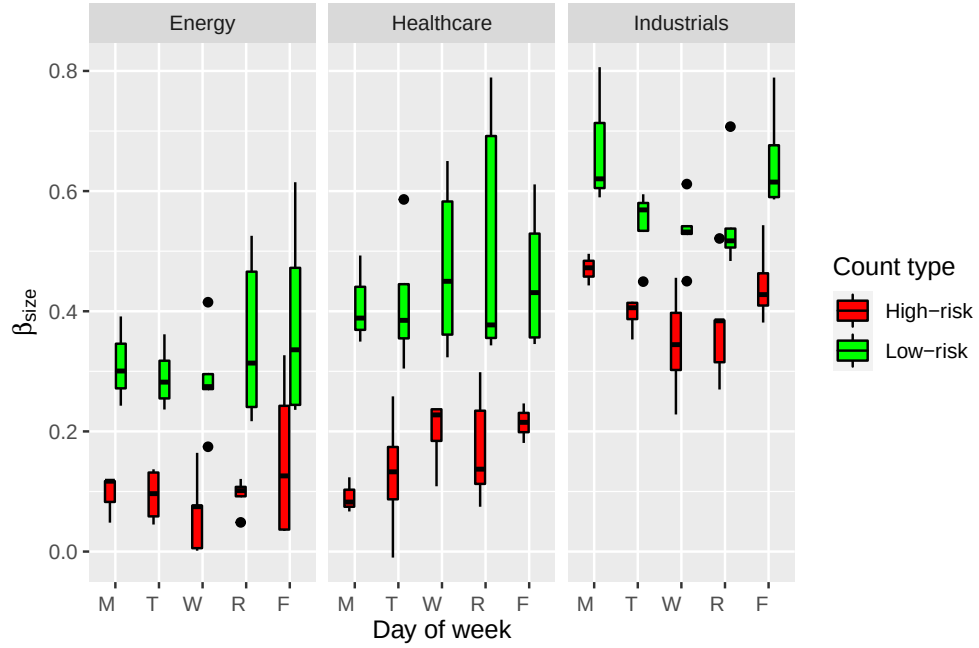


Figure 10: The association between averaged log trading size and counts

temporal effects for counts with different types. In the Energy sector, the correlations could be estimated at over .9 on some days such as 01/04/2013 (Day 3). A higher correlation ρ_{ω} indicates that the count data share some similarities regarding the market behavior. Therefore, the assumption

of uncorrelated latent temporal effects between count data can cause biased model inference and introduce more uncertainty to model prediction at the significant existence of such correlation.

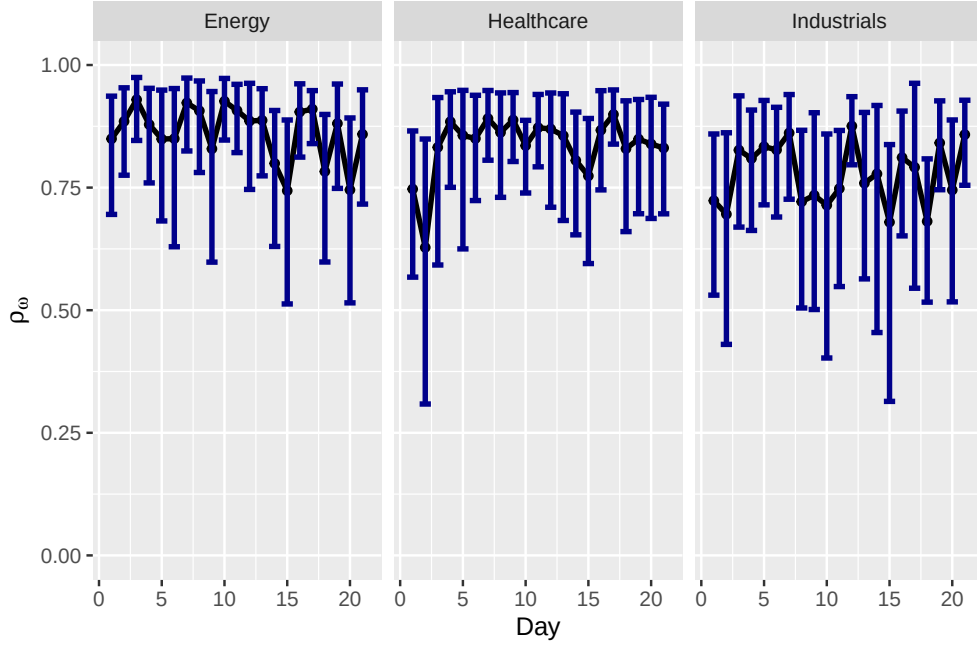


Figure 11: The estimated ρ_{ω} with their 95% credible interval across all three sectors in January 2013

The variances of the latent temporal effects have small estimated values in Figure 12. There are two possible reasons for the small variance estimation. First, due to the long memory property of an RW process, the cumulative variance of the marginal temporal effects throughout a trading day is increasingly significant as $Var(\gamma_{i,t}) = t^2 \sigma_{\omega_i}^2$ is dependent on the time index t . A large estimation of the variance is more likely to cause an inconsistent spread for the true time series. The second reason can be associated with the covariates included in the model. We have shown that the averaged durations and log trading size have significant fixed effects on the count data and a large proportion of data variability can be accounted for by the covariates, leaving less variation to be captured by the latent temporal effects. Even though the covariates are treated as the attributes for a specific type of count for a given asset in a time interval, the existent temporal dependence among the covariates also contributes to the overall temporal pattern, especially for the averaged durations. Between the two types of counts, the low-risk counts have more uncertain latent temporal effects than the ones at high-risk levels. Holding the covariates and the level-correlated random effect unchanged, the change in the ratios of the conditional mean of the counts at t -th interval can be

determined by $\exp(\gamma_{1,t} - \gamma_{2,t})$ with $Var(\gamma_{1,t} - \gamma_{2,t}) = t^2(\sigma_{\omega_1}^2 + \sigma_{\omega_2}^2 - \rho_{\omega}\sigma_{\omega_1}\sigma_{\omega_2})$. Even with the small estimation of variance on the temporal effects across all three sectors, the market behaviors for the Healthcare and Industrials are more stable than the Energy sector as the temporal effects vary less on the high-risk counts than on the low-risk counts, which implies the big pharmaceutical companies in the Healthcare sector and giant conglomerate corporations in the Industrials sector are less volatile in a sense of the common environment in January 2013. On the other hand, the Energy sector is considered more volatile regarding its market behavior, which could be related to the oscillation in the crude oil price.

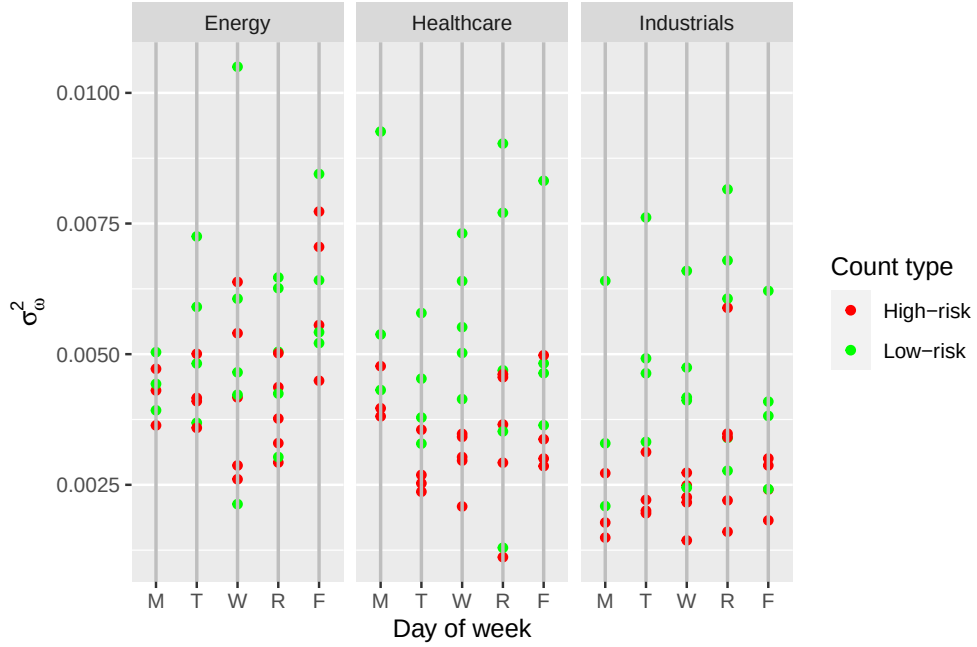


Figure 12: The dot plot of estimated variances, $\sigma_{\omega_1}^2$ and $\sigma_{\omega_2}^2$, of the latent temporal effects on different days of a week across all three sectors in January 2013

The number of covariates plays an important role in capturing the association between the response and explanatory variables. However, a small number of covariates are likely to explain a limited amount of data variation while a large number of covariates tend to cause an overfitting issue. Therefore, practitioners can select a moderate number of covariates of interest to be included in the model, and incorporate level-correlated random effect for the correlated count data to supplement the interpretation of the additional data variation that is not explained by the fixed effects. Such random effects can be regarded to come from unobservable financial factors or covariates that practitioners haven't taken into account. In Figure 13, the correlations between latent level effects

remain at a high level throughout the entire month and such strong association helps account for the correlation among the two different types of counts as well. The variance estimation for the level effects is shown in Figure 14. Similar to the pattern for the temporal effect, the latent level effects on the low-risk counts have a larger variance, leading to higher variability in the corresponding counts. Even though the market behavior mentioned above describes the Industrials sector as stabler than the Energy sector, a higher variance of the high-risk latent level effects indicates more volatility in the Industrials sector than in the Energy sector, which means the unobservable factors such as financial news and regulatory policies are idiosyncratic in different sectors. In Figure 15, the variances in the raw counts are shown to be in a similar range in all three sectors, which means that the variance comparison on the observational level does not provide distinctive guidance on the customized investing strategy for different sectors. Using our framework, practitioners will be able to uncover more latent patterns among different sectors even though the inference on the observational level doesn't differ significantly.

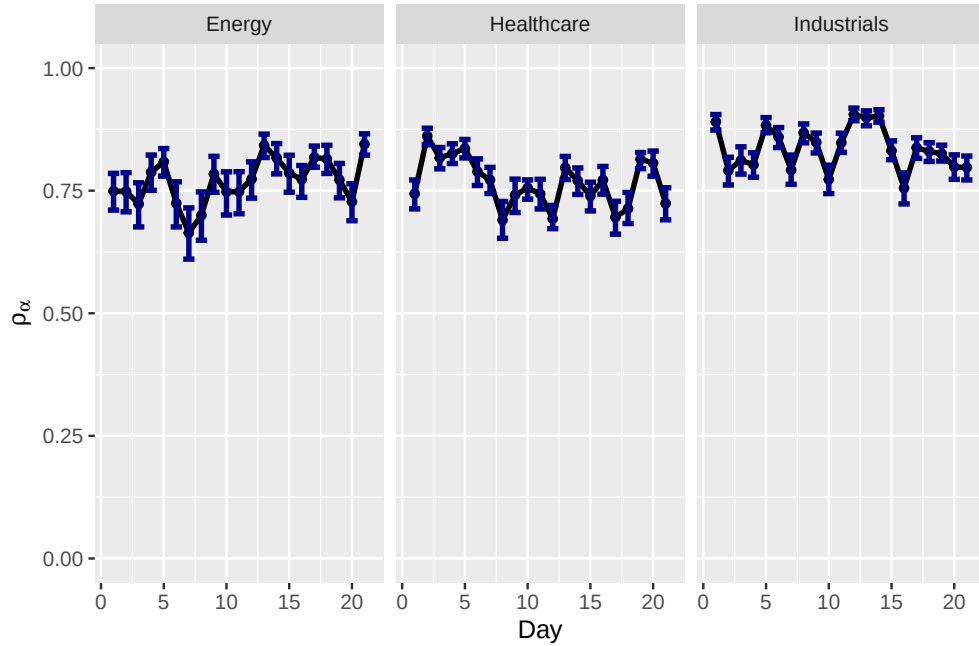


Figure 13: The estimated ρ_{α} with their 95% credible interval across all three sectors in January 2013

Correlation between observed counts Based on the inference of the fixed effects and random effects, we are also able to find the conditional relationship between the two types of counts. According to (3.11), the conditional posterior correlation between the two counts can be

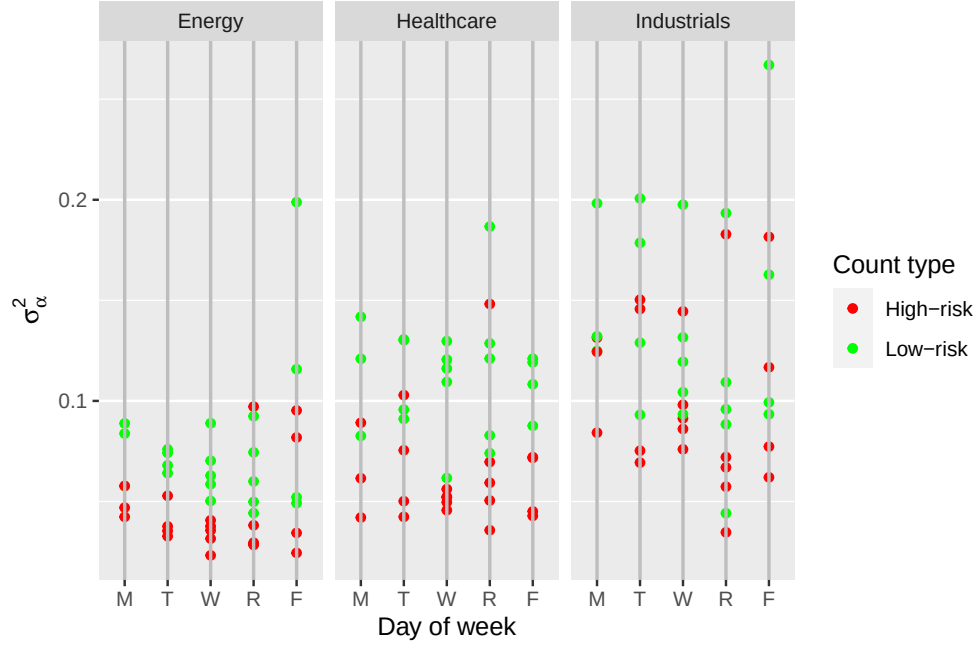


Figure 14: The dot plot of estimated variances of the latent level effects on different days of week across all three sectors in January 2013

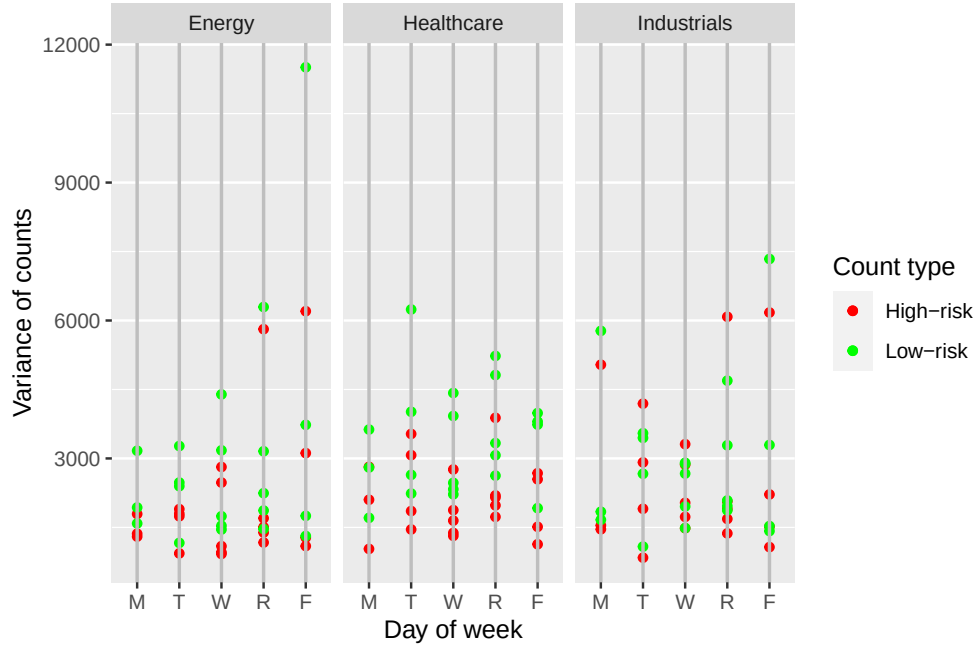


Figure 15: The variances of the observed counts on different days of week across all three sectors in January 2013

calculated given the conditional posterior marginal mean and variance of each type of count. The derivation of the correlation between the two count variables can be found in the Appendix section.

It is not appropriate to build a linear regression model between two count variables, especially when the observed counts are not large enough. However, it is not uncommon to observe a large number of transactions within a short time interval thus leading to a large number of counts at different risk levels, and the Poisson distributions with a large mean can be still approximated by normal distributions. If we use simple linear regression to describe the relationship between $Y_{H,st}$ (counts with non-zero log returns) and $Y_{L,st}$ ((counts with zero log returns)), we can build a linear model on the raw counts,

$$\hat{Y}_{H,st} = \hat{b}_{0,s} + \hat{b}_{1,s}Y_{L,st}. \quad (3.31)$$

For each individual asset, the R-squared value can be calculated as,

$$R_s^2 = \left(\frac{\sum(Y_{H,st} - \bar{Y}_{H,s})(Y_{L,st} - \bar{Y}_{L,s})}{\sqrt{(\sum(Y_{H,st} - \bar{Y}_{H,s})^2)(\sum(Y_{L,st} - \bar{Y}_{L,s})^2)}} \right)^2, \quad (3.32)$$

where $\bar{Y}_{H,s}$ and $\bar{Y}_{L,s}$ are the sample mean of the counts for asset s . In a simple linear regression model, the R-squared value stands for the proportion of the variation in the response variable explained by the regressor, and it is equivalent to the squared correlation coefficient. A high R-squared value is associated with a higher correlation coefficient.

Based on the inference of the fixed effects and random effects, we are also able to find the model-based conditional relationship. In Figure 16, the conditional correlation between counts derived by the latent effects varies throughout a trading day compared with the unconditional correlation between the counts. The unconditional correlations are directly computed based on the count data on the observational level as empirical correlations, and they are universally higher than the conditional posterior correlation computed from our model (BVRW(1)-LCM) in the Energy sector. The solid lines and curves stand for the aggregated mean of the estimated unconditional/conditional correlation over one month while the bands stand for a margin with ± 2 standard errors for the estimated correlations. Similar patterns are also displayed in the Healthcare and Industrials sectors, see Figures 31 and 32 in the Appendix section.

We can also compare the RMSE between the SLR prediction (3.31) and BVRW(1)-LCM-based conditional regression,

$$Y_{H,st}^* = b_{0,st}^* + b_{1,st}^*Y_{L,st}, \quad (3.33)$$

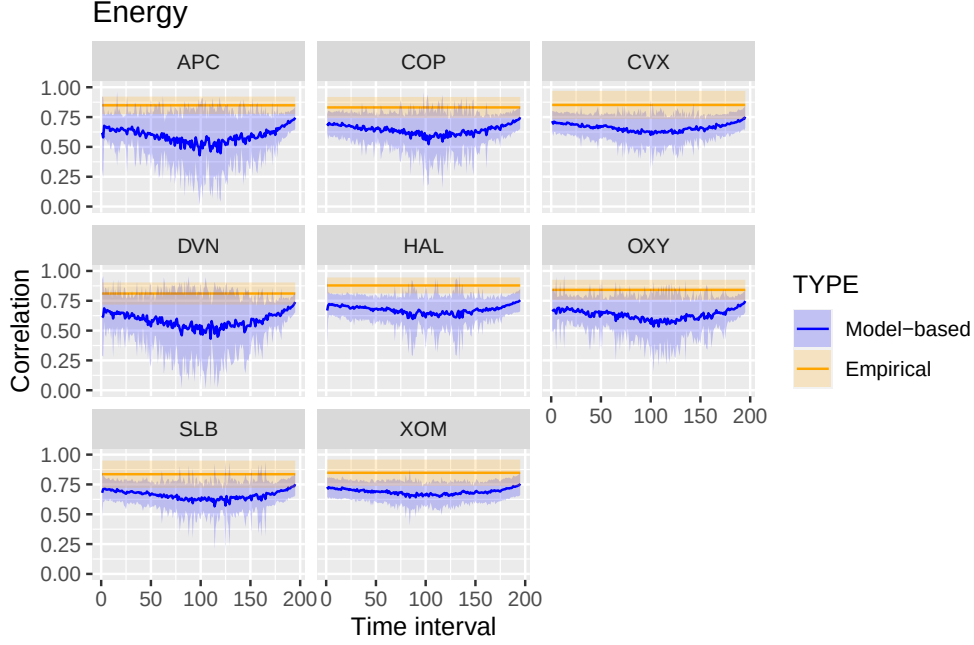


Figure 16: The trace plot of daily aggregated model-based and empirical correlations in the Healthcare sector throughout January 2023

where $b_{1,st}^*$ and $b_{0,st}^*$ are approximated by,

$$\begin{aligned} b_{1,st}^* &= \text{Corr}(Y_{H,st}, Y_{L,st} | \text{Data}) \frac{sd(Y_{H,st} | \text{Data})}{sd(Y_{L,st} | \text{Data})}, \\ b_{0,st}^* &= E(Y_{H,st} | \text{Data}) - b_{1,st}^* \cdot E(Y_{L,st} | \text{Data}). \end{aligned} \quad (3.34)$$

In the Energy sector shown in Figure 17, the RMSE based on the approximated conditional regression based on BVRW(1)-LCM is significantly lower than the RMSE of the SLR model. Therefore, imposing a constant correlation between the count data can be a strong assumption for real applications. BVRW(1)-LCM-based regression also has a better performance in the other two sectors, see Figures 33 and 34 in the Appendix section.

Across all sectors, the two types of counts have a stronger correlation at the opening and closing hours on a trading day and are less correlated in the middle of the day. Some assets such as APC, DVN in the Energy sector, MDT, BAX, and BIIB in the Healthcare sector, and FDX, UNP, and RTN in the industrials sector, their middle-day correlation estimation remains stable throughout the month because the standard errors are smaller compared with other time points and other assets. Such correlation estimation with reduced uncertainty can be useful for the prac-

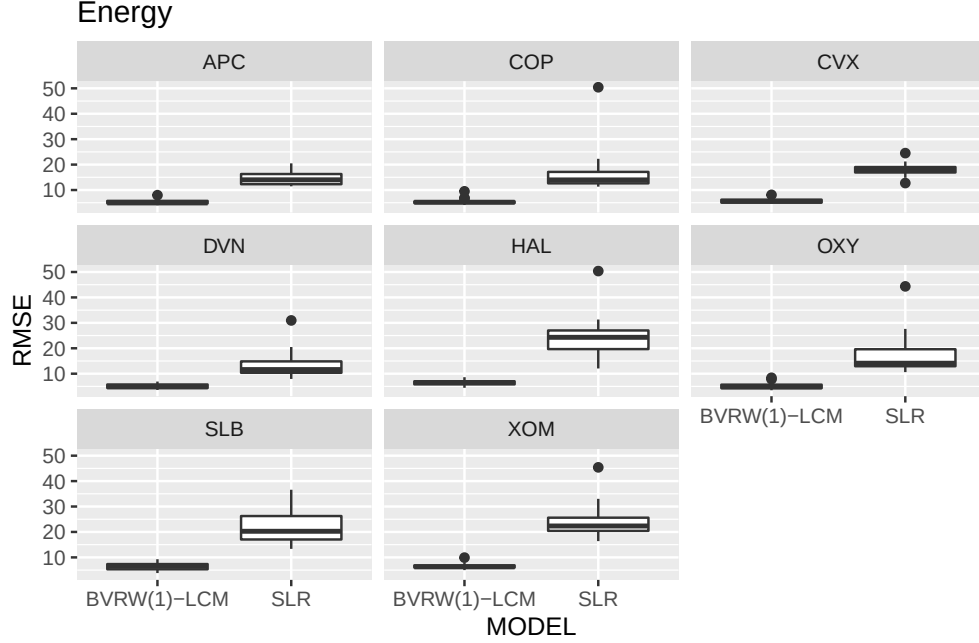


Figure 17: Box plots for the square root of MSE comparison in the Energy sector between BVRW(1)-LCM and Simple Linear Regression in January 2023

tioners to specifically adjust their investment scheme for these stocks in these time intervals. For example, when the correlation between these two counts is estimated to be high, the ratio of the estimated counts will be more reliable for the corresponding time interval thus the practitioners can evaluate whether or not the underlying asset price is actively fluctuating instead of staying steady with a large proportion of zero log returns. As a measurement for the association between count data, correlation describes the linear relationship between two variables and has a satisfactory performance when the count data are large but it is sensitive to outliers or non-linearity and has a poor performance when the counts are small or even close to zero. The unconditional correlations obtained from the raw count are also overestimated and unrealistic for practice. Therefore, we've also investigated Spearman's rank correlation between the two types of counts to measure the monotonic association between the count data. In Figure 18, the empirical Spearman's rank correlations on the observational level are aggregated means based on different stocks within the same sector. Although it indicates that the two types of counts have a fairly strong monotone relationship (correlation around .75), practitioners won't be able to make additional inferences barely with Spearman's rank correlation. Nonetheless, the estimated latent level correlation ρ_{α} from the BVRW(1)-LCM model can account for the correlation on the observational level as a hidden factor.

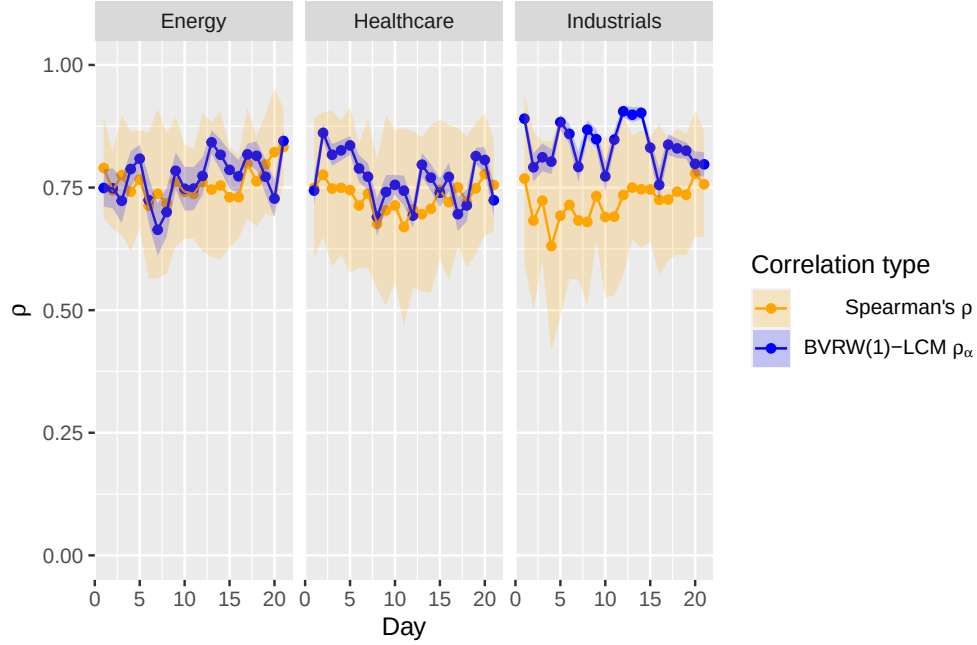


Figure 18: The comparison between Spearman's rank correlation and the latent level correlation across three sectors.

Although it is meaningless to compare the magnitude between Spearman's rank correlation and the latent level correlation as two different correlation measurements, their trajectories are observed to share some similarity and it will be more convincing to use the BVRW(1)-LCM model to make inference and prediction when the pattern of correlation between the counts is preserved.

Remarks Since liquidity reflects the ability of an asset to be traded frequently without a significant price change, the counts of transactions at a low-risk level (the magnitude of log return less than τ) can describe this property in a short-term manner. While volatility delineates how drastically the price of an asset fluctuates given a time interval, the counts of transactions at a high-risk level (the magnitude of log return greater than τ) displays the frequencies of trades with extreme price changes for which such extremity is pre-defined by the practitioners. In our study, we can observe positive correlations between these two types of counts as all conditional correlations are greater than zero with $\tau = 0$, which implies a positive association between liquidity and volatility in such a setting. However, the choice of $\tau = 0$ is just for the illustration of our model framework, and the definitions for the different types of counts can be customized by practitioners with their interests. The number of types depends on the number of thresholds and each individual asset can also be assigned separate thresholds as well because the price change patterns of different assets

can differ a lot from each other and a common threshold applied to define the counts can cause data imbalance issues with excessive zeros for some assets so that the unified conditional Poisson model framework needs to be adapted by some potential zero-inflated models. In practice, the prediction of the count data at different risk levels is not directly related to a profit-making trading algorithm as the model focuses more on the variability of the price fluctuation in a discrete manner, but the instant ratio between different counts within a short term provides statistical evidence on the volatility of the market. For investors in the HFT market, such information facilitates their decision on the strategies of trade and hold within a short period. A detailed trading algorithm can be a combination of other price-movement prediction models and our framework, but this is beyond the scope of this dissertation.

3.5 Summary

In this chapter, We proposed a Bayesian Poisson lognormal hierarchical model for multivariate count time series in the paper. The incorporation of latent temporal effects and level random effects helps to account for different sources of variation as well as the correlation among the counts. The model inference is achieved by the approximated Bayesian inference approach, INLA, of which the performances of model parameter recovery and computation cost are shown to be competitive with the traditional MCMC method in the simulation study. Although our model framework targets bivariate count data in this paper, it is feasible to extend our framework to count data with a higher dimension by correctly specifying the precision matrix of high dimensional latent temporal effects. Since such precision matrices are usually sparse, their inverse computation can enjoy an efficient algorithm via sparse representation. Due to its computational efficiency, the scalability of the model inference can also be achieved through parallel computing, which can save much more computational time than a fully Bayesian approach. In real applications, we use our framework to account for the microstructure of the HFT data, including the interpretation of the covariates, hyperparameters of latent effects, and conditional correlation between count data with the latent inference, which are not straightforward based on the counts on the observational level. Finally, We also provide our opinions on the potential usage of the framework in the real financial market.

4 Sequential Bayesian spatio-temporal outbreak detection

The first two topics discussed in Chapters 2 and 3 are related to statistical research problems from the high-frequency financial market. Online structural break detection via a CUSUM-like quasi-score detector statistic in financial durations helps monitor the trading intensity in the market and multivariate count time series modeling establishes a connection between observed counts via latent interdependent random effect. Motivated by relevant techniques from uni-variate online change point detection and efficient multivariate count data modeling, we are going to extend the online change point detection framework to the multivariate level. For Chapter 4, we aim at developing online outbreak detection framework for public health surveillance data with spatio-temporal interdependence.

4.1 Background

Public health plays an important role in protecting and improving the health of individuals and communities. One of the public health functions is disease prevention and control. Through vaccination programs, disease surveillance, and infection control measures, public health efforts can help control and prevent the spread of infectious diseases, such as the flu and COVID-19. Public health authorities use surveillance systems to monitor and track the occurrence of diseases, health conditions, and other health-related events in a population. Surveillance systems provide statistical insights into the trends, patterns, and risk factors for disease, as well as the evaluation of the effectiveness of public health interventions. For infectious diseases, early outbreak detection allows public health officials to take timely action to prevent the spread of the disease. A practical implementation of surveillance is the disease dashboard such as the one for COVID-19. Providing real-time tracking of the pandemic, such dashboards inform early warning of outbreaks with improved transparency and communication. Despite the concern about data quality and accuracy due to data collection or reporting processes, dashboards still give people access to a wealth of data. As long as practitioners handle such surveillance data with careful interpretation and bias adjustment, a dashboard can be a valuable resource, to facilitate public health decision-making on a disease outbreak from one of the various aspects. Tsui et al. (2008) gave a comprehensive review of different types of surveillance systems and Li et al. (2022a) summarized modern challenges and

opportunities in public health data surveillance and forecasting. The motivation of this chapter is to deliver an accurate early outbreak detection framework to monitor the occurrence of infectious disease outbreaks based on the multivariate discrete dashboard data for an ongoing pandemic and contribute to the collaboration with researchers in many different fields including epidemiology, computer science, and public health.

4.2 Literature review

4.2.1 Change point detection in public health surveillance system

The early outbreak detection problem in surveillance systems can be regarded as the research topic of online change point detection, also known as sequential or quickest change point detection because an outbreak of an infectious data can be represented as a data pattern change, such as the changes of trend, mean level, and variation. The main goal of online change point detection is to detect the change from sequential data in real-time as soon as possible. Online change point detection procedure deals with sequential data or streaming data and detects any change point shortly after its occurrence and stops at the detection while the offline procedure has access to the full data and aims at identifying and localizing the changes in data sequence in a retrospective manner. Although online change detection procedure is better aligned with the need for early outbreak detection in surveillance systems, statistical methodologies from both research lines provide an abundance of inspiration for our paper.

All the methods are usually designed based on the average detection delay, probability of false alarm, false alarm rate, and etc. Existing sequential change point detection methods can be divided into two categories, such as Bayesian in which the distribution of the change-point time is known and non-Bayesian (minimax) methods in which the change-point time is non-random and unknown. Johnson et al. (2017) provided a review of optimal change-point detection theory in both Bayesian and non-Bayesian settings recently. Regression methods of outbreak detection have been widely used, both for detecting outbreaks in surveillance systems based on laboratory reports and notified infections, and for syndromic surveillance. Farrington et al. (1996) described the detection of outbreaks to boost other more intensive surveillance methods by routinely scanned data using linear regression model but its weakness was insensitivity when the baseline values on

which the threshold calculation is based coincide with past outbreaks. Some methodological issues involved in outbreak detection using examples from different statistical techniques which are focused on infectious diseases such as monitoring birth defects were described by Farrington and Andrews (2003) while Diggle et al. (2009) illustrated how spatial statistical methods can be used on developing online surveillance systems for common diseases by the nature of the data. A systematic and comprehensive review of the advancement of aberration detection algorithms used in public health surveillance for the last decade was given by Yuan et al. (2019). Beside the classical stochastic process control methods and regression-based methods, modern surveillance system evolves to adopt more sophisticated monitoring regimens via such as Bayesian hierarchical models and machine learning frameworks as solutions to handle the increased complexity and volume of surveillance data. Another important line of related research, offline detection, handles the change point detection as well, but in a completely different fashion. What differentiates the online and offline change point detection problems most is type of data availability, sequential update versus complete retrospection.

4.2.2 Traditional outbreak detection methods

Traditional outbreak detection methods include stochastic process control (SPC)-related approaches and regression-based models. Early statistical applications in public health surveillance systems were based on the analysis of the reported data from clinicians or laboratories. Therefore, statistical methods stemming from stochastic process control in industrial manufacturing gained their popularity. Traditional control charts are common tools for change point detection problems. Shewhart control chart (Shewhart, 1929) as one of the earliest change point detection tools, uses previous data to calculate a threshold based upon normality assumption (Montgomery, 2020). Due to its easy-to-compute-and-interpret feature, Early Aberration Reporting System (EARS) software in Centers for Disease Control (CDC) of the United States employed Shewhart control chart as one of the surveillance tools to conduct near real-time monitoring. Instead of possessing congruence and stationarity as the production data from a stable manufacturing process, surveillance data can exhibit strong time trends, cyclic patterns, and other time-dependent effects, depending on the data aggregation, population behaviors as well as other environmental factors. To take into account these factors for valid applications, Shewhart control chart was adjusted to accommodate the mean

and standard deviation with a short sliding window of historical data, such as C and W algorithms, which capture the recent data pattern by consecutive days. However, Shewhart control chart and its extensions can have compromised performance because of inappropriate data preprocessing and violation of methodological assumptions. For example, the normality assumption is a strong one for surveillance time series especially the observations are small count data. Although Shore (2000) implemented an inverse normal transformation approach to handle non-Gaussian data, such approach may suffer from different application where larger data sets are required for a more accurate estimation for higher-order moment.

As another well-adopted SPC method for detecting change-points, CUSUM chart introduced by Page (1954) is able to detect small shift from mean for industrial quality control data more quickly than Shewhart chart. The idea of using cumulative information for detection is further extended or modified to solve various research problems. When dealing with discrete surveillance data, such as count data of health events, Rossi et al. (1999) proposed a approximation CUSUM procedure for a Poisson process for practicality and convenience, of which the main idea is to transform a Poisson variate through standardization and get an approximated Gaussian variate. When surveillance data are in the form of correlated count data, He et al. (2014) discussed the implementation of Multivariate Poisson CUSUM chart, and provided the control chart design based on log-likelihood ratios with in-control parameters for baseline and design parameters for the shift. As the common limitation, the SPC and SPC-related methods are more related to industry quality control, where a in-control process can be defined in full detail and usually don't need additional covariate information to enhance accountability. The stationary assumption of public health data may not be valid, and integration of various data sources facilitates a better understanding of the baseline process and a better performance of early outbreak detection. Nevertheless, SPC methods can be synthesized into regression-based methods to monitor the pattern of model residuals, scores, or quasi-scores (Berkes et al., 2004).

Regression-based methods enable the incorporation of external covariates and explain temporal dependence from past observations. Generalized linear models (GLMs) with distribution-specific links to the mean function grant the versatility to handle a variety of surveillance data. Apart from the well-known linear regression with independent Gaussian residuals for continuous data, count data can also be modeled by GLM with corresponding link functions to various discrete

distributions such as Poisson, negative binomial as well as their zero-inflated variants etc. The incorporation of additional factors of interest into GLMs enhances model forecast performance as well as interpretability. Temporal dynamics such as day of week effect or seasonality are usually elucidated via a dummy variable specifying different kinds of days or trigonometric term resembling the sinusoidal data pattern (Serfling, 1963). Farrington et al. (1996) developed a regression algorithm to assist outbreak detection by using the threshold predicted from the modeled baseline. Adaptive GLM frameworks with a short sliding window of historical data address the limitations of assuming static model parameters via sequential updating parameter estimates according to a sliding baseline pattern (Burkom et al., 2007; Xing et al., 2011).

Auto-regressive integrated moving average (ARIMA) models (Box et al., 2015) and integer-valued auto-regressive (INAR) models (Alzaid and Al-Osh, 1988), as special regression-based methods, account for the time dependence through a combination of past observations. For continuous-valued time series, the time-dependence is illustrated by a ARIMA(p,d,q) model,

$$\left(1 - \sum_{i=1}^p \phi_i B^i\right) (1-B)^d (Y_t - \mu_Y) = \left(1 + \sum_{j=1}^q \theta_j B^j\right) \epsilon_t,$$

$$\epsilon_t \sim N(0, \sigma_\epsilon^2),$$

where Y_t and μ_Y are the observation at time t and its expectation, and ϕ_i, θ_i , and σ_ϵ^2 are respectively the auto-regressive (AR) coefficients, moving average (MA) coefficients, and the variance of the white noise ϵ_t . B is a backshift operator, i.e., $B^i Y_t = Y_{t-i}$. p and q denotes the AR and MA orders, indicating the number of lagged historical observations included in the model, while d is the order of differencing to stabilize non-stationary time series. For integer-valued time series, the temporal association between counts can be modeled via INAR(p) model with a binomial thinning technique,

$$Y_t = \sum_{i=1}^p \alpha_i \circ Y_{t-i} + \epsilon_t,$$

where the binomial thinning operator $\alpha_i \circ Y_{t-i} = \sum_{k=1}^{Y_{t-i}} B_k, B_k \stackrel{i.i.d}{\sim} \text{Bernoulli}(\alpha_i)$. ϵ_t is an independent non-negative integer-valued random error. Besides the flexibility of using seasonally lagged observations and exogenous variable to explain the seasonality and external impact, the

auto-regressive framework alleviates the reliance on independent assumption and appreciates the accountability pertaining to the unobserved and latent factors contained in historical data. In addition, smoothing methods originated from the moving average (MA) methods with exponential weights such as Exponential Weighted Moving Average (EWMA) and Holt-Winters method (Winters, 1960). Such smoothing methods are commonly as part of data preprocessing to smooth out seasonal effect and trend by applying other methods.

However, the aforementioned methods have different limitations. For example, SPC-related methods rely on normality and stationary assumptions for their valid implementation. When surveillance data are discrete, a common choice of model fitting is still the standard Poisson model, even though it has been suggested in numerous literature that the data overdispersion issue can be remedied by imposing a gamma prior on the Poisson mean or using a negative binomial model.

4.2.3 Modern outbreak detection methods

Over the last decade, statistical surveillance algorithms for online outbreak detection have a substantial evolution with the complexity of surveillance data and the advancement of computational technology. To expedite early outbreak detection, public health authorities have already begun to take advantage of pre-diagnostic or syndromic data such as patient counts with disease-related symptoms recorded by hospitals or healthcare-related search information via electronic online platforms. The expanding volume and variety of such data sources place both more valuable pre-pandemic information and more challenging problems for efficient statistical models than traditional laboratory diagnostics. Therefore, a burgeoning research line of Bayesian models in public health surveillance offers several advantages over traditional methods including historical information borrowing, domain knowledge a priori incorporation, computational solution to complex hierarchical models, and natural interpretations of posterior probability for outbreak detection.

Bayesian Hidden Markov Models (HMM) enable practitioners to classify the surveillance data into a non-epidemic phase and an epidemic phase via a binary latent state variable Z (Martínez-Beneito et al., 2008; Watkins et al., 2009; Conesa et al., 2015) and a suspiciously anomalous observation with a large value of $Pr(Z = 1)$ can trigger further investigation before finalizing a public outbreak announcement. Since it is common to observe spatio-temporal patterns in surveillance areal-type data, a common approach to introduce temporal dependence in the model is via an

ARMA association among latent temporal effects. Meanwhile, to account for the spatial correlation, data from neighboring locations are assumed to behave similarly. Such spatial patterns can be explained through the incorporation of latent spatial effects from different locations. It would become natural for practitioners to account for the spatial association of the areal unit data. Conditional auto-regressive model (Besag, 1974) is a prevalent choice as a prior distribution for random spatial effect. Besag et al. (1991), Gelfand and Vounatsou (2003) and Jin et al. (2005) have additional discussions on the extension of CAR model, including the discussion of the special case of the CAR model, Intrinsic Auto-regressive (IAR) model with an extra constraint on the random effects, efficient computation for the determinant of the precision matrix, as well as the multivariate CAR model which is though more computationally demanding.

Uni-variate HMMs can be extended their multivariate version by incorporating spatio-temporal random effects (Heaton et al., 2012; Zou et al., 2012, 2014) and iterative update formula of $Pr(Z = 1)$ was also provided to circumvent parameter re-estimation when new observations arrive. Dimension reduction also needs to be considered when the number of spatial locations is large and there are many areas with small counts. Zou et al. (2018) addressed this issue via a semi-parametric Dirichlet process for clustering similar regions and applied the particle filter approach to make inferences on latent variables. A real-time Bayesian spatio-temporal syndromic surveillance framework was also applied to small companion animals as an auxiliary source for public health surveillance (Hale et al., 2019). Other Bayesian approaches such as Bayesian scan statistics (Neill, 2011), Bayesian networks (Cooper et al., 2015) and Bayesian disease mapping (Anderson et al., 2017) are also prevalent choices in public health surveillance systems to handle such as point-reference spatial data Banerjee et al. (2003), individual-level medical records, etc. Point-referenced data are also known as geostatistical data containing precise location information for individual data points, and the public health surveillance systems may usually aggregate the individual data points over a set of disjoint areas due to privacy concerns.

One important concern regarding the Bayesian frameworks is their computational cost. Most of public health surveillance literature using Bayesian frameworks adopted MCMC method as the computational tool, which can be significantly time-consuming if data dimension increases or the complex model structure is poorly specified. Although Rue et al. (2009) proposed Integrated Nested Laplace Approximation as an efficient computational tool, there is limited literature to our

knowledge that employed INLA implementation in public health surveillance (Manitz and Höhle, 2013; Salmon et al., 2015).

With the enhancement of computational power, machine learn methods in disease monitoring provides another solution to address the complex dynamics involving both social and biological systems. During the pandemic of COVID-19, it becomes more challenging and numerous early outbreak detection applications are delivered in this research area. Deep Learning-based models were also used. Saqib (2021) proposed a hybrid machine learning model that is not only predicted with good accuracy but also takes care of uncertainty of predictions using Bayesian Ridge Regression. Radev et al. (2021) presented a simulation-based Bayesian inference framework for complex epidemiological models using neural network which utilized short time-series samples to obtain early warning signals while Li et al. (2022b) introduced a model-based method. Most of these methods have limitations due to the number of data availability. According to Coughlin et al. (2021), they identified possible change or turning points as indicated by the dynamics of daily COVID-19 incidences which the identified change points were combined with the spline-fitted trend which interprets the behavior of the change points; it may have provided better prediction dates for the implementation of public restrictive interventions in each country but not appropriate for all nations. Even Guan et al. (2021) added prediction method using mobility data which helps to determine when and where outbreaks will occur, it has several limitations such as smoothing, missing data, and based on information about only known cases but not undetected cases. However, the practical limitations facing machine learning methods include the vast amount of training data, risk of over-fitting, and accurate labels for outbreaks in the training data.

4.3 BOSTON-PUPA procedure

To address complex and dynamic spatio-temporal patterns of streaming surveillance data, we propose a consolidated Bayesian Online Spatio-Temporal Outbreak detection framework with Prior Updating and P-value Adaptation (BOSTON-PUPA) techniques to effectively achieve both global and local sequential monitoring over a cluster of regions at risk. The streaming data of confirmed case counts take a spatio-temporal format as they are reported from various locations on a daily basis. As an iterative process with a fixed-size sliding window, the BOSTON-PUPA framework accounts for the spatio-temporal association and overdispersion of the count data via a Bayesian

hierarchical generalized Poisson distribution (GPD) model with Integrated Nested Laplace Approximation (INLA) method. The Prior Updating technique leverages past information and the current sliding window sequentially for integrated inference with a reduction of computational burdens. The P-value Adaptation technique enhances the significance of a true outbreak to address potential imbalance issues for spatial data, especially for the regions with small counts. The incorporation of eclectic techniques results in decent false detection control while preserving a robustly high detection power and efficiency against different signal-to-noise ratios.

Our proposed Bayesian Online Spatio-Temporal Outbreak detectionN with Prior Updating and P-value Adaptation (BOSTON-PUPA) framework is an iterative procedure with a sliding window for the training data modeling with window length $T + 1$. The streaming data are monitored consecutively with a one-day-ahead sliding pace and only the observed counts on the last day in the sliding window will be involved in the outbreak detection procedure. With the initial sliding window containing observed data from day 1 and day $T + 1$, the outbreak detection procedure aims at testing the null and the alternative hypotheses $H_{0,k}^{(s)}$ versus $H_{a,k}^{(s)}$, $k = 1, 2, \dots$, as below,

$$\left\{ \begin{array}{l} H_{0,k}^{(s)} : \text{No significant outbreak has occurred yet in the } s\text{-th location by the } (T + k)\text{-th day.} \\ H_{a,k}^{(s)} : \text{An outbreak has occurred significantly in the } s\text{-th location by the } (T + k)\text{-th day.} \end{array} \right. \quad (4.1)$$

The iterative outbreak detection procedure involves four steps: 1) Bayesian model inference and in-sample prediction. 2) Latent aberration assessment. 3) P-value adaptation. 4) Decision-making and algorithm update. To elucidate the detection procedure, we elaborate on each step at the k -th iteration for instance.

4.3.1 Step 1: Bayesian model inference and in-sample prediction

First, we will introduce the model framework for the surveillance data under the null hypothesis of no outbreak at any location. Let $\{Y_{s,t}\}$ denote the daily case count at day t at location s . We propose the following conditional generalized Poisson distribution (GPD) framework for the

multivariate count time series,

$$\begin{aligned}
Y_{s,t}|\theta_{s,t}, \lambda &\overset{ind}{\sim} GPD(\theta_{s,t}, \lambda), \\
\eta_{s,t} = \log \mu_{s,t} &= \log \left(\frac{\theta_{s,t}}{1-\lambda} \right) = P_s + \mathbf{X}_s \boldsymbol{\beta} + \phi_s + \varepsilon_t, \\
\phi_s | \boldsymbol{\phi}_{-s} &\sim N \left(\rho_\phi \sum_{j \neq s} \frac{w_{sj}}{w_{s+}} \phi_j, \frac{\sigma_\phi^2}{w_{s+}} \right), \\
\varepsilon_t &= \rho_\varepsilon \varepsilon_{t-1} + \omega_t, \omega_t \sim N(0, \sigma_\varepsilon^2), \\
\boldsymbol{\Theta} &= (\lambda, \boldsymbol{\beta}, \rho_\phi, \rho_\varepsilon, \sigma_\phi^2, \sigma_\varepsilon^2) \sim \pi(\boldsymbol{\Theta}).
\end{aligned} \tag{4.2}$$

In (4.2), $s = 1, 2, \dots, S, t = 1, 2, \dots, T$, where S is the total number of locations and T is the total number of days in the sliding window. P_s is the offset term for location s and it accounts for differences in expected values or exposures at each location. In our study, we choose the logarithmic population as the offset term. $\mathbf{X}_s$ is the covariate vector of interest for location s and $\boldsymbol{\beta}$ is the corresponding fixed effect vector. $\boldsymbol{\phi} = (\phi_1, \phi_2, \dots, \phi_s)$ are the spatial effects and account for the neighboring association among all the spatial units. $\mathbf{W} = (w_{sj})_{S \times S}$ is the adjacency matrix for these locations,

$$w_{sj} = \begin{cases} 1, & \text{if location } s \text{ and location } j \text{ are adjacent} \\ 0, & \text{otherwise.} \end{cases}$$

The diagonal elements of $\mathbf{W}$ are zeros and $w_{s+} = \sum_{j=1}^S w_{sj}$. ε_t is the common temporal latent effect, describing the shared dynamics by the mean functions of the counts from different locations. We assign the Conditional Autoregressive (CAR) prior and an AR(1) process prior to the spatial and temporal latent effects correspondingly. $\boldsymbol{\Theta}$ contains all the parameters in the Bayesian hierarchical framework. The prior distribution for $\boldsymbol{\Theta}$ is $\pi(\boldsymbol{\Theta})$, which will be discussed shortly in detail.

Considering the potential dispersion of count data, We select a conditional GPD for the count data. The point mass function of a conditional GPD (Consul and Jain, 1973) given as

$$Pr(Y_{s,t} = y | \theta_{s,t}, \lambda) = \begin{cases} \frac{\theta_{s,t}(\theta_{s,t} + \lambda y)^{y-1}}{y!} \cdot \exp(-(\theta_{s,t} + \lambda y)), & y = 0, 1, 2, \dots \\ 0 & \text{for } y > m, \text{ when } \lambda < 0 \end{cases} \tag{4.3}$$

with mean $\mu_{s,t} = \theta_{s,t}/(1 - \lambda)$ and variance $\sigma_{s,t}^2 = \mu_{s,t} \cdot (1 - \lambda)^{-2}$, $\max(-1, -\theta_{s,t}/m) < \lambda < 1$ and $m(\geq 4)$ is the largest positive integer for which $\theta_{s,t} + m\lambda > 0$ when λ is negative. Since our framework focuses on the overdispersion pattern in the count data, the range of λ is restricted within $[0, 1)$. When $\lambda = 0$, the GPD becomes a standard Poisson distribution. Such flexibility can better accommodate the small counts, especially when $\theta_{s,t}$ is close to zero. When the mean parameter $\theta_{s,t}$ approaches zero, the underlying distribution will face a degeneracy issue, with a dominant probability on $P(Y_{s,t} = 0|\theta_{s,t}, \lambda) \approx 1$. For the observed small counts at some locations, a detection procedure attributed to such a point mass distribution will deliver an extreme test statistic under $H_{0,k}^{(s)}$ even if the magnitude of the counts doesn't qualify for an outbreak claim. Compared with the standard Poisson distribution, a GPD with an overdispersion parameter will be less likely to reach a false alarm of an outbreak because it can account for larger variations in the counts than a standard Poisson distribution does.

In the INLA implementation of (4.2), all the parameters with restricted support will be internally transformed into the ones with support on $\mathcal{R}$ as described in Table 13. Such a reparameterization technique grants efficient computation of the internal parameters on their unrestricted supports. Therefore, we assign the internal hyperparameters, $\boldsymbol{\vartheta} = (\xi, \boldsymbol{\beta}, \vartheta_{\rho_\phi}, \vartheta_{\rho_\varepsilon}, \vartheta_{\sigma_\phi^2}, \vartheta_{\sigma_\varepsilon^2})$, in (4.2) with Gaussian priors. The variances of the Gaussian priors quantify the uncertainty about the internal parameters. In order not to cause further confusion in the rest of the paper, we use the parameters/internal parameters to include the covariate coefficients and the hyperparameters.

Hyperparameter	Support	Internal representation
λ	$(0, 1)$	$\xi = \log\left(\frac{\lambda}{1-\lambda}\right)$
ρ_ϕ	$(0, 1)$	$\vartheta_{\rho_\phi} = \log\left(\frac{\rho_\phi}{1-\rho_\phi}\right)$
ρ_ε	$(-1, 1)$	$\vartheta_{\rho_\varepsilon} = \log\left(\frac{1+\rho_\varepsilon}{1-\rho_\varepsilon}\right)$
σ_ϕ^2	$(0, \infty)$	$\vartheta_{\sigma_\phi^2} = \log(\sigma_\phi^2)$
σ_ε^2	$(0, \infty)$	$\vartheta_{\sigma_\varepsilon^2} = \log(\sigma_\varepsilon^2)$

Table 13: Hyperparameters with restricted support and their internal representation

In a sliding window of size $T + 1$ at the k -th iteration, (4.2) is fitted on the training data

$\mathcal{D}_k = \{\mathbf{Y}_{k+t}\}_{t=0}^T$, with $\mathbf{Y}_{k+t} = \{Y_{1,k+t}, Y_{2,k+t}, \dots, Y_{S,k+t}\}$ with a prior $\pi(\boldsymbol{\Theta}) = \pi(\boldsymbol{\vartheta})|\frac{\partial \boldsymbol{\vartheta}}{\partial \boldsymbol{\Theta}}|$,

$$\pi(\boldsymbol{\vartheta}) = \begin{cases} \pi_G(\boldsymbol{\vartheta}), 1 \leq k \leq T+1 \\ \pi_G^{a_0}(\boldsymbol{\vartheta}|\mathcal{D}_{k-T-1}^*), k \geq T+2 \end{cases} \quad (4.4)$$

where $\pi_G(\boldsymbol{\vartheta})$ corresponds to a vague internal Gaussian prior with a large variance, used in the first $(T+1)$ iterations. After the $(T+1)$ -th iteration, $\pi(\boldsymbol{\vartheta}) = \pi_G^{a_0}(\boldsymbol{\vartheta}|\mathcal{D}_{k-T-1}^*)$ becomes a historical-data-related Gaussian prior raised to a discounting factor $a_0 (0 < a_0 \leq 1)$, where $\mathcal{D}_{k-T-1}^*$ is the past cumulative information set up to the $(k-T-1)$ -th iteration and has no overlap with the current training data $\mathcal{D}_k$. Since (4.2) doesn't have a closed form for the posteriors of its parameters or hyperparameters, we intend to only borrow the historical information via a Gaussian prior, $\pi_G(\boldsymbol{\vartheta}|\mathcal{D}_{k-T-1}^*)$, which shares the first two moments with the corresponding posterior as in (4.5) for the purpose of prior updating in our iterative detection procedure,

$$\begin{aligned} \int \vartheta^* \pi_G(\vartheta^*|\mathcal{D}_{k-T-1}^*) d\vartheta^* &= \int \vartheta^* \pi(\vartheta^*|\mathcal{D}_{k-T-1}^*) d\vartheta^* \\ \int (\vartheta^*)^2 \pi_G(\vartheta^*|\mathcal{D}_{k-T-1}^*) d\vartheta^* &= \int (\vartheta^*)^2 \pi(\vartheta^*|\mathcal{D}_{k-T-1}^*) d\vartheta^*, \end{aligned} \quad (4.5)$$

where $\pi_G(\cdot)$ denotes a Gaussian density and $\pi(\vartheta^*|\mathcal{D}_{k-T-1}^*)$ corresponds to the actual posterior density for the internal parameter ϑ^* in $\boldsymbol{\vartheta}$. After the k -th iteration of model fitting, three pieces of information will be tracked, $\hat{\boldsymbol{\eta}}_{T+k}$, $(\hat{\xi}_k, \hat{\sigma}_{\xi_k}^2)$, and the posterior distributions for internal parameters. $\hat{\boldsymbol{\eta}}_{T+k} = (\hat{\eta}_{1,T+k}, \hat{\eta}_{2,T+k}, \dots, \hat{\eta}_{S,T+k})$ denotes the transformed posterior mean vector of $\mathbf{Y}_{T+k}|\mathcal{D}_k$ in (4.2). $\hat{\xi}_k$ and $\hat{\sigma}_{\xi_k}^2$ are the internal posterior mean and variance for the overdispersion parameter in Table 13. An illustration of Step 1 is summarized from Table 14 in which the model inference and in-sample prediction refer to the columns, “Mean”, “Overdispersion”, and “Posterior”. The “Posterior” column in Table 14 refers to the posterior distribution for the internal parameter $\boldsymbol{\vartheta}$ and $\pi(\boldsymbol{\vartheta}|\mathcal{D}_{k-T-1}^*) = \pi(\boldsymbol{\vartheta}|\mathcal{D}_{k-T-1}, \mathcal{D}_{k-2T-2}, \mathcal{D}_{k-3T-3}, \dots)$, where $\mathcal{D}_{k-T-1} \cap \mathcal{D}_{k-2T-2} \cap \mathcal{D}_{k-3T-3} \cap \dots = \emptyset$.

Iteration (k)	Training data	Prior	Mean	Overdispersion	Posterior
1	$\mathcal{D}_1$	$\pi_G(\boldsymbol{\vartheta})$	$\hat{\boldsymbol{\eta}}_{T+1}$	$(\hat{\xi}_1, \hat{\sigma}_{\xi_1}^2)$	$\pi(\boldsymbol{\vartheta} \mathcal{D}_1)$
2	$\mathcal{D}_2$	$\pi_G(\boldsymbol{\vartheta})$	$\hat{\boldsymbol{\eta}}_{T+2}$	$(\hat{\xi}_2, \hat{\sigma}_{\xi_2}^2)$	$\pi(\boldsymbol{\vartheta} \mathcal{D}_2)$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$T+1$	$\mathcal{D}_{T+1}$	$\pi_G(\boldsymbol{\vartheta})$	$\hat{\boldsymbol{\eta}}_{2T+1}$	$(\hat{\xi}_{T+1}, \hat{\sigma}_{\xi_{T+1}}^2)$	$\pi(\boldsymbol{\vartheta} \mathcal{D}_{T+1})$
$T+2$	$\mathcal{D}_{T+2}$	$\pi_G^{a_0}(\boldsymbol{\vartheta} \mathcal{D}_1)$	$\hat{\boldsymbol{\eta}}_{2T+2}$	$(\hat{\xi}_{T+2}, \hat{\sigma}_{\xi_{T+2}}^2)$	$\pi(\boldsymbol{\vartheta} \mathcal{D}_{T+2}, \mathcal{D}_1)$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$2T+2$	$\mathcal{D}_{2T+2}$	$\pi_G^{a_0}(\boldsymbol{\vartheta} \mathcal{D}_{T+1})$	$\hat{\boldsymbol{\eta}}_{3T+2}$	$(\hat{\xi}_{2T+2}, \hat{\sigma}_{\xi_{2T+2}}^2)$	$\pi(\boldsymbol{\vartheta} \mathcal{D}_{2T+2}, \mathcal{D}_{T+1})$
$2T+3$	$\mathcal{D}_{2T+3}$	$\pi_G^{a_0}(\boldsymbol{\vartheta} \mathcal{D}_{T+2}, \mathcal{D}_1)$	$\hat{\boldsymbol{\eta}}_{3T+3}$	$(\hat{\xi}_{2T+3}, \hat{\sigma}_{\xi_{2T+3}}^2)$	$\pi(\boldsymbol{\vartheta} \mathcal{D}_{2T+3}, \mathcal{D}_{T+2}, \mathcal{D}_1)$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Table 14: Iterative prior updating procedure with tracked mean and overdispersion

4.3.2 Step 2: Latent aberration assessment

During the iterative procedure with a sliding window, two types of latent aberrations are monitored simultaneously: global latent aberration and local latent aberration. Latent aberrations in the detection procedure are assessed via binary indicators stemming from comparisons between Z-score-based statistics and a user-specified threshold. The aberration indicators calculated in this step facilitate the subsequent p-value adaptation process.

To indicate the latent aberration at the k -th iteration, two latent estimates are of interest: the estimates of overdispersion parameter $\hat{\lambda}_k$ and location-wise GPD mean $\hat{\mu}_{s,T+k}$. The global latent aberration is associated with the overdispersion parameter λ in (4.2) because the spatio-temporal data share the common overdispersion parameter and any extreme counts observed from either location will contribute to a larger estimate for the overdispersion parameter, signaling a global latent aberration that certain location(s) can be experiencing unusual case counts. To specify the local latent aberrations, $\hat{\mu}_{s,T+k}$ will be compared with its historical estimates, and a larger value of $\hat{\mu}_{s,T+k}$ tend to deviate from its historical pattern thus raising an alarm. Since binary aberration indicators are Z-score-based, the corresponding internal representations, $\hat{\xi}_k$ and $\hat{\eta}_{s,T+k}$, are used in the calculation due to their unrestricted supports.

There are two sorts of collections involved to keep track of those two types of historical estimates. The collection of the estimates for the overdispersion, $\Xi_k = \left\{ (\hat{\xi}_1, \hat{\sigma}_{\xi_1}^2), (\hat{\xi}_2, \hat{\sigma}_{\xi_2}^2), \dots, (\hat{\xi}_k, \hat{\sigma}_{\xi_k}^2) \right\}$, refers to the ‘‘Overdispersion’’ column in Table 14, where the posterior estimates of mean and variance are

gathered sequentially. Meanwhile, the collection of posterior mean estimates refers to the “Mean” column in Table 14. $\mathbf{H}_{s,k} = \{\hat{\eta}_{s,T+1}, \hat{\eta}_{s,T+2}, \dots, \hat{\eta}_{s,T+k}\}$ denotes the historical estimates set for the s -th location. When $k = 1$, $\Xi_k = \mathbf{H}_{1,k} = \dots = \mathbf{H}_{S,k} = \emptyset$. When $k \geq 3$, the global and local latent aberration indicators, $\delta_{\hat{\xi}_k}$ and $\delta_{\hat{\eta}_{s,T+k}}$, $s = 1, 2, \dots, S$, are calculated respectively in (4.6),

$$\delta_{\hat{\xi}_k} = I\left(\frac{\hat{\xi}_k - \bar{\xi}_{k-1}}{\bar{\sigma}_{\xi_{k-1}}} > z\right), \bar{\xi}_{k-1} = \frac{1}{k-1} \sum_{i=1}^{k-1} \hat{\xi}_i, \bar{\sigma}_{\xi_{k-1}} = \sqrt{\frac{\sum_{i=1}^{k-1} \hat{\sigma}_{\xi_i}^2}{k-1}}. \quad (4.6)$$

and (4.7),

$$\begin{aligned} \delta_{\hat{\eta}_{s,T+k}} &= I\left(\frac{\hat{\eta}_{s,T+k} - \bar{\eta}_{s,T+k-1}}{\sigma_{\mathbf{H}_{s,k-1}}} > z\right), \bar{\eta}_{s,T+k-1} = \frac{1}{k-1} \sum_{i=1}^{k-1} \hat{\eta}_{s,T+i}, \\ \sigma_{\mathbf{H}_{s,k-1}} &= \sqrt{\frac{\sum_{i=1}^{k-1} (\hat{\eta}_{s,T+i} - \bar{\eta}_{s,T+k-1})^2}{k-2}}. \end{aligned} \quad (4.7)$$

In (4.6) and (4.7), $I(\cdot)$ denotes an indicator function, and z was chosen to be 2 with empirical normal approximations for the internal parameters. When $k = 1, 2$, $\delta_{\hat{\xi}_k} = \delta_{\hat{\eta}_{s,T+k}}$. The collection of the historical estimates for the overdispersion parameter and GPD means aims at providing a baseline pattern when no outbreaks have happened during the monitoring procedure. The true posteriors for those latent parameters don't have analytic expression thus the normal approximation technique acts as a rudimentary screening for potential outbreaks. Nonetheless, such preliminary Z-score-based aberration indicators contribute to a refined p-value-based outbreak detection approach in the next step.

4.3.3 Step 3: P-value adaptation

Since an outbreak is associated with the occurrence in a region of cases of an illness or health-related events clearly in excess of normal expectancy, public health practitioners are more likely to claim a disease outbreak when the observed incidence count in a region is unusually high or takes a drastic jump, regarding the early outbreak detection application. It would be meaningless to claim an outbreak when no significant deviation takes place from the past data pattern. Therefore, one-sided p-values become a natural measurement for the outbreak evidence against $H_{0,k}^{(s)}$. Not only

can the p-values be converted to binary outcomes given a user-defined threshold as in Farrington's algorithm, but also provide the magnitude of extremeness for the observations under the null hypotheses. However, p-values will have reduced significance at a contaminated baseline estimate when the sliding window contains a mixture of non-epidemic and epidemic data. To boost the signal of a potential outbreak, we propose an adapted p-value computation with the latent aberration indicators obtained in Step 2. In the k -th iteration, the adapted p-value at location s with the observed count $y_{s,T+k}$ is calculated as in (4.8),

$$p_{s,T+k}^* = \begin{cases} \tilde{p}_{s,T+k}, & \text{if } \delta_{\hat{\xi}_k} = 1. \\ p_{s,T+k}, & \text{if } \delta_{\hat{\xi}_k} = 0. \end{cases} \quad (4.8)$$

where

$$\tilde{p}_{s,T+k} = \begin{cases} Pr \left(Y_{s,T+k} > y_{s,T+k} | \bar{\eta}_{s,T+k-1}, \bar{\xi}_{k-1}, H_{0,k}^{(s)} \right), & \text{if } \delta_{\hat{\eta}_{s,T+k}} = 1. \\ p_{s,T+k}, & \text{if } \delta_{\hat{\eta}_{s,T+k}} = 0. \end{cases}$$

The adaptation for the p-value in (4.8) is determined by $\delta_{\hat{\xi}_k}$ and $\delta_{\hat{\eta}_{s,T+k}}$. When both the global and local latent aberrations for location s are triggered, the corresponding p-value will be adaptively calculated using smaller estimates of baseline GPD mean and overdispersion, in order to increase the significance of a potential outbreak. Otherwise, the regular Bayesian posterior predictive p-values are computed to quantify the evidence against $H_{0,k}^{(s)}$ as in (4.9),

$$\begin{aligned} p_{s,T+k} &= Pr \left(Y_{s,T+k} > y_{s,T+k} | \mathcal{D}_k, \mathcal{D}_{k-T-1}^*, H_{0,k}^{(s)} \right) \\ &= \sum_{Y_{s,T+k}} I(Y_{s,T+k} > y_{s,T+k}) p(Y_{s,T+k} | \mathcal{D}_k, \mathcal{D}_{k-T-1}^*, H_{0,k}^{(s)}) \\ &= \int \left(\sum_{Y_{s,T+k}} I(Y_{s,T+k} > y_{s,T+k}) p(Y_{s,T+k} | \boldsymbol{\Theta}) \right) p(\boldsymbol{\Theta} | \mathcal{D}_k, \mathcal{D}_{k-T-1}^*, H_{0,k}^{(s)}) d\boldsymbol{\Theta} \\ &= \int Pr(Y_{s,T+k} > y_{s,T+k} | \boldsymbol{\Theta}) p(\boldsymbol{\Theta} | \mathcal{D}_k, \mathcal{D}_{k-T-1}^*, H_{0,k}^{(s)}) d\boldsymbol{\Theta} \\ &= \int \int Pr(Y_{s,T+k} > y_{s,T+k} | \eta_{s,T+k}, \xi_k) p(\eta_{s,T+k}, \xi_k | \mathcal{D}_k, \mathcal{D}_{k-T-1}^*, H_{0,k}^{(s)}) d\eta_{s,T+k} d\xi_k. \end{aligned} \quad (4.9)$$

The p-value adaptation step has two major benefits in the detection procedure. First of all, it can avoid weakening signals of potential outbreaks if the sliding window contains a mixture of

baseline data and outbreak data. When the (4.2) is fitted on the count data contaminated by unusually large observations, the conditional GPD mean $\mu_{s,T+k}$ will be overestimated as well as its overdispersion parameter λ_k thus leading to an overestimated variance $\mu_{s,T+k}/(1 - \lambda_k)^2$, which yields a large p-value weakening the evidence against $H_{0,k}^{(s)}$. Secondly, When no aberration alarms are raised, the Bayesian posterior predictive p-value incorporates the uncertainty of the model parameters by integrating the conventional p-value over their joint posterior density. Hence, the Bayesian posterior predictive p-value provides a baseline significance level of rejecting $H_{0,k}^{(s)}$ when no outbreak occurs. In (4.9), $\eta_{s,T+k}$ and ξ_k are sufficient for the computation of Bayesian p-value because $\eta_{s,T+k}$ is a function of all other parameters and $p_{s,T+k}$ can be approximated by the average of p-values conditional on the joint posterior samples of $\eta_{s,T+k}$ and ξ_k . Next, the adapted p-values can be accumulated recursively for outbreak detection in the following step.

4.3.4 Step 4: Decision-making and algorithm update

The decision-making step is based on combined p-values for two reasons: 1) some individual p-values obtained with small magnitude during the iterative procedure will not be necessarily caused by an outbreak, especially if the monitored data have a highly overdispersed pattern and the procedure hasn't encountered any extreme observations due to overdispersion. 2) The successive modeling fitting procedure with a sliding window has training data set with overlaps, which introduces a dependent association among the sequential hypothesis tests. For example, the intersection of two consecutive training data sets, $\mathcal{D}_{k-1}$ and $\mathcal{D}_k$, is $\mathcal{D}_{k-1} \cap \mathcal{D}_k = \{\mathbf{Y}_{k+t}\}_{t=0}^{T-1}$. Therefore, such decision-making in the online outbreak detection procedure will integrate dependent historical information to assess the ongoing dynamics in the surveillance system via a combined p-value approach. The combined p-value approach is known as the global test (also named omnibus test) of p-values and it is a popular technique in high-dimensional genetic data analysis for multiple hypothesis tests. A plethora of research work has been done to combine independent and dependent p-values for the global test in the genetic study. Instead of combining all available p-values and performing a single global test, the online procedure keeps updating recent p-values sequentially until an outbreak is detected thus leading to a series of affiliated global tests. With the adapted p-value calculated in

the previous step, the k -th cumulative detector statistics will be updated as in (4.10),

$$Q_{s,T+k} = Q_{s,T+k-1} + g(p_{s,T+k}^*)$$

where

$$Q_{s,T+k-1} = \sum_{i=1}^{k-1} g(p_{s,T+i}^*) \text{ and } Q_{s,T+0} = 0. \quad (4.10)$$

In (4.10), $g(\cdot)$ is a transformation function of the p-values. In the conventional tests of combined p-values, the classical choices for $g(\cdot)$ include log transformation (Fisher, 1932), inverse Gaussian transformation (Stouffer et al., 1949) and inverse gamma transformation (Lancaster, 1961). When p-values follow the identical and independent uniform distribution under the global null hypothesis, $Q_{s,T+k}$ will have simple limiting distributions. However, When p-values are dependent, the resulting distribution of $Q_{s,T+k}$ becomes complicated and adjustments can be made to account for the dependence via using an effective number of tests (Cheverud, 2001; Nyholt, 2004; Li and Ji, 2005; Gao et al., 2008; Galwey, 2009), using re-sampling methods such as permutation tests for an empirical distributional approximation for the combined p-values (Westfall and Young, 1993; Good, 2013), or applying generalized multivariate theory to the classical choices of $g(\cdot)$ under dependence (Brown, 1975; Kost and McDermott, 2002; Yang et al., 2016). Recent powerful combined p-value tests include the Harmonic Mean P-value (HMP) approach (Wilson, 2019) and Cauchy Combination Test (CCT) approach (Liu and Xie, 2020), which can handle arbitrary correlation structure among the test statistics although HMP approach doesn't have the exact additive property shown in (4.10).

Therefore, the real-time outbreak detection indicator $\delta_{s,T+k}$ for county s at iteration k is determined as in (4.11),

$$\delta_{s,T+k} = \begin{cases} I(Q_{s,T+k} > q_{s,T+k}^\alpha), & \exists k^* \leq k, s.t. \delta_{\hat{\xi}_{k^*}} = 1, \\ 0, & \text{otherwise,} \end{cases} \quad (4.11)$$

where $q_{s,T+k}^\alpha$ is the threshold associated with the corresponding method and α is the nominal level for false detection control pre-determined by practitioners. Such a decision rule aims at controlling false detection rates due to data overdispersion and increasing detection power when outbreaks

happen. At the end of iteration k , The baseline-resembling sets for the overdispersion parameter and the conditional GDP mean are updated as follows,

$$\begin{aligned}\mathcal{H}_{s,k} &= \mathcal{H}_{s,k-1} \cup \left\{ (1 - \delta_{\hat{\eta}_{s,T+k}}) \hat{\eta}_{s,T+k} + \delta_{\hat{\eta}_{s,T+k}} \bar{\hat{\eta}}_{s,T+k-1} \right\} \\ \Xi_{s,k} &= \Xi_{s,k-1} \cup \left\{ \left((1 - \delta_{\hat{\xi}_k}) \hat{\xi}_k + \delta_{\hat{\xi}_k} \bar{\hat{\xi}}_{k-1}, (1 - \delta_{\hat{\xi}_k}) \hat{\sigma}_{\xi_k} + \delta_{\hat{\xi}_k} \bar{\hat{\sigma}}_{\xi_{k-1}} \right) \right\}\end{aligned}\tag{4.12}$$

The main purpose of (4.12) is to obviate contamination of the baseline collection by any aberrant estimates thus enhancing the sensitivity of the algorithm. Meanwhile, the posterior inference will also be collected in Table 14 for future use as prior information. In the next subsection, we are going to implement the eclectic framework via numerical studies and demonstrate its performance in different aspects.

4.4 Numerical study

4.4.1 Simulation study

In our simulation, we are going to illustrate the performance of the BOSTON-PUPA approach from two aspects: (1) Model parameter recovery and in-sample model prediction; (2) Online outbreak detection. For a better replication of the actual data pattern, the parameters in the model in (4.2) is based on the model inference on the partial real data of daily confirmed COVID-19 case count starting from June 15, 2020, which we assume to be in the non-outbreak phase before the news report of an outbreak in the fall. The Boston Globe reported on October 26, that COVID-19 cases in the state had risen sharply on October 22,

”After a sudden jump Thursday, Massachusetts coronavirus cases have been maintaining levels we haven’t seen in months, raising concern among experts that the state might need to consider rolling back some parts of its reopening process.”

A detailed real data description will be given in the real application section. The parameters include fixed effects β_{den} (related to county-wise population density, a ratio of its population to the corresponding area), β_{DOW} (related to the Day of Week effect), and β_{time} (related to a linear time trend), as well as other spatio-temporal hyperparameters for the random effects. They are summarized in Table 15. For constant terms, W is based on the border-sharing adjacency of 14

counties in Massachusetts. Due to the small populations of Dukes and Nantucket, we merged these two neighboring island counties into one to avoid extreme population imbalance, and P_s are logarithms of the county populations recorded in 2018 in Table 16.

Parameter	Symbol	Value	Hyperparameter	Symbol	Value
Intercept	β_0	-13.5752	Overdispersion	λ	.4448
Population density effect	β_{den}	.2905	Temporal correlation	ρ_ϵ	.5123
Day of Week effect	β_{DOW}	.1762	Temporal variance	σ_ϵ^2	.2609
Time trend effect	β_{time}	.3056	Spatial correlation	ρ_ϕ	.3588
			Spatial variance	σ_ϕ^2	.7275

Table 15: Parameter setup throughout the simulation study

Spatial ID	County name	Population	$\log(\text{Population})$ (P_s)
1	Plymouth	518132	13.1580
2	Berkshire	126348	11.7468
3	Barnstable	213413	12.2710
4	Norfolk	705388	13.4665
5	Bristol	564022	13.2428
6	Suffolk	807252	13.6014
7	Franklin	70963	11.1699
8	Hampshire	161355	11.9914
9	Essex	790638	13.5806
10	Hampden	470406	13.0614
11	Dukes and Nantucket	28679	10.2639
12	Middlesex	1614710	14.2947
13	Worcester	830839	13.6302

Table 16: County populations in Massachusetts in 2018

4.4.1.1 Model parameter recovery and in-sample model prediction Due to the iterative feature of the BOSTON-PUPA framework, the quality of model inference and prediction is influenced by choices of the sliding window size (T) and historical information discounting factor a_0 . For a proof of concept, we conducted different combinations of sliding window sizes $T = 14, 21, 28, 35, 42$ and discounting factors $a_0 = 0.25, 0.5, 1$. The total number of simulations is 200. For each simulation, the total number of simulated case counts at each location is $T_0 = 200$. To illustrate the performance of parameter recovery for the baseline data, we only simulated the data without any outbreak, and outbreak detection performance will be demonstrated in the next subsection. The

measurement of the parameter recovery is defined as a binary outcome of whether or not the true parameters from the setup are captured by their corresponding 95% posterior credible interval in the BOSTON-PUPA procedure. As for the measurement of in-sample prediction, we measure the Mean-Squared-Error (MSE) between the true observation and in-sample prediction for the last day in the sliding window, because latent aberration assessment in step 2 is dependent on the last-day estimates $\hat{\eta}_{s,T+k}$ and historical baseline estimates $\bar{\eta}_{s,T+k-1}$ in (4.7). Therefore, there are in total $(T_0 - T) \times 200$ model fittings in the simulation study, from which we summarized the performance of parameter recovery and in-sample model prediction. The results are shown in Table 17 and 18,

From Table 17, we can see that all the fixed effect and hyperparameter recovery rates increase with a larger sliding window size (T) and a smaller discounting factor (a_0), except for the temporal variance (σ_ϵ^2). For the majority of the parameters, a sliding window with a larger width provides more information in the training data, which can enhance the model inference. Meanwhile, a smaller discounting factor dampens the certainty of historical information prior and grants more flexibility for the INLA solver to explore the posterior distribution for the current training data. Since all the internal parameter representations are assigned with a Gaussian prior, the original variance σ^2 in the prior turns into σ^2/a_0 after discounting and shows complete historical information ($a_0=1$) does not yield good model inference according to Table 17. However, since the logarithmic GDP mean in (4.2) contains a summation of separable spatial effect and temporal effect, a larger sliding window size tends to reduce more uncertainty about the spatial effect than the temporal effect, which can explain why there is an opposite relationship between parameter recovery rate and sliding window size for the temporal variance (σ_ϵ^2). Although the hyperparameter recovery rates for the spatial effects and the temporal effects are not as good as the other parameters, it is still satisfactory to get recovery rates above 70% for these second-level parameters with appropriate selections for T and a_0 . From Table 18, the choice of discounting factor has a negligible impact on the MSE. Although there tends to be a slight positive relationship between sliding window size and MSE, the increments are not significant and it is related to different total numbers of model fittings ($(T_0 - T) \times 200$) in the MSE calculation.

An appropriate selection of sliding window size is a trade-off between methodological efficiency and detection timeliness. As is demonstrated in Farrington and Andrews (2003), the size of training data to reflect a baseline pattern ranges from 2 to 8 weeks, depending on the specifics in the

Parameter	a_0	(T=)14	21	28	35	42
β_0	0.25	0.9910	0.9900	0.9907	0.9870	0.9847
	0.5	0.9703	0.9658	0.9692	0.9643	0.9673
	1	0.5159	0.6509	0.7458	0.8068	0.8458
β_{den}	0.25	0.9442	0.9439	0.9419	0.9403	0.9394
	0.5	0.9234	0.9209	0.9210	0.9194	0.9181
	1	0.6557	0.7469	0.7938	0.8240	0.8462
β_{DOW}	0.25	0.9485	0.9546	0.9544	0.9544	0.9509
	0.5	0.9620	0.9639	0.9616	0.9605	0.9580
	1	0.9088	0.9303	0.9352	0.9377	0.9348
β_{time}	0.25	0.9870	0.9855	0.9857	0.9824	0.9785
	0.5	0.9701	0.9703	0.9710	0.9671	0.9672
	1	0.5428	0.6732	0.7756	0.8363	0.8690
λ	0.25	0.9633	0.9650	0.9668	0.9642	0.9633
	0.5	0.9691	0.9702	0.9695	0.9663	0.9659
	1	0.9313	0.9413	0.9453	0.9502	0.9478
ρ_ϵ	0.25	0.8921	0.9163	0.9276	0.9328	0.9352
	0.5	0.9010	0.9122	0.9262	0.9297	0.9360
	1	0.6213	0.7358	0.8137	0.8592	0.8735
σ_ϵ^2	0.25	0.8448	0.8340	0.8178	0.7934	0.7643
	0.5	0.8269	0.7998	0.7711	0.7452	0.7206
	1	0.4845	0.5213	0.5471	0.5658	0.5722
ρ_ϕ	0.25	0.5702	0.6649	0.7343	0.7587	0.7935
	0.5	0.5535	0.6348	0.7044	0.7501	0.8020
	1	0.5157	0.6242	0.6912	0.7542	0.7947
σ_ϕ^2	0.25	0.8796	0.8963	0.9128	0.9206	0.9267
	0.5	0.3681	0.4372	0.5365	0.6554	0.7435
	1	0.1349	0.1983	0.2714	0.3561	0.4446

Table 17: Parameter recovery rate under different combinations of sliding window size and prior discounting factor

$a_0 \backslash T$	14	21	28	35	42
0.25	9.1479	9.3548	9.4971	9.6074	9.7028
0.5	9.1456	9.3497	9.4958	9.6064	9.7019
1	9.1395	9.3471	9.4942	9.6051	9.7009

Table 18: MSE for the last-day prediction in the sliding window

surveillance system such as the disease outbreak span or data format from regulatory authorities. The computational cost is another criterion to choose a better combination of tuning parameters. However, due to the initial value self-correction feature of the internal rinla program, the time consumption can have a wide range in different simulations, which is less reliable to quantify the model computational speed. In general, a larger sliding window tends to have a higher computational cost. Taking into account all the combinations of sliding window sizes and discounting factors, we can moderately select $T = 28$ and $a_0 = 0.25$ for the procedure BOSTON-PUPA. Since the optimal tuning parameter selection for our proposed framework is not the focus of this dissertation, we will just proceed with the outbreak detection performance evaluation with a reasonable choice for T and a_0 . As a comparison, we also considered using all historical data in a sequential approach in which the window size increases with newly observed data and weakly informative priors are maintained throughout the procedure. In Table 19, our Prior Updating (PU) approach with $T = 28$ and $a_0 = 0.25$ have an overall better performance than the cumulative fitting (CF) approach in terms of parameter recovery and in-sample prediction accuracy.

Performance	PU	CF	Performance	PU	CF
β_0 recovery rate	.9907	.9450	σ_ϵ^2 recovery rate	.8178	.5631
β_{den} recovery rate	.9419	.9408	ρ_ϕ recovery rate	.7343	.9715
β_{DOW} recovery rate	.9544	.9505	σ_ϕ^2 recovery rate	.9128	.9646
β_{time} recovery rate	.9857	.9315	MSE	9.4971	27.3535
λ recovery rate	.9668	.9473	Computational time	74.3770 (s)	133.0322 (s)
ρ_ϵ recovery rate	.9276	.9159			

Table 19: Performance comparison between Prior Updating (PU) approach ($T = 28, a_0 = .25$) and Cumulative Fitting (CF) approach. Computation time is calculated as the average computation time in seconds for individual model fittings in the iterative process.

4.4.1.2 Outbreak detection performance Next, we will demonstrate the outbreak detection performance of the BOSTON-PUPA framework under different scenarios of signal-to-noise ratio (SNR). We keep the parameter setup in the previous section and introduce disease outbreaks at different locations.

In the data generation process, the outbreak occurring time stamps $\tau_s, s = 1, 2, \dots, 13$, were randomly generated from a Poisson distribution with mean 100 as 103, 99, 95, 82, 102, 102, 102, 91, 89, 97, 109, 100 and 87, which divided the 13 locations into non-outbreak and outbreak phases

correspondingly. Baseline counts were generated for each location throughout $T_0 = 200$ days and outbreaks were introduced after the outbreak time stamps accordingly given the SNR in (4.13),

$$\begin{aligned} Y_{s,t}|\theta_{s,t}, \lambda &\sim GPD(\theta_{s,t} + \delta_{s,t}\theta_{s,t}^*, \lambda), \\ \theta_{s,t}^* &= (r - 1)\theta_{s,t}, \end{aligned} \tag{4.13}$$

where SNR is $r = Var(Y_{s,t}|r\theta_{s,t}, \lambda)/Var(Y_{s,t}|\theta_{s,t}, \lambda)$ and $\delta_{s,t}$ is a binary outbreak indicator for location s on day t . We also considered numerical adjustment by adding 5 to $\theta_{s,t}^*$ when introducing outbreak into baseline data, because the multiplicative association between $\theta_{s,t}$ and $\theta_{s,t}^*$ won't add meaningful outbreak to the baseline data when $\theta_{s,t}$ is close to zero. For real application in surveillance systems, practitioners will also set a minimum bar for the observed counts before claiming a reasonable outbreak.

In Step 4 of the BOSTON-PUPA procedure, we chose both classical and recent combined p-value for dependent hypothesis tests: Fisher's, Lancaster's, and Stouffer's approaches adjusted with the effective number of hypothesis tests (Li and Ji, 2005), P-value (HMP) and Cauchy Combination Test (CCT) approach from existing R packages, *poolr*, *ACAT*, and *harmonicmeanp*. Four measurements were taken to evaluate the outbreak detection performance among these methods with/without p-value adaptation: Sensitivity = $TP/(TP+FN)$, Specificity = $TN/(TN+FP)$, Proportion of False Positive (PropFP) = $FP/(TP+FP)$, and Global Error (GE) = $(FP+FN)/(TN+TP+FP+FN)$. We set $\alpha = 0.05$ for the false detection control and investigate the detection performance when SNR = 1 (No outbreak), 1.25, 1.5, 1.75, and 2 with 200 simulations for each scenario. We will summarize the outbreak detection performance of the BOSTON-PUPA procedure from the following four perspectives:

1. ***Performance enhancement by P-value Adaptation (PA)***. To illustrate the detection performance enhancement by PA technique, four evaluation measurements are aggregated by different combined p-value methods as well as all spatial locations, compared only regarding whether incorporation of PA technique is applied in the outbreak detection procedure. In Figure 19, there is a significant improvement in increasing Sensitivity and GE with PA technique. When SNR ranges from 1.25 to 2, a larger SNR leads to less variation on those four perfor-

mance measurements especially for Sensitivity and GE, as the outbreak signal strengthens. Although PA technique doesn't give a superior Specificity and PropFP over its exclusion, the benefit of PA technique can still be asserted. According to the formulae of the Specificity and PropFP, a high value of Specificity or a low value of PropFP implies a small portion of false positives are produced in the detection procedure, which is expected for an ideal outbreak detection procedure. However, with comparable Specificity and PropFP, the incorporation of PA technique with higher Sensitivity measurement has a higher power than No-PA technique. Instead of being overly conservative in the detection procedure, PA technique can boost the outbreak signal and actively facilitate accurate outbreak claims, which is a crucial attribute of an efficient online outbreak detection procedure. It is worth mentioning that PA technique grants an average of Sensitivity as high as about 75% even when the SNR = 1.25.

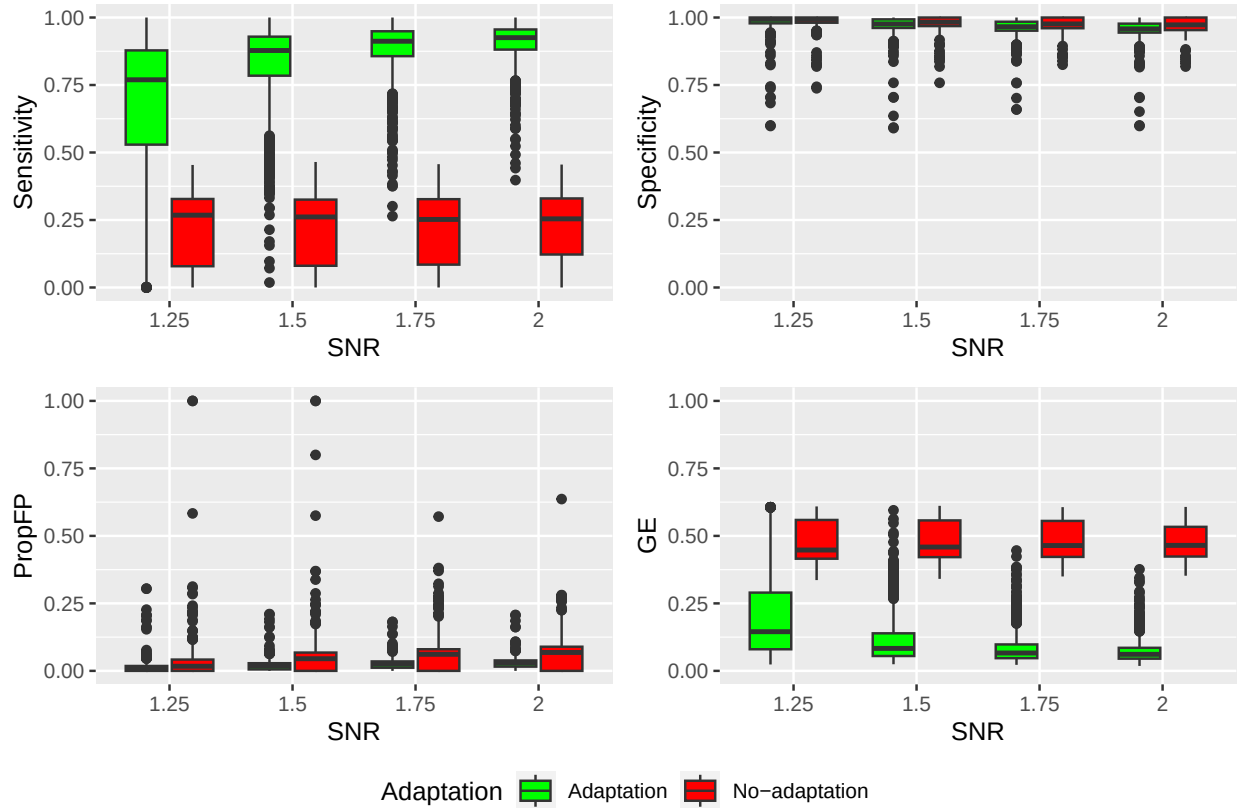


Figure 19: Signal-to-ratio vs Aggregated Performance Measurements

2. **Empirical detection power and delay.** Regarding the empirical detection power, the relative frequencies of detecting an outbreak out of 200 simulations using the BOSTON-PUPA

procedure are collected in Table 20. When $\text{SNR} = 1$, the relative frequencies can be regarded as empirical false detection rates accordingly for each location. For all of combined p-value methods, the empirical false detection rates are controlled at $\alpha = 5\%$ or with slight inflation ($5.5\% \sim 6.5\%$) except for locations with spatial IDs, 2, 3, 7, 8, and 11, which correspond to the ones with the first five smallest populations in the study. HMP and CCT methods have a better false detection control ($< 5\%$) for these less populated locations than the other methods ($4\% \sim 16\%$) using the adjustment of effective numbers of tests and Stouffer's method has the most inflated false detection rates ($12.5\% \sim 16\%$).

When $\text{SNR} = 1.25, 1.5, 1.75$, and 2, the relative frequencies can be interpreted as empirical detection powers. BOSTON-PUPA procedure demonstrates its powerful detection performance for all methods for outbreaks with different signal-to-ratios. The detection powers reach as high as over 80% when SNR starts at 1.25 and increase to nearly 100% for larger SNRs. HMP and CCT methods are uniformly more powerful than Fisher's, Stouffer's, and Lancaster's methods for locations with large populations. Although Fisher's, Stouffer's and Lancaster's methods demonstrate their high power in those less populated locations, such high powers also result from their excessive false detection rates so that outbreaks detected by HMP and CCT methods have more credibility.

When outbreaks are detected for the streaming data, it is also important to study the distributions of the detected outbreak time points for detection delay evaluation. Figure 20 presents empirical distributions for the detection delays. For the locations with large populations, the outbreak indicators are triggered around the actual outbreak time points given in the simulation study. As SNR increases, the empirical distributions of detected outbreaks tend to concentrate more on the true values. For the less populated locations, 2, 3, 7, 8, and 11, outbreaks are detected before their true occurrences, while detected outbreaks have a wider spread for the rest of the locations. For the locations with small populations, the false detection was made almost at the earliest true break, $\tau_4 = 82$, because the combined p-values based on Fisher's, Stouffer's, and Lancaster's methods had already been below α before the global aberration indicator was triggered at $\tau_4 = 82$. According to the decision rule in Step 4, these regional outbreaks will be claimed immediately when the global aberration

alarm is triggered, which leads to the regional outbreak indicators $\delta_{s,T+k}$ dominated by the global aberration indicator $\delta_{\hat{\xi}_k}$. We have also run simulations with decision rule not involving the global indicator but the false outbreaks were way too early detected and could even be claimed at the beginning of the monitoring process. Therefore, such domination helps avoid false detection too early before true outbreaks for those three methods in small areas. From this perspective, we need to acknowledge the limitation that controlled false detection rates for the less populated areas only remain until a global aberration alarm is raised, because the baseline GPD expectations have low or close to zero estimates, false outbreak detection is very sensitive to small counts. We have done the simulations as well for early (around day 50) and late (around day 150) outbreaks across 13 locations and observed similar performance among these methods. As a comparison, HMP and CCT methods show more robustness against random fluctuations caused by data overdispersion.

Based on the empirical detection delays, it is also flexible for practitioners to customize decision-making rules for individual locations. Rather than a proactive outbreak detection strategy with the first non-zero $\delta_{s,T+k}$, a scheme via aggregating the real-time outbreak indicators using a screening window can help reduce the false detection but unavoidably prolong the detection delay. Such trade-off strategies between low false detection rate and short detection delay are more related to decision theory, and it is beyond the scope of this dissertation.

3. ***Recommendations for combined p-value methods*** We would also like to compare different existing combined p-value methods and make recommendations to implement the BOSTON-PUPA procedure. Figure 21 demonstrates the spatially aggregated detection performance among those five methods when PA technique is incorporated. While Stouffer's method slightly falls short of the performance, the other four methods present competitive performance. Fisher's and Lancaster's methods perform better than HMP and CCT methods regarding Sensitivity and GE, whereas the latter two methods have better Specificity and PropFP. Before a final recommendation is made, the spatial detection performance can be taken into consideration due to potential population imbalance for online spatio-temporal data.

Method	SNR	Spatial ID												
		1	2	3	4	5	6	7	8	9	10	11	12	13
Fisher	1	0.035	0.14	0.045	0.04	0.04	0.05	0.16	0.095	0.04	0.055	0.16	0.035	0.045
	1.25	0.765	0.99	0.93	0.815	0.78	0.73	0.99	0.985	0.84	0.83	0.99	0.75	0.8
	1.5	0.895	1	0.985	0.985	0.92	0.89	1	1	0.96	0.945	1	0.9	0.975
	1.75	0.97	1	1	1	0.96	0.97	1	1	0.99	0.98	1	0.97	0.995
	2	0.98	1	1	1	0.985	0.985	1	1	0.995	1	1	0.99	0.995
Stouffer	1	0.02	0.16	0.125	0.03	0.04	0.025	0.16	0.155	0.035	0.065	0.16	0.02	0.025
	1.25	0.745	0.99	0.98	0.79	0.695	0.465	0.99	0.99	0.685	0.85	0.99	0.45	0.745
	1.5	0.905	1	1	0.97	0.875	0.725	1	1	0.895	0.95	1	0.745	0.93
	1.75	0.955	1	1	1	0.95	0.87	1	1	0.985	0.97	1	0.895	0.995
	2	0.99	1	1	1	0.96	0.945	1	1	0.995	0.99	1	0.94	1
Lancaster	1	0.04	0.125	0.04	0.04	0.045	0.05	0.155	0.06	0.04	0.045	0.16	0.05	0.05
	1.25	0.765	0.985	0.915	0.835	0.79	0.765	0.99	0.985	0.84	0.825	0.99	0.785	0.805
	1.5	0.9	1	0.98	0.98	0.925	0.93	1	1	0.97	0.95	1	0.925	0.975
	1.75	0.97	1	1	1	0.965	0.975	1	1	0.99	0.975	1	0.98	0.995
	2	0.98	1	1	1	0.985	0.985	1	1	0.995	0.995	1	0.995	0.995
HMP	1	0.065	0.015	0.035	0.05	0.065	0.06	0.02	0.035	0.05	0.05	0.025	0.06	0.06
	1.25	0.83	0.83	0.8	0.87	0.845	0.865	0.885	0.81	0.865	0.855	0.935	0.86	0.875
	1.5	0.95	0.88	0.93	0.99	0.97	0.975	0.925	0.955	0.985	0.965	0.95	0.98	0.99
	1.75	0.985	0.945	0.97	1	0.985	0.99	0.97	0.975	0.995	0.99	0.975	0.99	0.995
	2	0.99	0.97	0.98	1	1	0.995	0.97	0.985	0.995	1	0.98	1	0.995
CCT	1	0.065	0.025	0.035	0.05	0.065	0.06	0.025	0.045	0.05	0.05	0.03	0.06	0.06
	1.25	0.84	0.855	0.84	0.87	0.85	0.865	0.91	0.855	0.87	0.86	0.955	0.86	0.88
	1.5	0.955	0.915	0.945	0.99	0.97	0.975	0.955	0.97	0.985	0.97	0.96	0.98	0.99
	1.75	0.985	0.96	0.98	1	0.985	0.99	0.985	0.975	0.995	0.99	0.985	0.99	0.995
	2	0.99	0.98	0.99	1	1	0.995	0.995	0.99	0.995	1	0.99	1	0.995

Table 20: Relative frequencies of detecting an outbreak out of 200 simulations for each location using different methods for different signal-to-noise ratios.

Figure 22 and 23 demonstrate the performance of BOSTON-PUPA across different regions with/without outbreaks. Since there was no outbreak added in the simulation study when $\text{SNR} = 1$, the detection probabilities in Figure 24 only stand for false outbreak detection or type I error. While the detection procedure maintains a controlled false detection rate for regions 1, 4, 5, 6, 9, 10, 12, and 13, the false detection rates are inflated (greater than 5% but less than 20%) in the rest of the regions for the classical methods: Fisher’s, Stouffer’s, and Lancaster’s. The regions with excessive false positives correspond to the ones with the first five smallest populations in Table 16, which implies that small populations can have an impact on these classical combined p-value methods. Detection probabilities in Figure

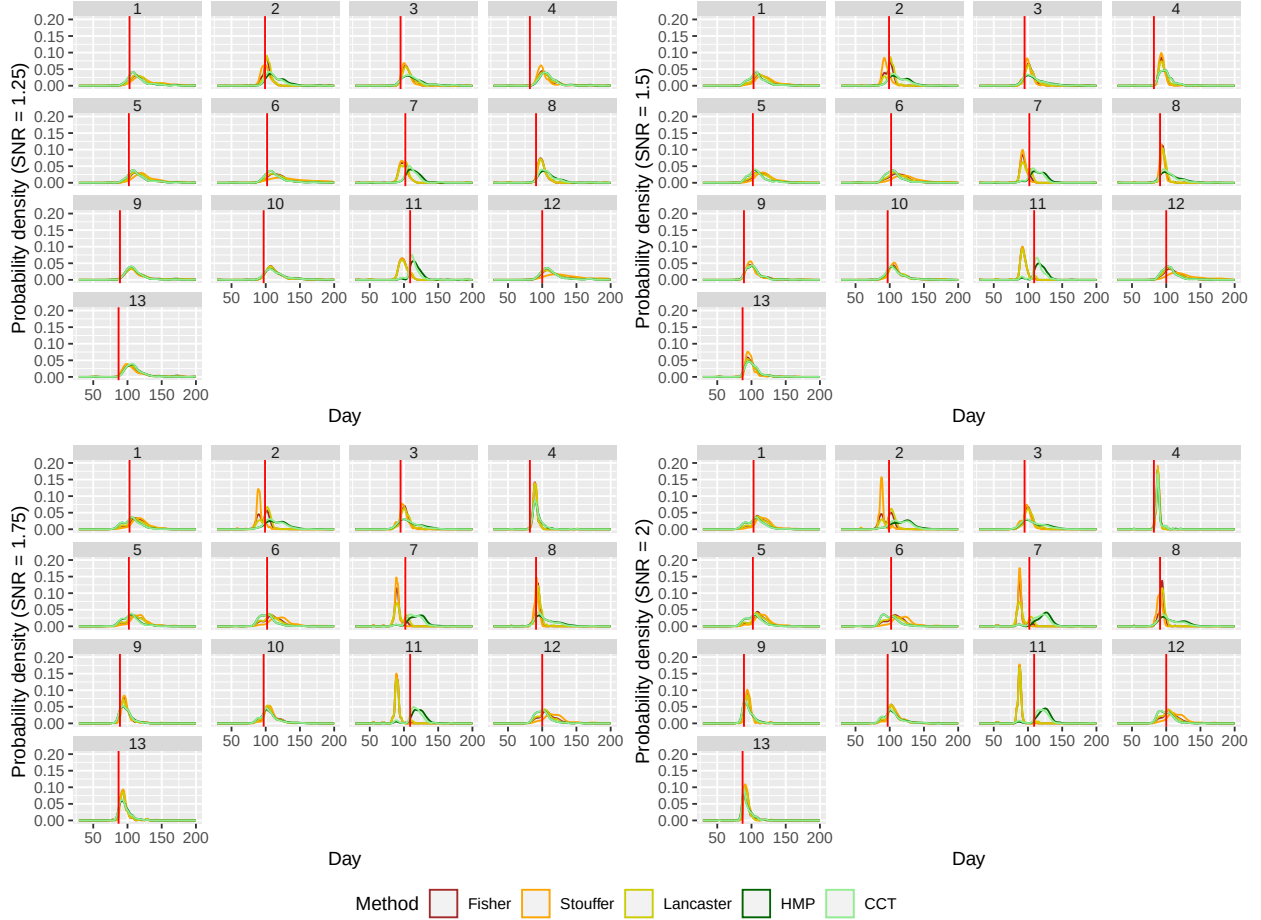


Figure 20: Empirical density plots of BOSTON-PUPA detected outbreak time point where the first non-zero $\delta_{s,T+k}$ occurs in county s for different methods with different signal-to-noise ratios (> 1). The red vertical lines represent the true outbreak time points given in the simulation study, 103, 99, 95, 82, 102, 102, 91, 89, 97, 109, 100, and 87 for each location respectively.

23 can be interpreted as false detection rate before the true outbreak timing and detection power after the true outbreak. The red lines portray the expected performance of an ideal detection procedure, which has controlled false detection rates at level α before any outbreaks happen and enjoys a timely spike-up in detection power when there are any outbreaks. For different regions and signal-to-noise ratios, HMP and CCT methods have better controlled false detection rates and provide a more better resemblance of the ideal detection pattern. Therefore, we recommend HMP or CCT method be implemented in the detection procedure to handle inflated false detection rates in the regions with smaller populations.

4. **Remarks** We would also like to make two remarks on our detection framework regarding its

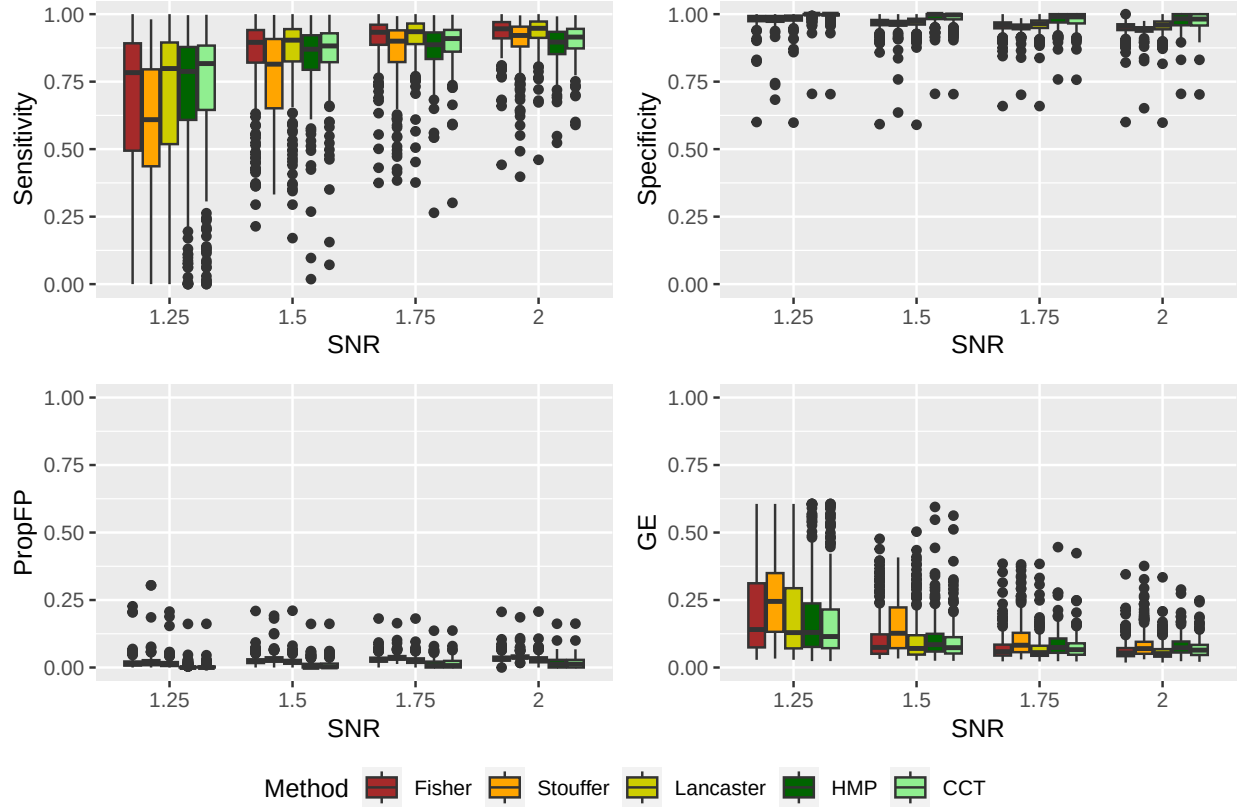


Figure 21: Signal-to-ratio vs Evaluation Metrics for different combined p-value methods: Fisher's, Stouffer's, Lancaster's, HMP and CCT when P-value Adaptation is implemented.

implementation and property: (1) Correlation matrix calculation for dependent hypotheses testing; (2) Sequential estimation of the overdispersion parameter λ .

- *Correlation matrix calculation.* As is pointed out in Cinar and Viechtbauer (2022), the adjustment for k dependent hypothesis tests the Fisher's, Stouffer's, and Lancaster's combined p-value methods stems from the scenario where a smaller number ($< k$) of independent hypothesis tests are conducted, known as the effective number of tests. The computation effective number of tests is mainly achieved through PCA of the $k \times k$ correlation matrix R_t for the dependent hypothesis tests, which is based on test statistics $t_1, t_2, \dots, t_k$ under multivariate normality assumption. However, R_t can be consistently approximated by R_Y under the regularity of the multivariate central limit theorem (Van der Vaart, 2000; Härdle and Simar, 2019), where R_Y is the $k \times k$ correlation matrix of the observations, $Y_1, Y_2, \dots, Y_k$ even if the observations are not normal. Hence, we

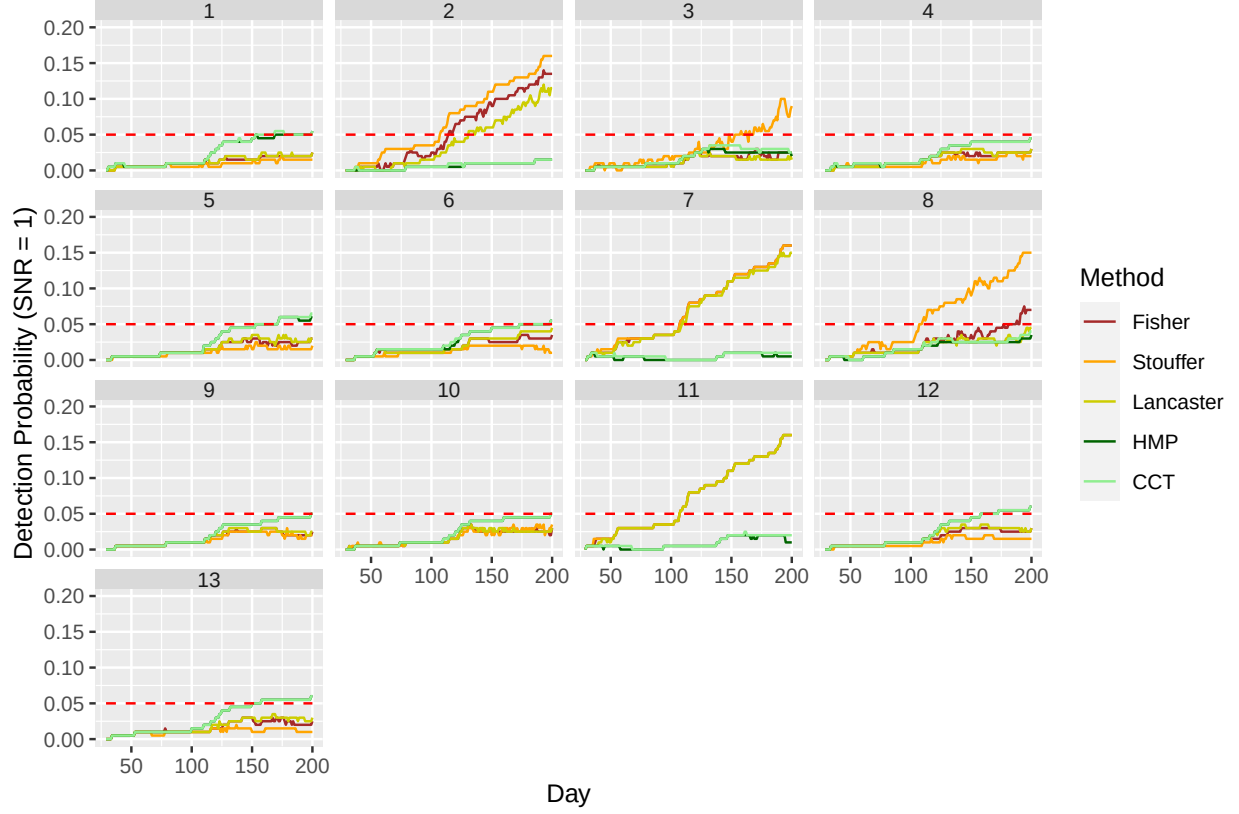


Figure 22: Traceplot of outbreak detection probabilities from day 29 to day 200 across 13 regions. Five combined p-value methods are compared at the nominal level $\alpha = 0.05$, represented by the red dashed horizontal lines.

supplied R_{Y_s} by calculating the $(T_0 - T) \times (T_0 - T)$ correlation matrix of standardized $Y_{s,t}$, $t = T + 1, T + 2, \dots, T_0$ for region s with adapted GPD mean and overdispersion respectively from the simulation study without outbreak added, which was thought to represent the baseline dependency among the hypothesis tests.

Although it has been shown the GPD model is almost symmetrical in shape, resembling a normal distribution, when the values of $\theta_{s,t}$ are as high as 8 and $0 < \lambda < 0.5$ (Consul and Jain, 1973), the dynamic adaptation on the GPD mean and overdispersion can distort the asymptotic normality of standardized $Y_{s,t}$ and the efficient number of simulations for a good baseline estimation of R_t is yet to be determined. Therefore, we need to acknowledge from another four aspects the limitation of calculating the correlation matrix of R_Y in the BOSTON-PUPA procedure if one uses either adjusted Fisher's, Stouffer's, or Lancaster's method: (1) The normality assumption for the test statis-

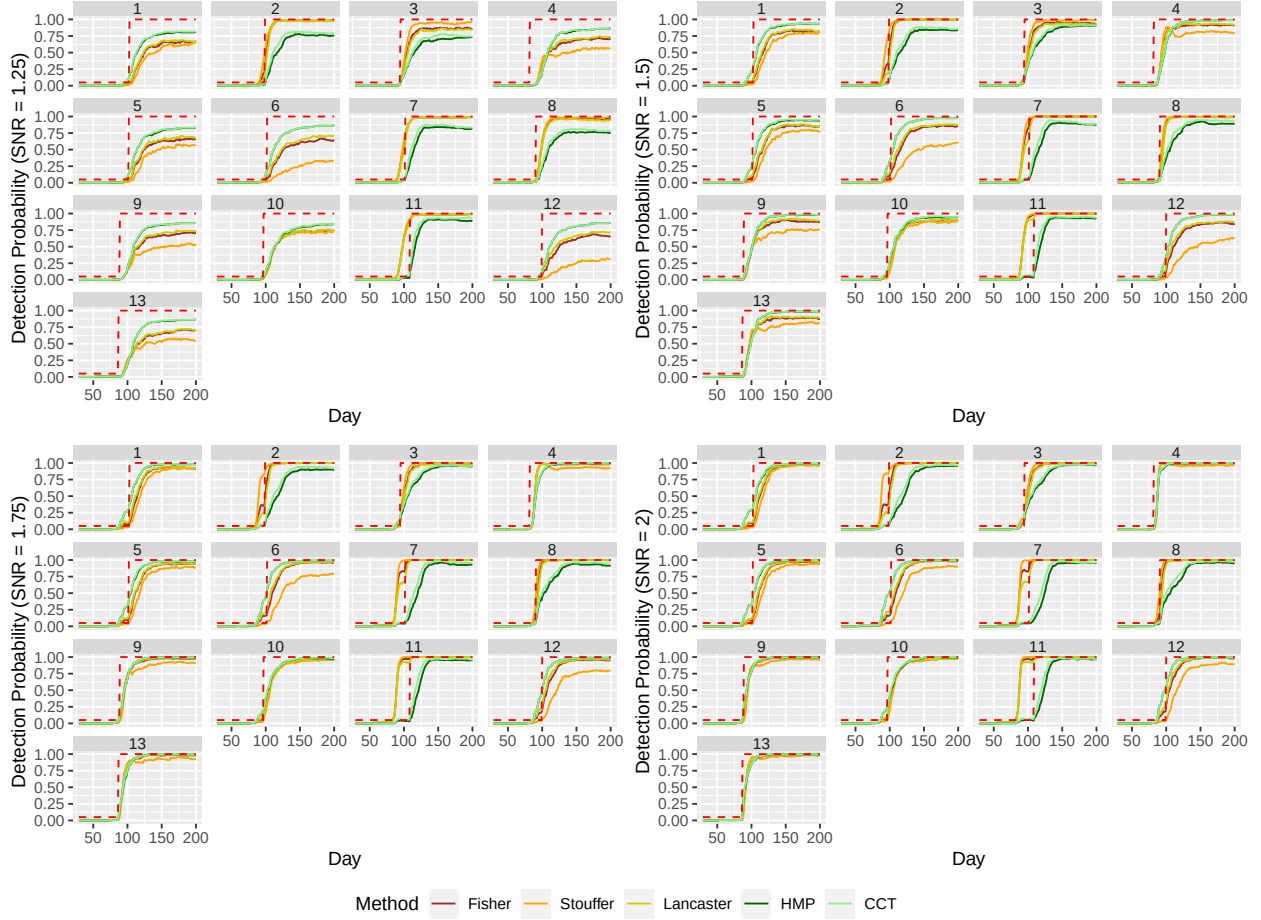


Figure 23: Traceplot of outbreak detection probabilities from day 29 to day 200 across 13 regions. Five combined p-value methods are compared at the nominal level $\alpha = 0.05$. The red dashed lines stand for an ideal detection probability pattern of maintaining at the nominal level before any outbreaks occur and spiking up promptly to 1 when there are any outbreaks.

tics is compensated by the approximated normality of the GPD model. (2) Estimation accuracy of R_Y is dependent on the simulation size, which is not the scope of this dissertation. (3) Practical concern. In the simulation study, all the parameters are provided and fixed beforehand but they will be estimated sequentially in the online detection procedure, which requires further criteria to select reasonable parameter estimates for a good baseline representation for R_Y . (4) The approaches calculating effective numbers of hypothesis tests are rather ad hoc than principled techniques and should be applied with caution (Dudbridge and Koeleman, 2004; Salyakina et al., 2005). These limitations can convince practitioners to adopt the combined p-value methods handling arbitrary correlation structures such as HMP or CCT method without dependency on R_t .

- *Sequential estimation of the overdispersion parameter* The decision rule described in Step 4 of the BOSTON-PUPA procedure involves the latent global aberration indicator δ_{ξ_k} to ensure better control of potential false detection caused by data overdispersion. To illustrate the importance of the global aberration indicator, we also collected from the simulation study the sequential estimation of $\hat{\lambda}_k = e^{\hat{\xi}_k} / (1 + e^{\hat{\xi}_k})$ and its adapted version $\bar{\lambda}_k := e^{\bar{\xi}_k} / (1 + e^{\bar{\xi}_k})$ in PA technique in Figure 24 and 25.

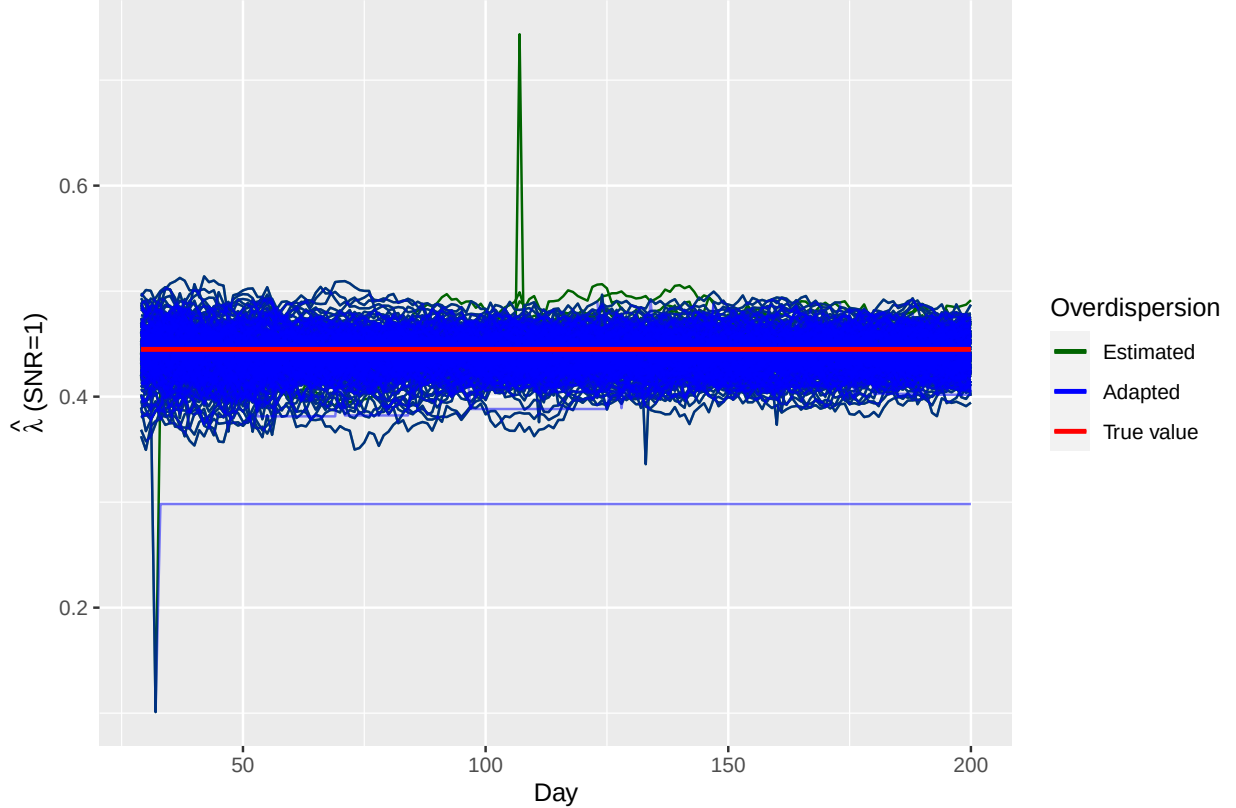


Figure 24: Trajectories of sequential estimation of overdispersion parameter λ without any outbreaks introduced ($\text{SNR} = 1$) in the simulation study. Green lines stand for $\hat{\lambda}_k$, blue lines stand for $\bar{\lambda}_k$, and the red line represents the actual value of the overdispersion parameter $\lambda = .4448$

Apart from a few numerical outliers in the computation, $\hat{\lambda}_k$ shares common trajectories with $\bar{\lambda}_k$ when $\text{SNR} = 1$ in Figure 24, which implies that $\delta_{\xi_k} = 1$ for most of the time and agrees with the underlying fact of no outbreak. The aforementioned benefit of the false detection control using the BOSTON-PUPA procedure can be supported by the pattern of the estimated and adapted λ because small p-values caused by overdispersion will not lead to an outbreak claim unless the first occurrence of a global aberration in

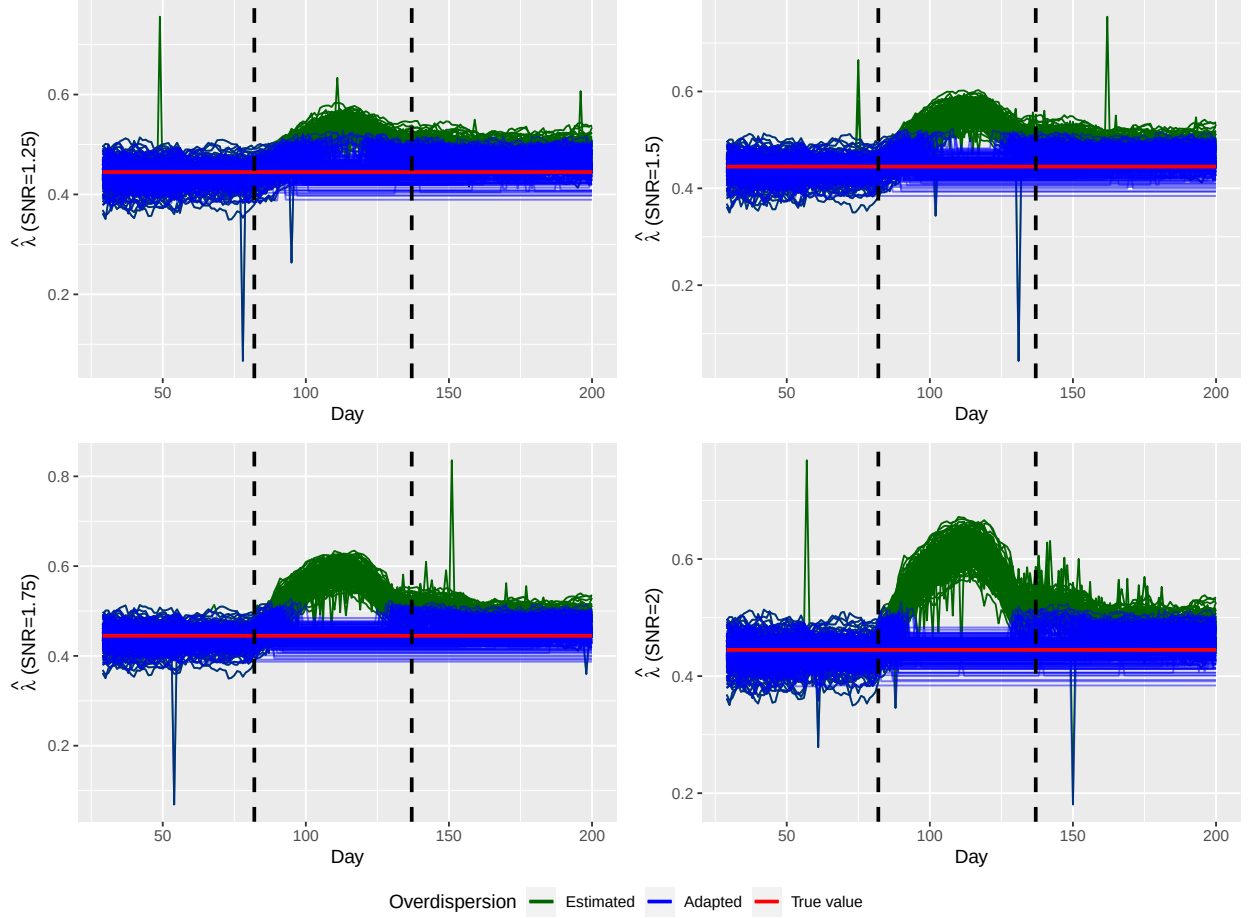


Figure 25: Trajectories of sequential estimation of overdispersion parameter λ with different SNRs in the simulation study. Green lines stand for $\hat{\lambda}_k$, blue lines stand for $\tilde{\lambda}_k$, and the red line represents the actual value of the overdispersion parameter $\lambda = .4448$. The two black vertical dashed lines stand for $\tau_4 = 82$ and $\tau_{11} + T = 109 + 28 = 137$ accordingly.

the process. Therefore, such a setup in the decision rule guarantees a long monitoring horizon where the false detection rate is controlled before any true outbreaks occur. In Figure 25, there are humps in the trajectories of $\hat{\lambda}_k$ for different SNRs, starting at around the earliest outbreak day ($\tau_4 = 82$) in the simulation study, which indicates that at least one of the regions is experiencing abnormally large confirmed case counts and triggers a global aberration alarm for any subsequent outbreak detection for the regions at risk. When the green trajectories deviate from the blue, region-specific Bayesian p-values get boosted significance from adaptations for both the abnormal overdispersion and GPD means.

The U-shape pattern of $\hat{\lambda}$ implies another property of modeling a mixture of GPD data

with identical λ but different θ in the sliding window. The humps in Figure 25 die down at around the day $\tau_{11} + T = 137$, where $\tau_{11} = 109$ is the last outbreak onset day in the simulation study. Empirically, it can be observed that the overdispersion parameter (1) is estimated consistently around its true value when the sliding window only contains non-outbreak data. (2) is overestimated when the sliding window contains a mixture of non-outbreak and outbreak data even though they are generated by identical λ , which provides helpful information for outbreak detection. (3) returns to a consistent estimate slightly larger than the true value when the sliding window contains only the outbreak data. Such property can be further investigated with theoretical evidence but it won't be discussed in this dissertation.

4.4.2 Real application

For the real application part, we will implement the BOSTON-PUPA procedure for COVID-19 daily case count data in Massachusetts with HMP and CCT methods for p-value combination. Firstly, we will conduct empirical data analysis to assert the validity of using a spatio-temporal model framework. Secondly, we implement the BOSTON-PUPA procedure to real data and report the detected outbreaks for each county.

4.4.2.1 COVID-19 Data description The daily COVID-19 confirmed case count data is available in the COVID-19 Dashboard Data Repository created by the Center for Systems Science and Engineering (CSSE) at Johns Hopkins University (Dong et al., 2020). In the evolution of the pandemic, the frequency of the case count report has shifted from a daily basis to a weekly basis. We focus on the daily case count data for the different counties in Massachusetts. To avoid excessive zero daily counts, we combine the two adjacent island counties, Dukes and Nantucket into one. The period of raw data we chose is between June 15th, 2020, and Dec 31st, 2020, with only missing values on two days, Nov 31st (Thanksgiving) and Dec 25th (Christmas), while the daily case counts on the following days Nov 27th and Dec 26th are nearly twice as usual. The total number of days in our study will be $T = 200$ from summer to the end of the year 2020. Let $Y_{s,t}$ denote the daily case count of county s on day t . Our imputation for the missing values $\hat{Y}_{s,t_1}$ and

$\hat{Y}_{s,t_2}$ and data adjustment are as follows ²:

$$\begin{cases} \hat{Y}_{s,t_1} = \lfloor Y_{s,t_1+1}/2 \rfloor & ; \hat{Y}_{s,t_1+1} = \lfloor Y_{s,t_1+1}/2 \rfloor \\ \hat{Y}_{s,t_2} = \lfloor Y_{s,t_2+1}/2 \rfloor & ; \hat{Y}_{s,t_2+1} = \lfloor Y_{s,t_2+1}/2 \rfloor \end{cases}$$

Figure 26 presents the evolution of daily confirmed case counts. Across all the regions, the daily count data have remained roughly stable or a slightly positive time trend on the mean level throughout the summer months from June to Sept until spikes happen around some time points in the fall. The positive time trend could be explained by the fluctuation of temperature, which is negatively associated with the COVID-19 mortality rate (Quilodran et al., 2021).

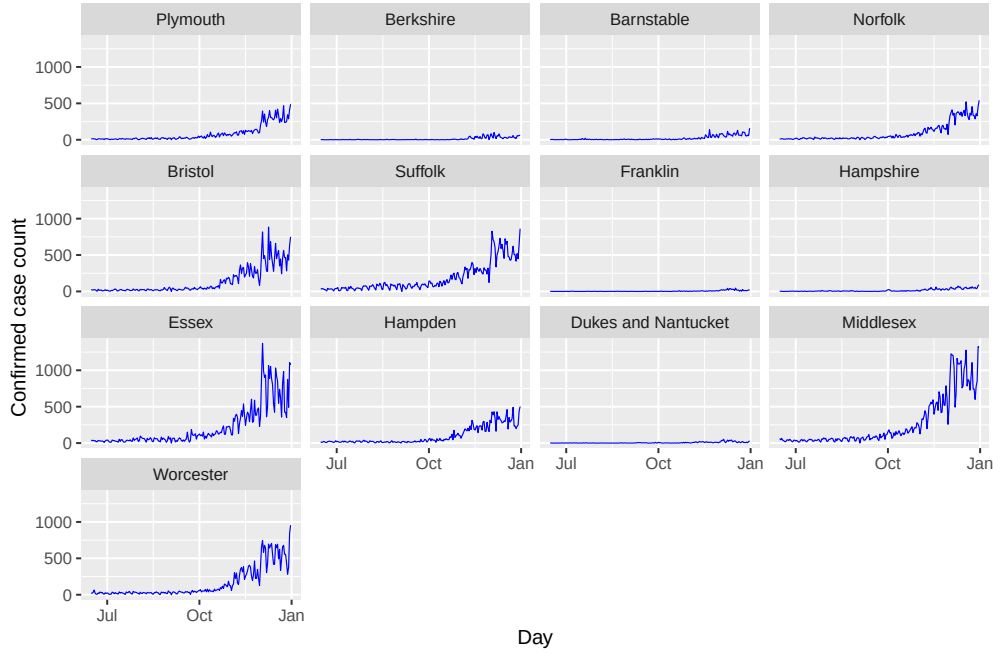


Figure 26: COVID-19 daily case count in different counties in Massachusetts from the dashboard

The spatio-temporal pattern of the count data can be verified through empirical data analysis. For the areal type of spatial data, there are two standard statistics to measure the spatial correlation, Moran's I and Geary's C (Banerjee et al., 2003) which take the forms as shown in equations (2.1) and (2.2) respectively,

² $\lfloor \cdot \rfloor$ denotes rounding to the nearest integer.

$$I = \frac{n \sum_i \sum_j w_{ij} (Y_i - \bar{Y})(Y_j - \bar{Y})}{\left(\sum_{i \neq j} w_{ij} \right) \sum_i (Y_i - \bar{Y})^2}, \quad (4.14)$$

while Geary's C takes the form,

$$C = \frac{(n-1) \sum_i \sum_j w_{ij} (Y_i - Y_j)^2}{2 \left(\sum_{i \neq j} w_{ij} \right) \sum_i (Y_i - \bar{Y})^2}, \quad (4.15)$$

where w_{ij} are the entries of the adjacency matrix W . We computed Moran's I and Geary's C as well as their p-values throughout all 200 days using *moran.test()* and *geary.test()* from the *spdep* package in R. Figure 27 shows that among the spatial data from 200 days, 95% of Moran's statistics and 92.5% of Geary's statistics have a p-value less than the default significance level 0.05. Therefore, it is reasonable to account for the spatial association among the count data in a statistical model.

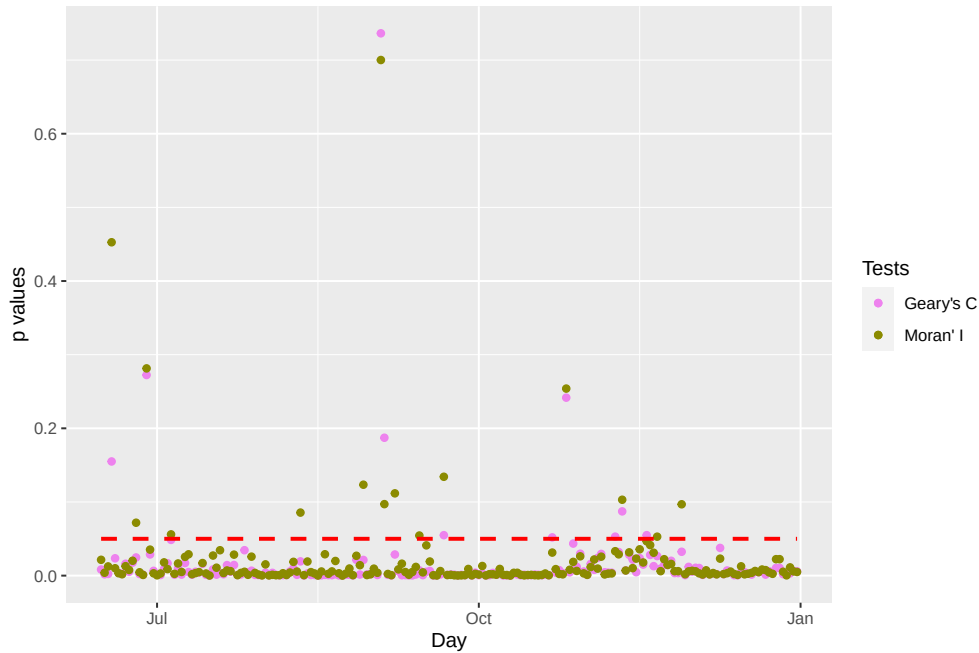


Figure 27: P-values of Moran's I and Geary's C at every time stamp. The red dashed line represents the significance level of 0.05.

Figure 28 shows the estimates of the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) across different regions. The red dashed line represents the 95% boundaries for the correlation functions. Based on the sample estimates, both types of correlation functions indicate a significant temporal dependence in the case data. Therefore, The temporal dependence

among lagged data is also important to be incorporated for the multivariate time series data.

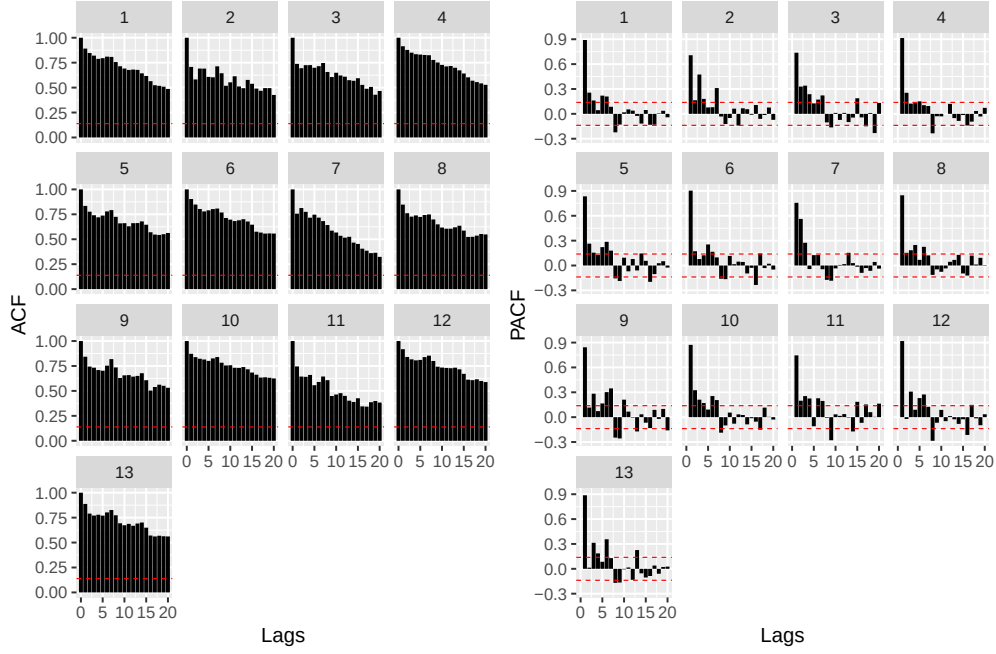


Figure 28: ACF and PACF estimates for different counties in Massachusetts. Spatial IDs 1~13 correspond to the following counties in order: Plymouth, Berkshire, Barnstable, Norfolk, Bristol, Suffolk, Franklin, Hampshire, Essex, Hampden, Dukes and Nantucket, Middlesex, and Worcester. The red dashed lines represent the boundaries of a 95% confidence interval for ACF and PACF.

4.4.2.2 Implementation of BOSTON-PUPA For the implementation of BOSTON-PUPA on the real data, we adopted the findings from the simulation study by choosing the sliding window size $T = 28$ and discounting factor $a_0 = 0.25$. We chose HMP and CCT methods in Step 4 to compute combined p-values. The first iteration of model fitting starts on 06/15/2020 (Day 1) and the first binary outbreak detection indicator on 07/13/2020 (Day 29) for all the counties. Figure 29 shows the calculated outbreak detection indicators in (4.11) in the iterative procedure. Table 21 summarizes the earliest outbreak detection days $\hat{\tau}_s$ for each county. The results are very similar between HMP and CCT methods except for Suffolk County.

As a reference, the news report of a sharp case count increase was released on 10/22/2020, which corresponds to $\tau = 130$ in the data. To assess the credibility of $\hat{\tau}_s$, we also provided trace plots of daily case counts with both $\hat{\tau}_s$ and τ in Figure 30, and the plots were separated by population size to avoid imbalanced observation scales. In Figure 30, the BOSTON-PUPA procedure detected the anomalous counts in counties with large populations such as Bristol, Suffolk, Essex, and Middlesex

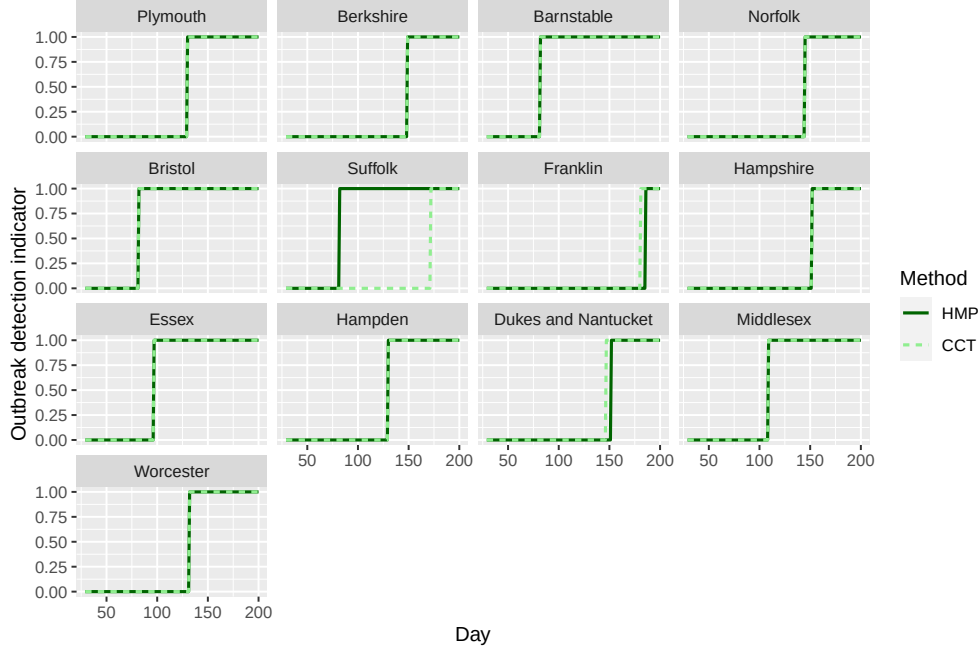


Figure 29: Calculated outbreak detection indicators $\delta_{s,T+k}$ using HMP and CCT across all counties in the BOSTON-PUPA procedure.

Spatial ID	County name	Earliest detection days	
		HMP	CCT
1	Plymouth	130	130
2	Berkshire	149	149
3	Barnstable	82	82
4	Norfolk	145	145
5	Bristol	82	82
6	Suffolk	82	172
7	Franklin	186	181
8	Hampshire	152	152
9	Essex	97	97
10	Hampden	130	130
11	Dukes and Nantucket	152	147
12	Middlesex	109	109
13	Worcester	132	132

Table 21: Earliest detection days across different counties using HMP and CCT method.

several weeks before the day of the reported outbreak with the HMP method, while CCT method reported another significant spike after τ as the detected outbreak. The anomalous escalation of case counts in Plymouth, Norfolk, Hampden, and Worcester was captured right at or near the time of the reported outbreak. As for the counties with small populations, Berkshire, Hampshire, Dukes,

and Nantucket had the detected outbreaks of about $2 \sim 4$ weeks later than $\tau = 130$. Meanwhile, Barnstable and Franklin had an early and a more delayed detection correspondingly.

4.4.2.3 Remarks There is no ground truth available about the exact outbreak timing even though a news report may be chosen as a reference for the state-wise declared outbreak. However, one important takeaway from this real application is that instead of making an outbreak claim based on the entire state, public health surveillance systems can account for the spatio-temporal pattern of case count in each county and monitor the online data spatially. From this perspective, the BOTSON-PUPA procedure enjoys an improvement in the granularity and timeliness of disease surveillance. A detected outbreak in one of the counties can stimulate further investigation for the suspicious outbreak pattern before the onset of a real pandemic. This can help public health officials customize region-specific strategies to combat the pandemic, protect people at risk, and preserve normal societal functions in the regions which have not been impacted by the disease outbreak from elsewhere. In addition, we also would like to point out that the BOSTON-PUPA procedure only handles the anomalous data dynamics from the presumed baseline pattern. Nonetheless, statistical significance cannot determine the final policy decision if the statistically detected outbreak does not align with domain knowledge such as epidemiology, public health, etc.

4.5 Summary

In this chapter, we proposed a 4-step online outbreak detection framework, BOSTON-PUPA. This iterative procedure accounts for spatio-temporal data dependence and overdispersion via a generalized Poisson distribution model. Prior Updating (PU) technique ensures good-quality statistical inference drawn from a fixed-size sliding window by leveraging historical information. P-value Adaptation (PA) technique is able to boost the significance of a true outbreak thus leading to timely online outbreak detection. The global aberration indicator, stemming from the sequential estimation of the overdispersion, provides reliable outbreak surveillance for all involved locations and facilitates a strict false detection control locally. Implementation of combined p-values in sequential analysis enables (1) Better information preservation about outbreak severity than dichotomized detection strategies solely based on upper boundaries for data prediction. (2) Dependence among individual p-values, whereas residual-combining approaches are often under the assumption of residuals being

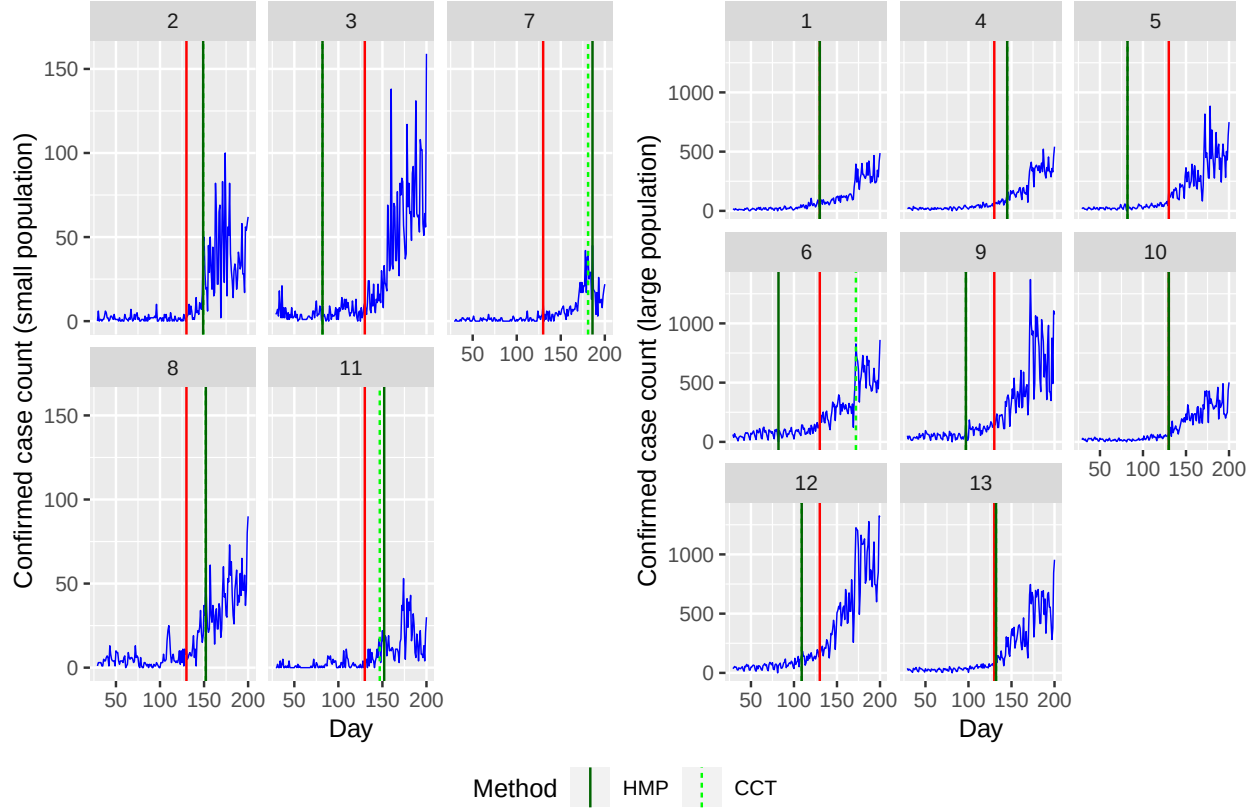


Figure 30: Trace plot of daily COVID-19 case counts in MA, 2020, with detected outbreaks using BOSTON-PUPA procedure. Red line represents a state-wise outbreak indicator from the news

independent or uncorrelated. As a holistic framework, the BOSTON-PUPA procedure is able to perform robust outbreak detection of the count data against different signal-to-noise ratios on spatially imbalanced scales. In addition, the inherent computational efficiency of the INLA method grants practitioners the flexibility to straightforwardly implement the BOSTON-PUPA procedure for real application and to conduct relevant research via large-scale simulations.

5 Discussion and Future Work

This dissertation contributes new methodological developments in online change point detection and spatio-temporal analysis. Chapter 2 and 3 focus on the innovative statistical methods with applications for high-frequency financial market microstructure analysis. E-PEF method from Chapter 2 aims at robustly detecting structural breaks in financial duration time series with a fixed-size of training data via a semi-parametric detection algorithm, and detected structural breaks can inform the practitioners the change of market trading intensity so that corresponding transactional strategies can be deployed promptly to adjust for an advantageous asset position. BVAR(1)-LCM Bayesian hierarchical model from Chapter 3 accounts for the association between transaction counts at different risk levels across multiple assets. The data interdependence on the observational level can be further elaborated via correlated latent random effects. Specified Gaussian Markov Random Field with sparsity enables fast parametric inference via INLA method thus leading to scaled-up analyses for a large number of assets in a single model. By synthesizing relevant techniques involved in univariate online change point detection and multivariate count time series modeling, BOSTON-PUPA procedure from Chapter 4 took a further step to deal with spatio-temporal online outbreak detection problems in public health surveillance. This iterative detection procedure enjoys fast computational speed with fundamental historical information retaining via Prior Updating (PU) technique. Meanwhile, streaming surveillance data are monitored by a global and location aberration indicators with P-value Adaptation (PA) technique, which enforces a satisfactory false detection control and provide timeliness and granularity for the powerful disease surveillance over multiple geographically related locations even if there is a data imbalance issue with the count observations.

As future work based on the three proposed frameworks, three potential directions are worth extended research attention.

- First, as a univariate online structural break detection procedure, the E-PEF method is able to achieve a single structural break detection given a fixed amount of training data. However, in the volatile high-frequency financial market, a fixed amount of data doesn't provide a sustainable representation of the baseline pattern for change point detection thus the monitoring horizon is limited and monitoring longer time series requires updates of training

data. In order to develop a multiple structural break detection framework, one needs to study the monitoring horizon of the E-PEF method to ensure a controlled false detection rate in a monitoring window of reasonable length. Subsequently, a clock resetting regimen can be introduced to recursively update the latest training data to capture the recent baseline pattern when (1) The incoming observations are beyond the monitoring horizon, or (2) A structural break is detected within the monitoring horizon.

- Second, BOSTON-PUPA procedure involves a sliding window as a baseline pattern without the concern about monitoring horizon for online outbreak detection, but for its multiple change point detection, a resetting rule is also very helpful because of the findings of the sequential estimation for the overdispersion parameter for generalized Poisson distribution. The sequential estimation of the overdispersion is stationary when the sliding window contains only non-epidemic or epidemic data, so that changes over the trajectory of the overdispersion parameter estimates can be good indicators when the procedure enters a transition stage between two phases. Therefore, the resetting rule can be placed over the baseline behavior of the sequential overdispersion estimation when the estimate begins to deviate from its baseline pattern. With appropriately specified p-value calculation, BOSTON-PUPA procedure with a resetting rule will have promising performance to detect not only the onset of a pandemic but also its sign of ending.
- Third, BVAR(1)-LCM framework is involved in financial application but the main idea is based on the correct specification of precision matrix for the random effects with convoluted association and numerous future research can be considered when an innovative data inter-dependent structure is developed. By taking advantage of computational efficiency, one can use the INLA method to make high-quality inference on the parameters and latent effects from sparse Gaussian Markov Random Field as long as their precision matrix is reasonably formulated. For example, the spatio-temporal framework in BOSTON-PUPA involve naive separable spatial and temporal effects also with a symmetric adjacency matrix establishing an equal influence between two neighboring regions. To demonstrate and model more complex space and time association, one can consider a non-separable spatio-temporal effects as in (Zou et al., 2012) and an asymmetrically weighted spatial-dependence matrix as in a

Simultaneous Auto-regressive (SAR) model in (Arab et al., 2008).

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6 Appendix

6.1 Derivation of conditional correlation

For a BVAR(1)-LCM model, we have

$$\begin{aligned}
Y_{j,st}|\lambda_{j,st} &\overset{ind}{\sim} Pois(\lambda_{j,st}), \\
\eta_{j,st} = \log \lambda_{j,st} &= \mathbf{Z}_j \boldsymbol{\beta}_j + \gamma_{j,t} + \alpha_{j,st}, \\
\gamma_{j,t} &= \phi_j \gamma_{j,t-1} + \omega_{j,t}, \\
\boldsymbol{\omega}_t = \begin{pmatrix} \omega_{1,t} \\ \omega_{2,t} \end{pmatrix} &\sim N(\mathbf{0}, \boldsymbol{\Sigma}_{\boldsymbol{\omega}}), \boldsymbol{\alpha}_{st} = \begin{pmatrix} \alpha_{1,st} \\ \alpha_{2,st} \end{pmatrix} \sim N(\mathbf{0}, \boldsymbol{\Sigma}_{\boldsymbol{\alpha}}), \\
\boldsymbol{\Sigma}_{\boldsymbol{\omega}} = \begin{pmatrix} \sigma_{\omega_1}^2 & \rho_{\boldsymbol{\omega}} \sigma_{\omega_1} \sigma_{\omega_2} \\ \rho_{\boldsymbol{\omega}} \sigma_{\omega_1} \sigma_{\omega_2} & \sigma_{\omega_2}^2 \end{pmatrix}, \boldsymbol{\Sigma}_{\boldsymbol{\alpha}} = \begin{pmatrix} \sigma_{\alpha_1}^2 & \rho_{\boldsymbol{\alpha}} \sigma_{\alpha_1} \sigma_{\alpha_2} \\ \rho_{\boldsymbol{\alpha}} \sigma_{\alpha_1} \sigma_{\alpha_2} & \sigma_{\alpha_2}^2 \end{pmatrix} \\
\gamma_{j,t} &\perp \alpha_{j,st}.
\end{aligned} \tag{6.1}$$

we are aiming at deriving the conditional correlation between the count data in the BVAR(1)-LCM model. We are going to derive the conditional mean, variance, and covariance in three subsections.

6.1.1 Conditional Mean of $Y_{j,st}$

The conditional mean of $Y_{j,st}$ given the covariates and covariance matrix of the latent effects is expressed as,

$$\begin{aligned}
E(Y_{j,st}|\mathbf{Z}_j \boldsymbol{\beta}_j, \boldsymbol{\Sigma}_{\boldsymbol{\omega}}, \boldsymbol{\Sigma}_{\boldsymbol{\alpha}}) &= E_{\gamma_{j,t}, \alpha_{j,st}|\boldsymbol{\Sigma}_{\boldsymbol{\omega}}, \boldsymbol{\Sigma}_{\boldsymbol{\alpha}}} [E(Y_{j,st}|\gamma_{j,t}, \alpha_{j,st}, \mathbf{Z}_j \boldsymbol{\beta}_j, \boldsymbol{\Sigma}_{\boldsymbol{\omega}}, \boldsymbol{\Sigma}_{\boldsymbol{\alpha}})] \\
&= E_{\gamma_{j,t}, \alpha_{j,st}|\boldsymbol{\Sigma}_{\boldsymbol{\omega}}, \boldsymbol{\Sigma}_{\boldsymbol{\alpha}}} [\lambda_{j,st}] = E_{\gamma_{j,t}, \alpha_{j,st}|\boldsymbol{\Sigma}_{\boldsymbol{\omega}}, \boldsymbol{\Sigma}_{\boldsymbol{\alpha}}} [\exp(\mathbf{Z}_j \boldsymbol{\beta}_j + \gamma_{j,t} + \alpha_{j,st})] \\
&= \exp(\mathbf{Z}_j \boldsymbol{\beta}_j) E_{\gamma_{j,t}|\boldsymbol{\Sigma}_{\boldsymbol{\omega}}} [\exp(\gamma_{j,t})] E_{\alpha_{j,st}|\boldsymbol{\Sigma}_{\boldsymbol{\alpha}}} [\exp(\alpha_{j,st})]
\end{aligned} \tag{6.2}$$

For an AR(1) process ($\phi_j \neq 1$), the conditional distribution of $\gamma_{j,t}$ given lag-one observation is,

$$\gamma_{j,t}|\gamma_{j,t-1}, \boldsymbol{\Sigma}_{\boldsymbol{\omega}} \sim N(\phi_j \gamma_{j,t-1}, \sigma_{\omega_j}^2) \tag{6.3}$$

The marginal distribution of $\gamma_{j,t}$ is,

$$\gamma_{j,t}|\mathbf{\Sigma}_{\omega} \sim N\left(0, \frac{\sigma_{\omega_j}^2}{1 - \phi_j^2}\right) \quad (6.4)$$

For a Poisson lognormal model, we have

Lemma 6.1 *If*

$$Y|\lambda \sim Pois(\lambda), \log(\lambda)|\mu, \sigma^2 \sim N(\mu, \sigma^2), \quad (6.5)$$

then,

$$\begin{aligned} E(Y|\mu, \sigma^2) &= \exp(\mu + \sigma^2/2) \\ Var(Y|\mu, \sigma^2) &= E(Y|\mu, \sigma^2) + \exp(2\mu) \exp(\sigma^2)(\exp(\sigma^2) - 1) \end{aligned} \quad (6.6)$$

From lemma 6.1, the conditional expectation of $Y_{j,st}$ given $\mathbf{Z}_j\boldsymbol{\beta}_j, \mathbf{\Sigma}_{\omega}, \mathbf{\Sigma}_{\alpha}$ in (6.2) will be,

$$E(Y_{j,st}|\mathbf{Z}_j\boldsymbol{\beta}_j, \mathbf{\Sigma}_{\omega}, \mathbf{\Sigma}_{\alpha}) = \exp(\mathbf{Z}_j\boldsymbol{\beta}_j) \exp\left(\frac{\sigma_{\omega_j}^2}{2(1 - \phi_j^2)}\right) \exp\left(\frac{\sigma_{\alpha_j}^2}{2}\right) = m_{j,st} \quad (6.7)$$

6.1.2 Conditional variance of $Y_{j,st}$

With (6.7), the conditional variance of $Y_{j,st}$ given $\mathbf{Z}_j\boldsymbol{\beta}_j, \mathbf{\Sigma}_{\omega}, \mathbf{\Sigma}_{\alpha}$ is expressed as,

$$\begin{aligned} Var(Y_{j,st}|\mathbf{Z}_j\boldsymbol{\beta}_j, \mathbf{\Sigma}_{\omega}, \mathbf{\Sigma}_{\alpha}) &= E(Y_{j,st}^2|\mathbf{Z}_j\boldsymbol{\beta}_j, \mathbf{\Sigma}_{\omega}, \mathbf{\Sigma}_{\alpha}) - (E[Y_{j,st}|\mathbf{Z}_j\boldsymbol{\beta}_j, \mathbf{\Sigma}_{\omega}, \mathbf{\Sigma}_{\alpha}])^2 \\ &= E(Y_{j,st}^2|\mathbf{Z}_j\boldsymbol{\beta}_j, \mathbf{\Sigma}_{\omega}, \mathbf{\Sigma}_{\alpha}) - m_{j,st}^2 \\ &= E_{\gamma_{j,t}, \alpha_{j,st}|\mathbf{\Sigma}_{\omega}, \mathbf{\Sigma}_{\alpha}} [E(Y_{j,st}^2|\gamma_{j,t}, \alpha_{j,st}, \mathbf{Z}_j\boldsymbol{\beta}_j, \mathbf{\Sigma}_{\omega}, \mathbf{\Sigma}_{\alpha})] - m_{j,st}^2 \\ &= E_{\gamma_{j,t}, \alpha_{j,st}|\mathbf{\Sigma}_{\omega}, \mathbf{\Sigma}_{\alpha}} [\lambda_{j,st}^2 + \lambda_{j,st}] - m_{j,st}^2 \\ &= E_{\gamma_{j,t}, \alpha_{j,st}|\mathbf{\Sigma}_{\omega}, \mathbf{\Sigma}_{\alpha}} [\exp(2\mathbf{Z}_j\boldsymbol{\beta}_j + 2\gamma_{j,t} + 2\alpha_{j,st}) + \exp(\mathbf{Z}_j\boldsymbol{\beta}_j + \gamma_{j,t} + \alpha_{j,st})] - m_{j,st}^2 \\ &= \exp(2\mathbf{Z}_j\boldsymbol{\beta}_j) \exp\left(\frac{2\sigma_{\omega_j}^2}{1 - \phi_j^2}\right) \exp(2\sigma_{\alpha_j}^2) + \exp(\mathbf{Z}_j\boldsymbol{\beta}_j) \exp\left(\frac{\sigma_{\omega_j}^2}{2(1 - \phi_j^2)}\right) \exp\left(\frac{\sigma_{\alpha_j}^2}{2}\right) - m_{j,st}^2 \\ &= m_{j,st} + m_{j,st}^2 \left(\exp\left(\frac{\sigma_{\omega_j}^2}{1 - \phi_j^2}\right) \exp(\sigma_{\alpha_j}^2) - 1 \right) \end{aligned} \quad (6.8)$$

6.1.3 Conditional covariance between $Y_{i,st}$ and $Y_{j,st}$

With (6.7), the conditional covariance between $Y_{i,st}$ and $Y_{j,st}$ is expressed as,

$$\begin{aligned}
Cov(Y_{i,st}, Y_{j,st} | \mathbf{Z}_i \boldsymbol{\beta}_i, \mathbf{Z}_j \boldsymbol{\beta}_j, \boldsymbol{\Sigma}_\omega, \boldsymbol{\Sigma}_\alpha) &= E_{\boldsymbol{\gamma}, \boldsymbol{\alpha} | \boldsymbol{\Sigma}_\omega, \boldsymbol{\Sigma}_\alpha} [Cov(Y_{i,st}, Y_{j,st} | \boldsymbol{\gamma}, \boldsymbol{\alpha}, \mathbf{Z}_i \boldsymbol{\beta}_i, \mathbf{Z}_j \boldsymbol{\beta}_j, \boldsymbol{\Sigma}_\omega, \boldsymbol{\Sigma}_\alpha)] \\
&+ Cov_{\boldsymbol{\gamma}, \boldsymbol{\alpha} | \boldsymbol{\Sigma}_\omega, \boldsymbol{\Sigma}_\alpha} (E[Y_{i,st} | \boldsymbol{\gamma}, \boldsymbol{\alpha}, \mathbf{Z}_i \boldsymbol{\beta}_i, \boldsymbol{\Sigma}_\omega, \boldsymbol{\Sigma}_\alpha], E[Y_{j,st} | \boldsymbol{\gamma}, \boldsymbol{\alpha}, \mathbf{Z}_j \boldsymbol{\beta}_j, \boldsymbol{\Sigma}_\omega, \boldsymbol{\Sigma}_\alpha]) \\
&= 0 + Cov_{\boldsymbol{\gamma}, \boldsymbol{\alpha} | \boldsymbol{\Sigma}_\omega, \boldsymbol{\Sigma}_\alpha} (\exp(\mathbf{Z}_i \boldsymbol{\beta}_i + \gamma_{i,t} + \alpha_{i,st}), \exp(\mathbf{Z}_j \boldsymbol{\beta}_j + \gamma_{j,t} + \alpha_{j,st})) \\
&= m_{i,st} m_{j,st} \left(\exp \left(\frac{\rho_\omega \sigma_{\omega_i} \sigma_{\omega_j}}{1 - \phi_i \phi_j} + \rho_\alpha \sigma_{\alpha_i} \sigma_{\alpha_j} \right) - 1 \right)
\end{aligned} \tag{6.9}$$

Since the joint marginal distribution of $\gamma_{i,t}$ and $\gamma_{j,t}$ is,

$$\begin{pmatrix} \gamma_{i,t} \\ \gamma_{j,t} \end{pmatrix} \sim N \left(\mathbf{0}, \begin{pmatrix} \frac{\sigma_{\omega_i}^2}{1 - \phi_i} & \frac{\rho_\omega \sigma_{\omega_i} \sigma_{\omega_j}}{1 - \phi_i \phi_j} \\ \frac{\rho_\omega \sigma_{\omega_i} \sigma_{\omega_j}}{1 - \phi_i \phi_j} & \frac{\sigma_{\omega_j}^2}{1 - \phi_j} \end{pmatrix} \right) \tag{6.10}$$

With (6.9) and (6.10), we have

$$\begin{aligned}
Cov(Y_{i,st}, Y_{j,st} | \mathbf{Z}_i \boldsymbol{\beta}_i, \mathbf{Z}_j \boldsymbol{\beta}_j, \boldsymbol{\Sigma}_\omega, \boldsymbol{\Sigma}_\alpha) &= Cov_{\boldsymbol{\gamma}, \boldsymbol{\alpha} | \boldsymbol{\Sigma}_\omega, \boldsymbol{\Sigma}_\alpha} (\exp(\mathbf{Z}_i \boldsymbol{\beta}_i + \gamma_{i,t} + \alpha_{i,st}), \exp(\mathbf{Z}_j \boldsymbol{\beta}_j + \gamma_{j,t} + \alpha_{j,st})) \\
&= m_{i,st} m_{j,st} \left(\exp \left(\frac{\rho_\omega \sigma_{\omega_i} \sigma_{\omega_j}}{1 - \phi_i \phi_j} + \rho_\alpha \sigma_{\alpha_i} \sigma_{\alpha_j} \right) - 1 \right)
\end{aligned} \tag{6.11}$$

6.1.4 Conditional correlation between counts

Combining the results from (6.7), (6.8) and (6.11), we will be able to compute the conditional correlation for the s -th stock and the t -th time interval,

$$\begin{aligned}
Corr(Y_{i,st}, Y_{j,st} | \mathbf{Z}_i \boldsymbol{\beta}_i, \mathbf{Z}_j \boldsymbol{\beta}_j, \boldsymbol{\Sigma}_\omega, \boldsymbol{\Sigma}_\alpha) &= \\
&\frac{m_{i,st} m_{j,st} \left(\exp \left(\frac{\rho_\omega \sigma_{\omega_i} \sigma_{\omega_j}}{1 - \phi_i \phi_j} + \rho_\alpha \sigma_{\alpha_i} \sigma_{\alpha_j} \right) - 1 \right)}{\sqrt{\left(m_{i,st} + m_{i,st}^2 \left(\exp \left(\frac{\sigma_{\omega_i}^2}{1 - \phi_i^2} + \sigma_{\alpha_i}^2 \right) - 1 \right) \right) \left(m_{j,st} + m_{j,st}^2 \left(\exp \left(\frac{\sigma_{\omega_j}^2}{1 - \phi_j^2} + \sigma_{\alpha_j}^2 \right) - 1 \right) \right)}}
\end{aligned} \tag{6.12}$$

6.2 Additional figures

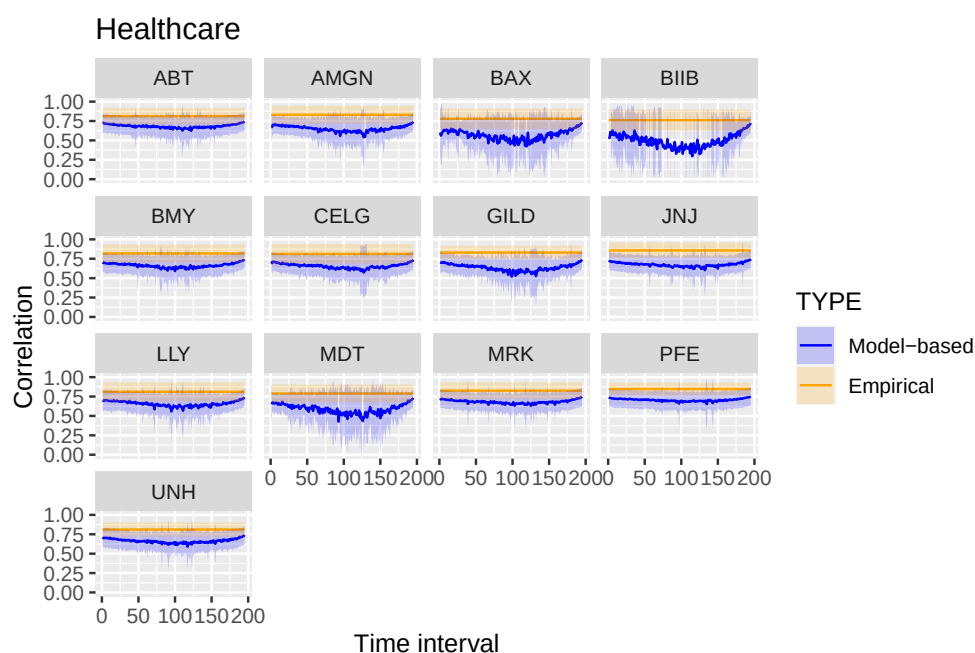


Figure 31: The trace plot of daily aggregated model-based and empirical correlations in the Healthcare sector throughout January 2023

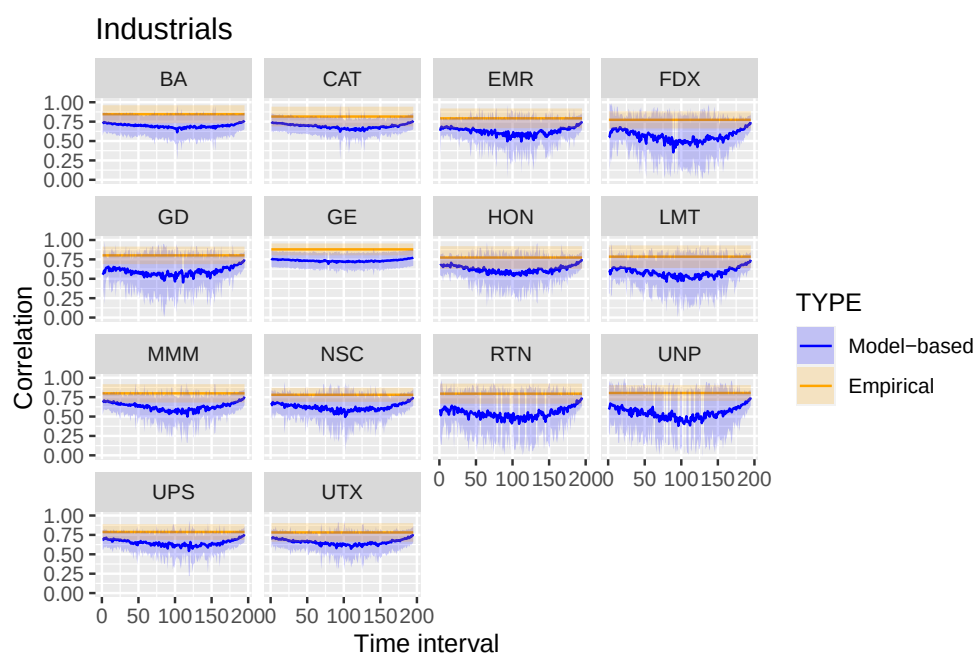


Figure 32: The trace plot of daily aggregated model-based and empirical correlations in the Industrials sector throughout January 2023

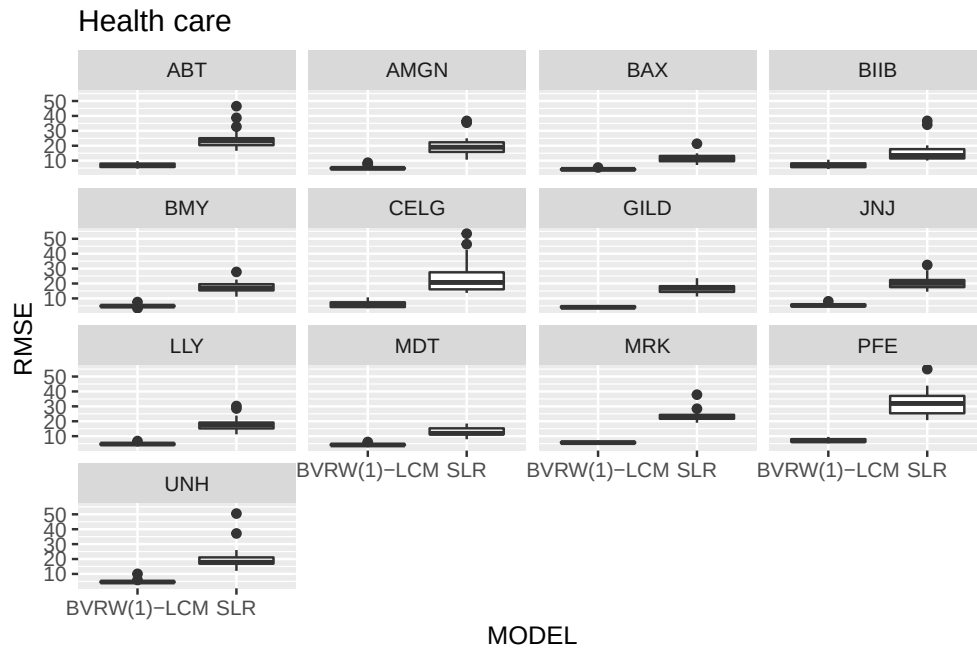


Figure 33: Box plots for the square root of MSE comparison in the Health care sector between BVRW(1)-LCM and Simple Linear Regression in January 2023

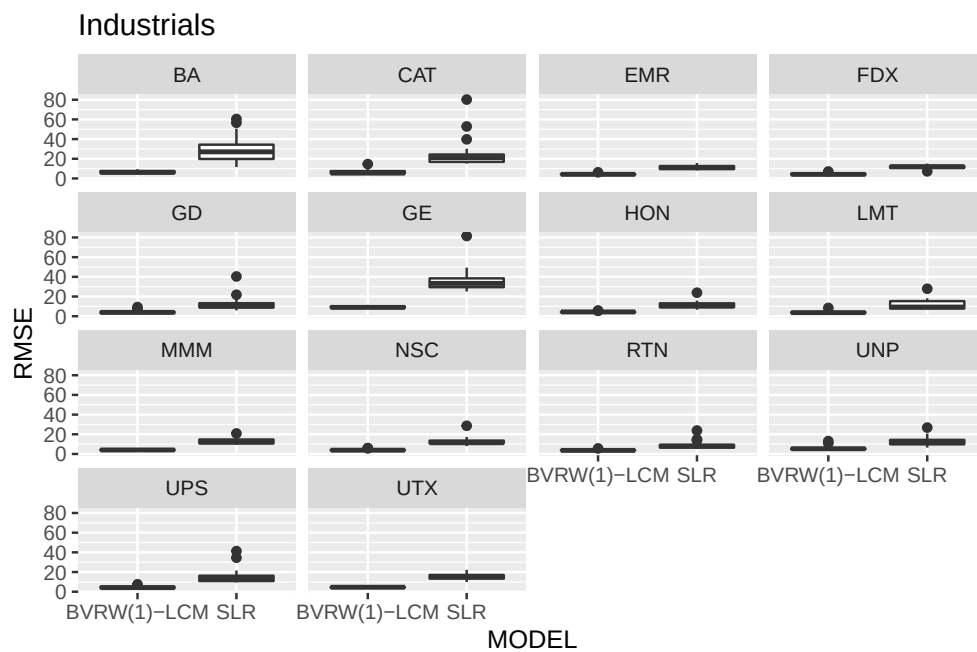


Figure 34: Box plots for the square root of MSE comparison in the Industrials sector between BVRW(1)-LCM and Simple Linear Regression in January 2023